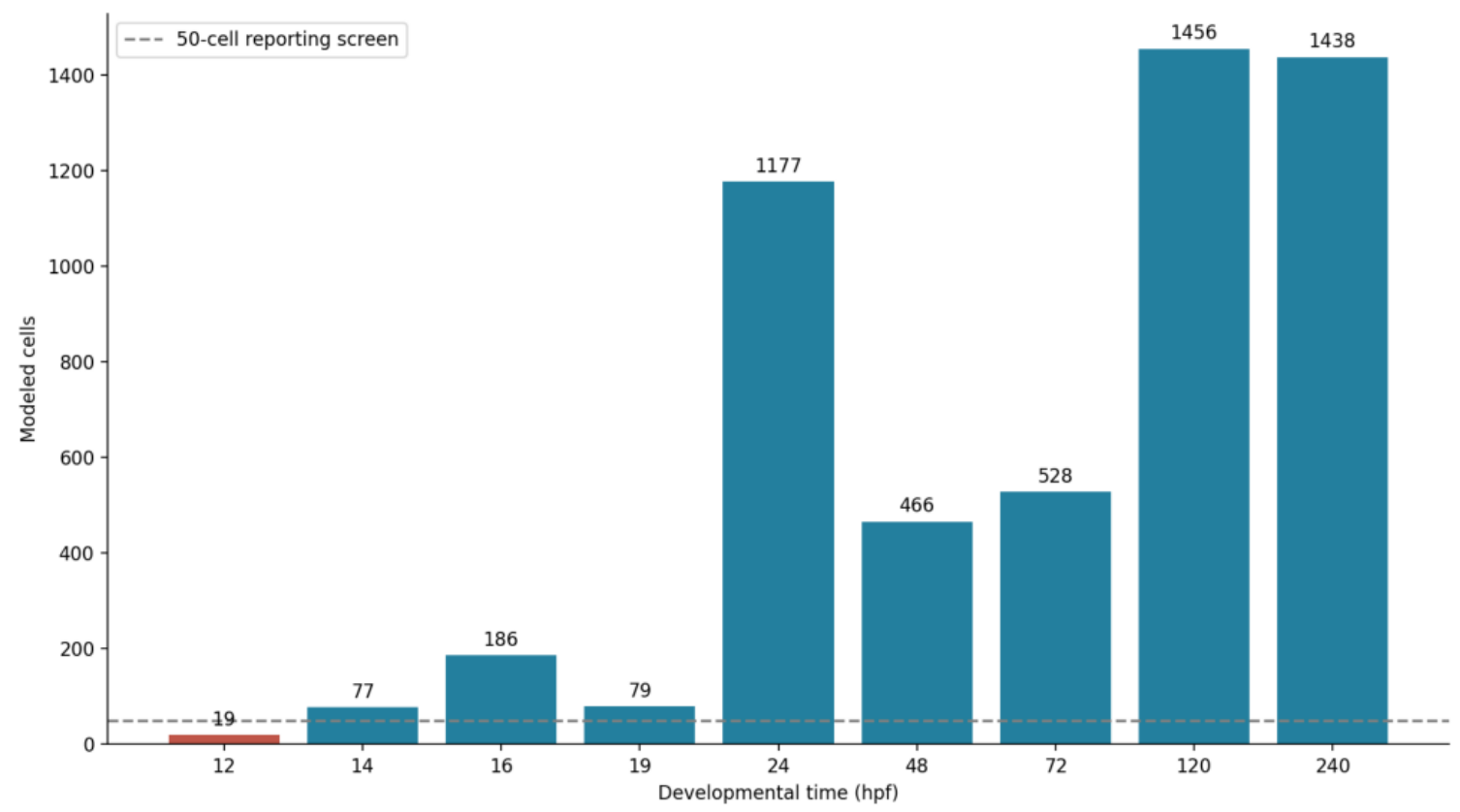


1. Cohort and sampling support



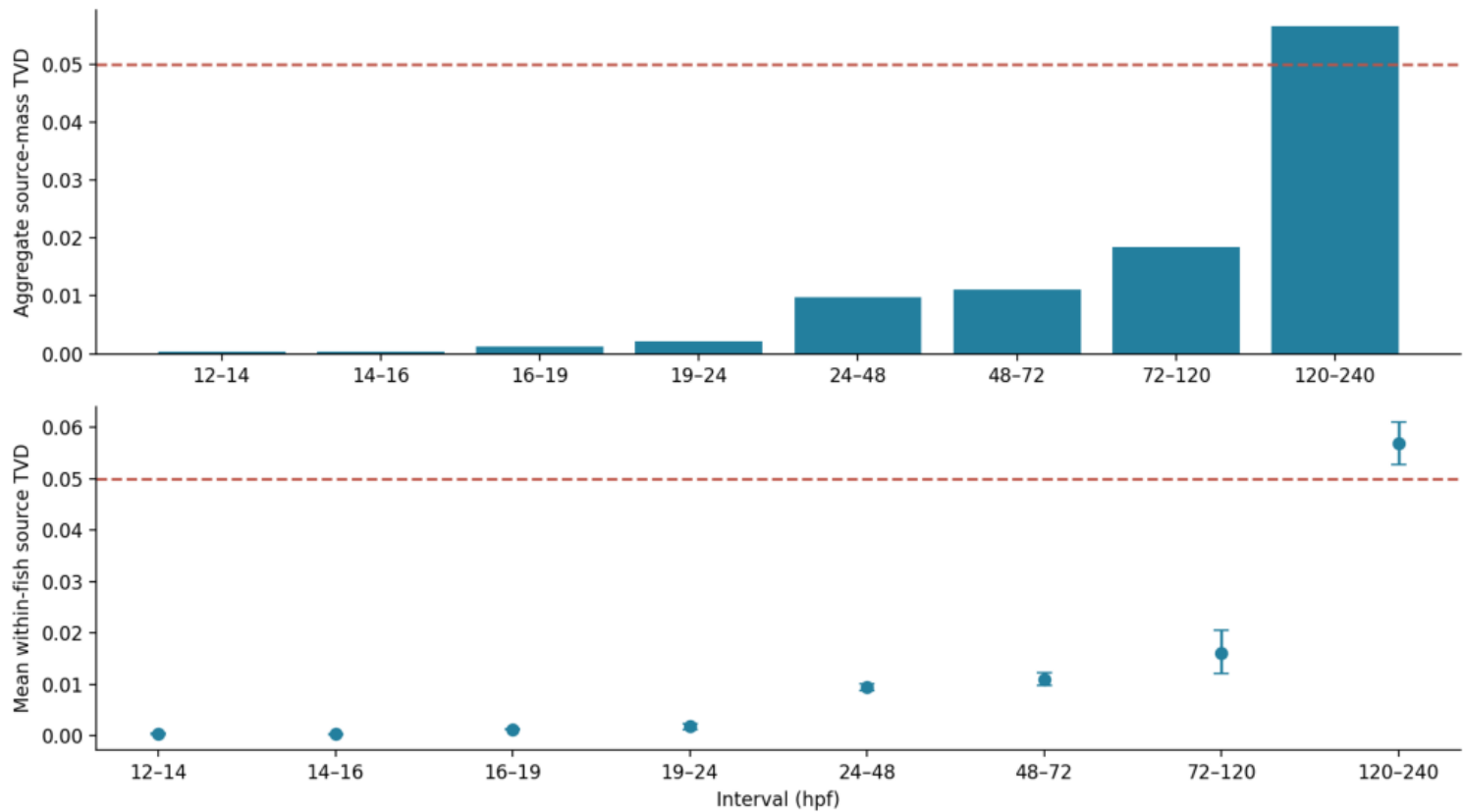
5,426 cells; 36 different fish; four fish per stage. The 12 hpf stage has 19 cells and cannot support type-level 12-14 hpf claims. Fish at different stages are different animals.

5,426 cells; 36 different fish; four fish per stage. The 12 hpf stage has 19 cells and cannot support type-level 12-14 hpf claims. Fish at different stages are different animals.

2. 2. Growth priors change late transport most

CURRENT SUMMARY / SAVED TABLES

2. Growth priors change late transport most



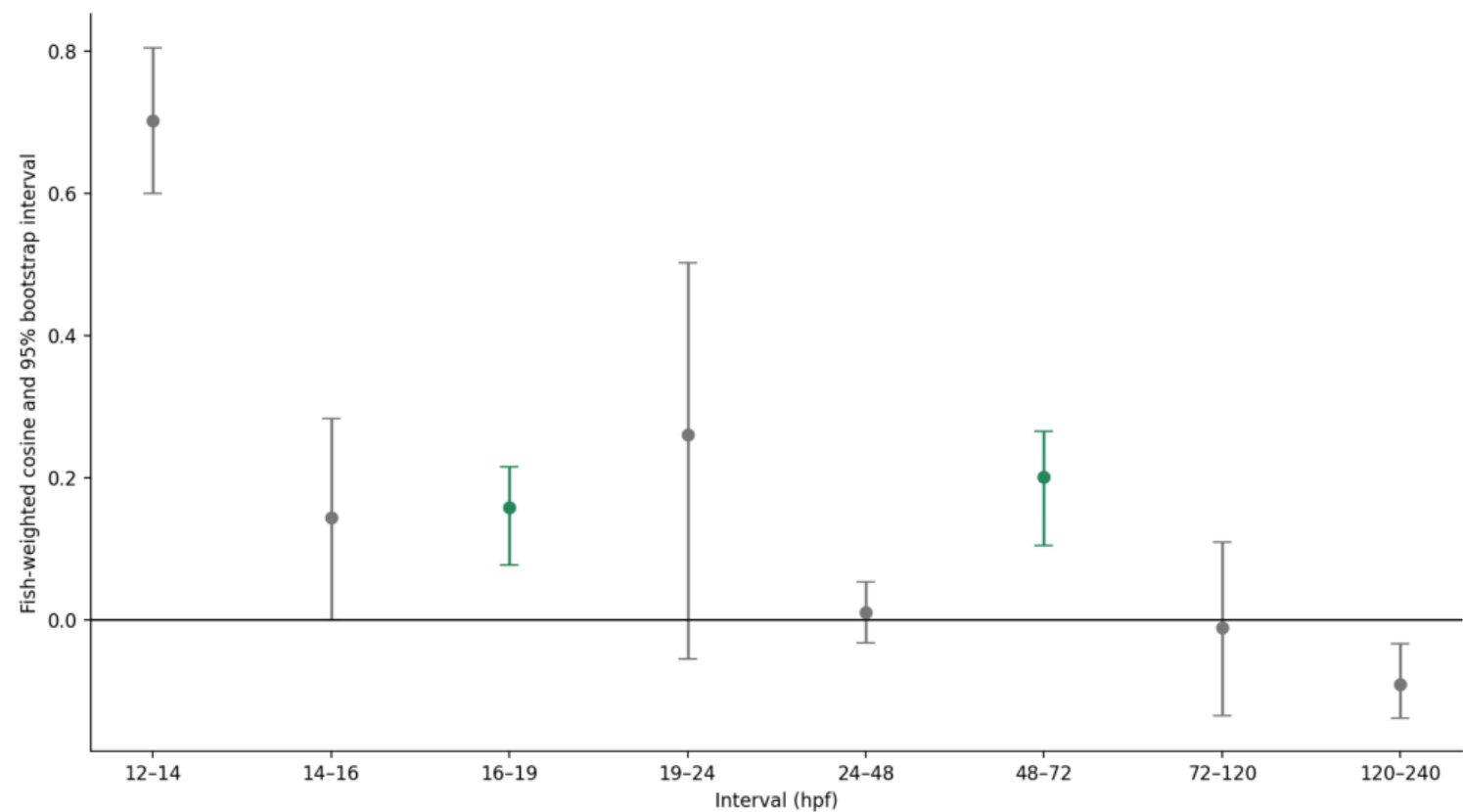
Top: aggregate source-mass effect. Bottom: mean within-fish normalized effect with descriptive 95% fish-bootstrap intervals; these are different estimands. No bootstrap refitting. The 0.05 line is an operational reporting rule, not biological significance.

Top: aggregate source-mass effect. Bottom: mean within-fish normalized effect with descriptive 95% fish-bootstrap intervals; these are different estimands. No bootstrap refitting. The 0.05 line is an operational reporting rule, not biological significance.

3. 3. Directional agreement is limited to two intervals

[CURRENT SUMMARY](#) / [SAVED TABLES](#)

3. Directional agreement is limited to two intervals



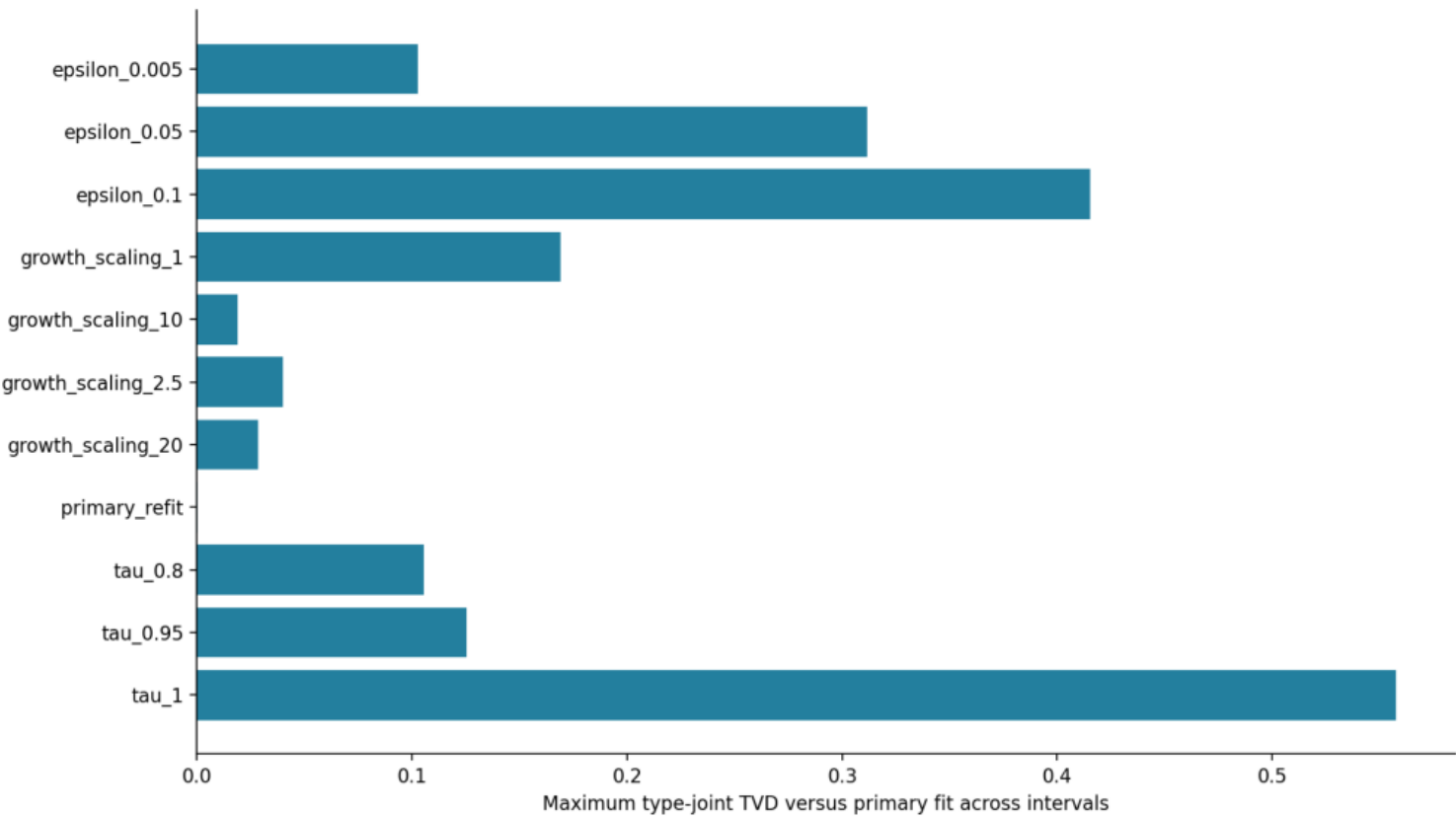
Green: positive absolute direction, positive fish-bootstrap lower bound, and within-fish permutation FDR <0.05 (16-19 and 48-72 hpf). Gray: criteria not met. Late negative cosine is not positive concordance even when better than the shuffled null. Shared expression/PCA means this is complementary computational evidence.

Green: positive absolute direction, positive fish-bootstrap lower bound, and within-fish permutation FDR <0.05 (16-19 and 48-72 hpf). Gray: criteria not met. Late negative cosine is not positive concordance even when better than the shuffled null. Shared expression/PCA means this is complementary computational evidence.

4. 4. Solver choices materially affect the coupling

CURRENT SUMMARY / SAVED TABLES

4. Solver choices materially affect the coupling

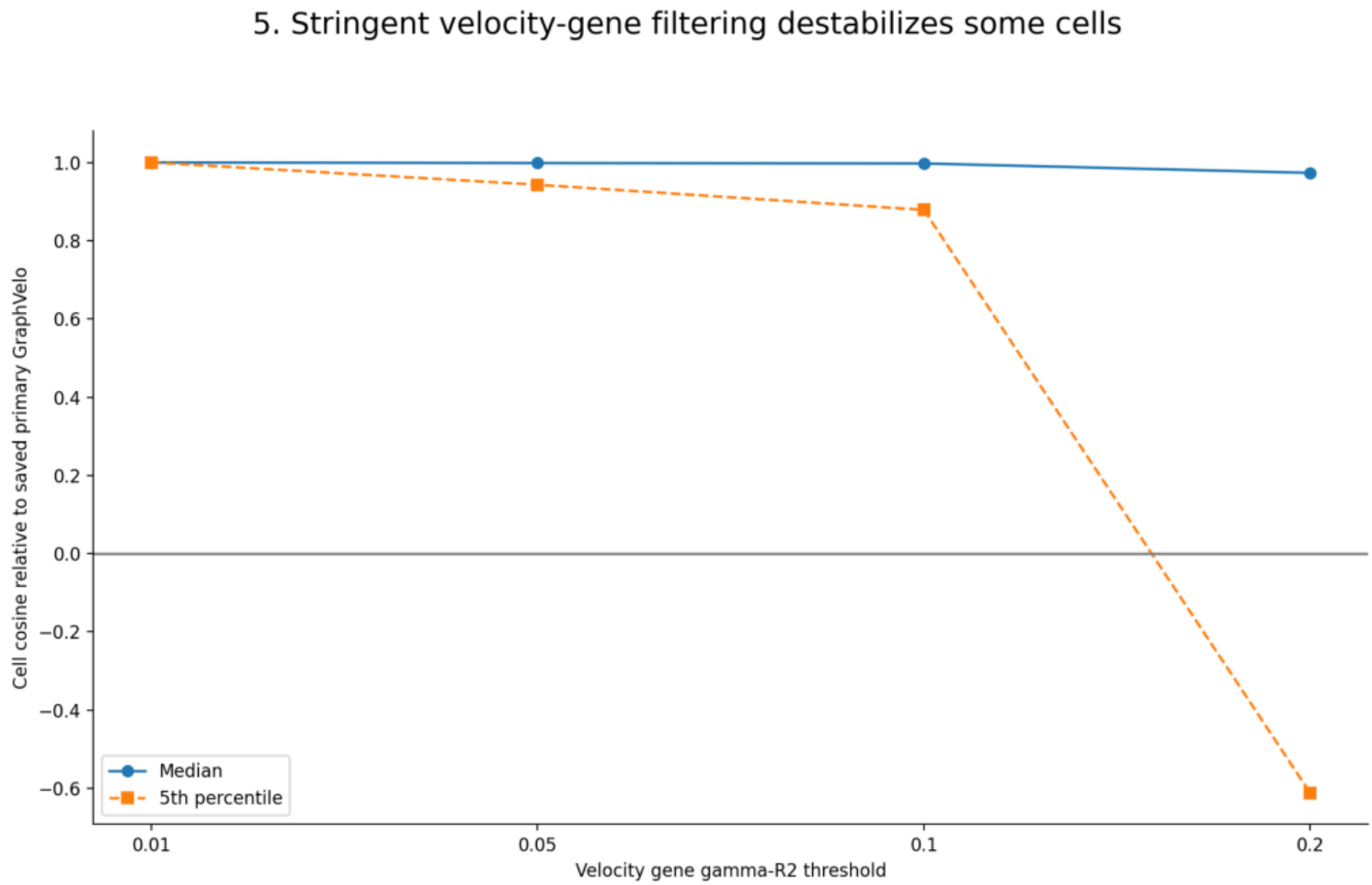


Plotted TVD is half the saved normalized type-joint L1 distance. Eleven one-factor configurations; all fits converged. Numerical convergence does not establish biological accuracy. Balanced tau=1 is a stress test. Source: native_moscot_parameter_sensitivity_summary.csv.

Plotted TVD is half the saved normalized type-joint L1 distance. Eleven one-factor configurations; all fits converged. Numerical convergence does not establish biological accuracy. Balanced tau=1 is a stress test. Source: native_moscot_parameter_sensitivity_summary.csv.

5. 5. Stringent velocity-gene filtering destabilizes some cells

[CURRENT SUMMARY](#) / [SAVED TABLES](#)



Thresholds 0.05 and 0.10 largely preserve primary directions. At 0.20, the lower tail becomes negative. Similarity to the primary fit measures stability, not correctness. Source: graphvelo_gene_qc_sensitivity_summary.csv.

Thresholds 0.05 and 0.10 largely preserve primary directions. At 0.20, the lower tail becomes negative. Similarity to the primary fit measures stability, not correctness. Source: graphvelo_gene_qc_sensitivity_summary.csv.

6. 6. What succeeded, what did not, and what is still unknown

CURRENT SUMMARY / SAVED TABLES

6. What succeeded, what did not, and what is still unknown

WHAT THE CURRENT WORK SUPPORTS

- A reproducible saved native M0/M1 run, plus fish-aware audits.
- A larger growth-prior source-mass effect at 120–240 hpf.
- Positive complementary direction evidence at two intervals.
- Modest human expression-program concordance, with negative specificity tests.

WHAT REMAINS UNRESOLVED

- Actual division/death rates and true parent–descendant relationships.
- Negative absolute late-interval velocity agreement.
- Strong epsilon/tau sensitivity and sparse early sampling.
- Whether growth priors improve independent predictive performance.

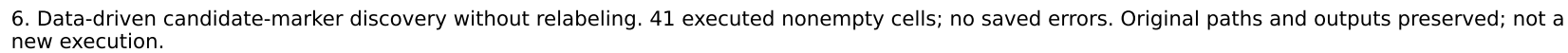
HISTORY IS INCLUDED, NOT HIDDEN

- Earlier combined/reweighted models and smaller-marker-panel results.
- Cluster-23 exploratory trajectories, including mixed lineage markers.
- Partial notebooks and saved errors, explicitly labeled as historical.

Historical plots are a record of work, not additional validation of the current model. The separate non-OT project and unapproved integrated proposal are outside this sharing package.

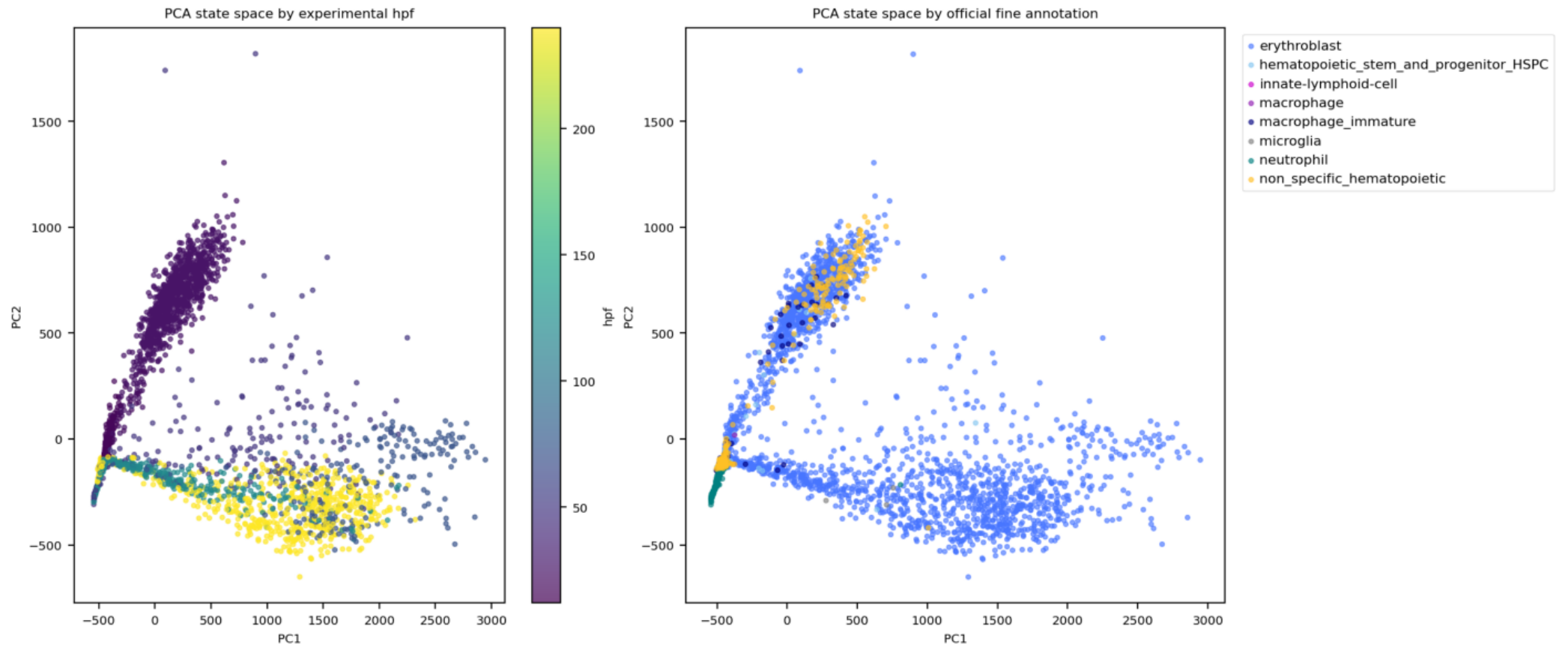
Historical plots are a record of work, not additional validation of the current model. The separate non-OT project and unapproved integrated proposal are outside this sharing package.

CURRENT SAVED RUN



8. cell 023 output 01

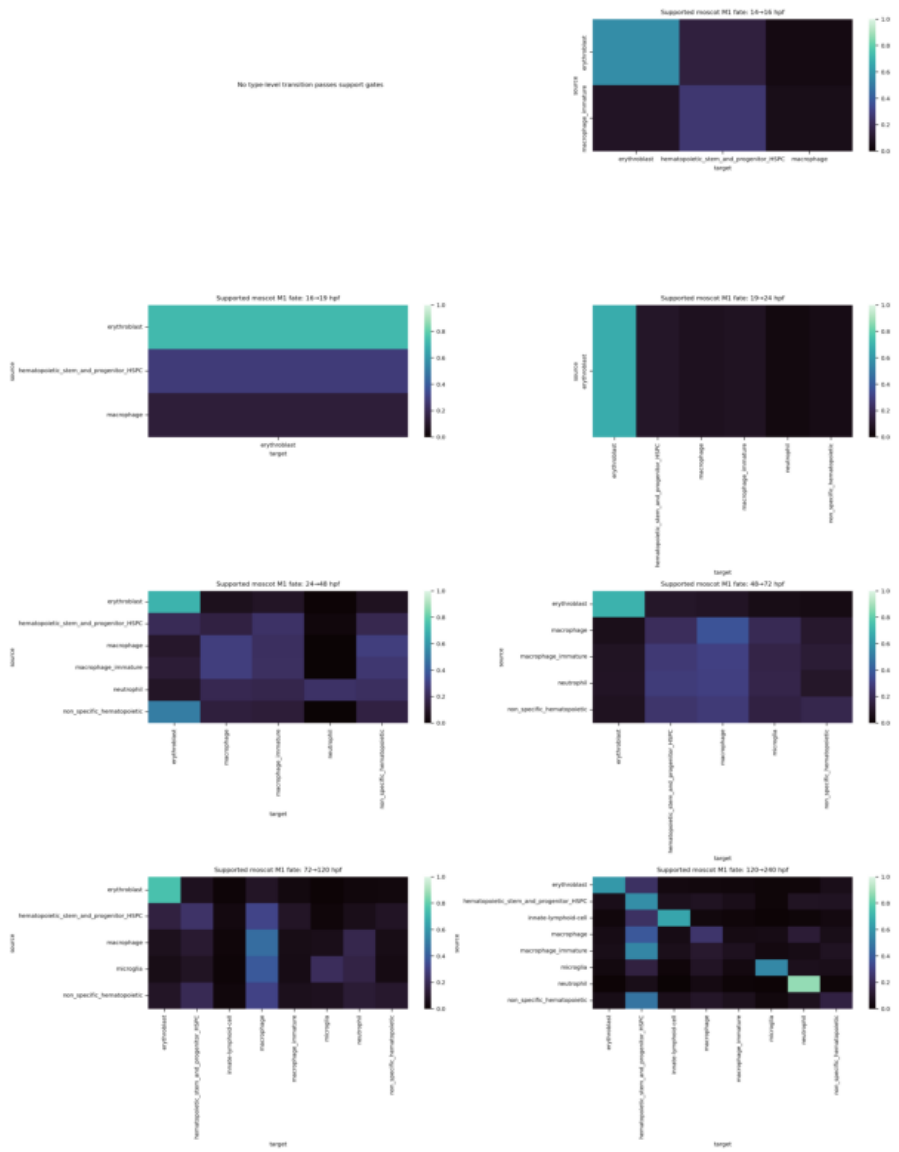
CURRENT SAVED RUN



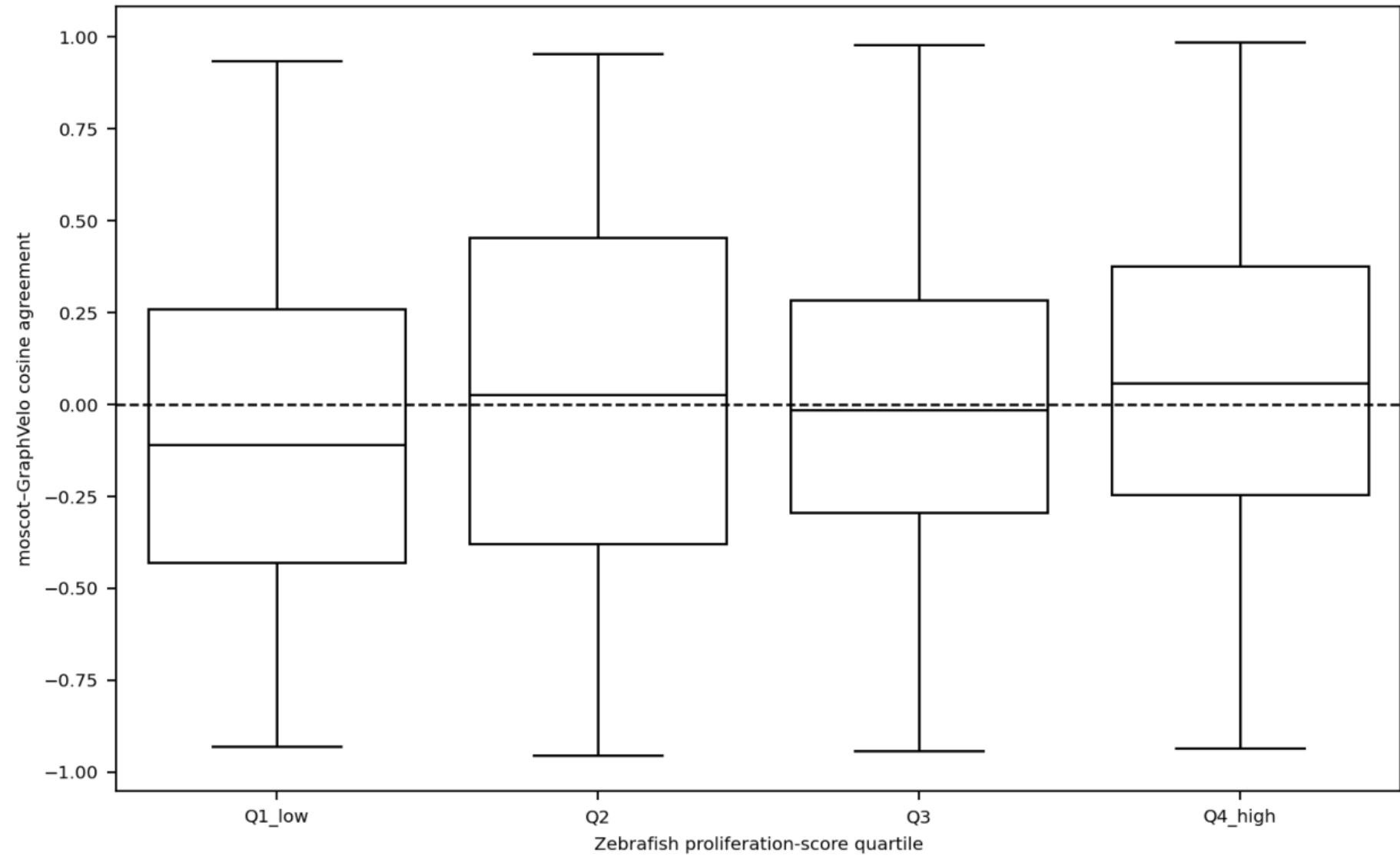
7. Restrict to official labels and run Dynamo preprocessing once. 41 executed nonempty cells; no saved errors. Original paths and outputs preserved; not a new execution.

9. cell 066 output 02

CURRENT SAVED RUN



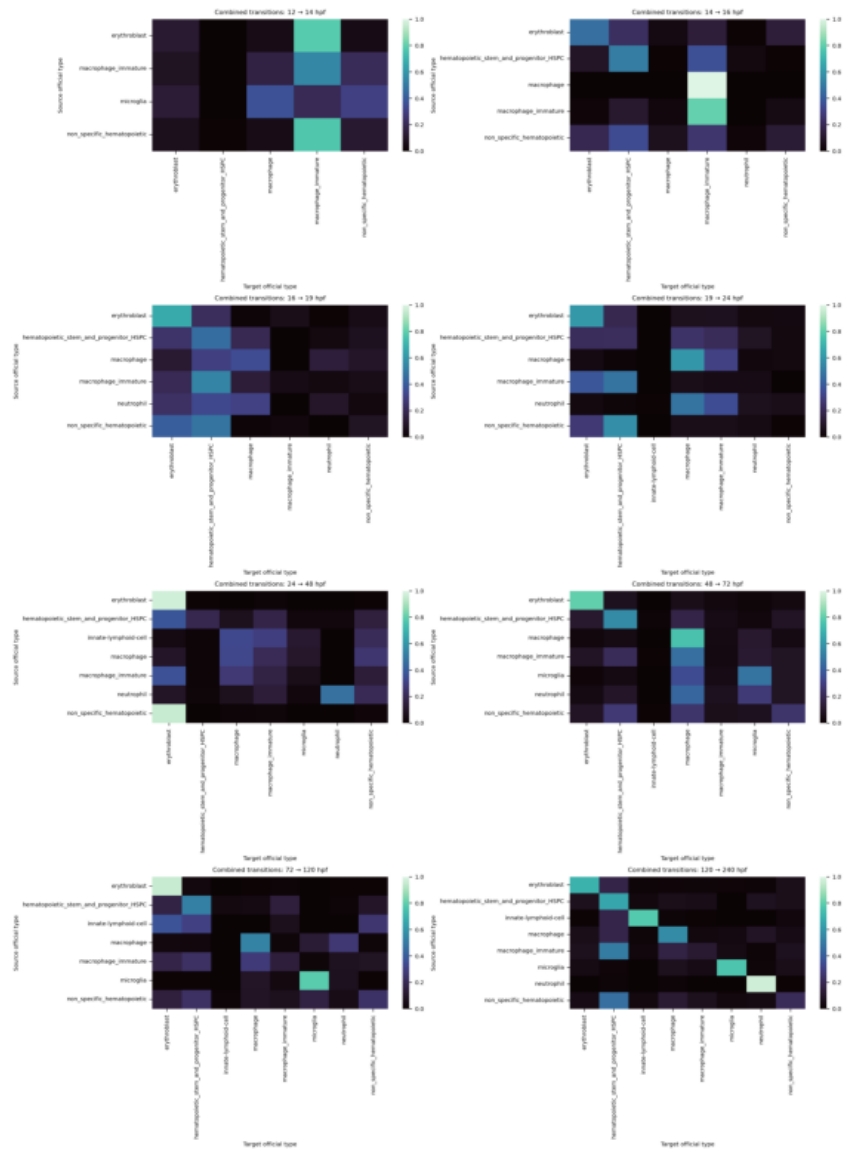
24. Visualize native moscot growth, fate, and velocity agreement. 41 executed nonempty cells; no saved errors. Original paths and outputs preserved; not a new execution.



24. Visualize native moscot growth, fate, and velocity agreement. 41 executed nonempty cells; no saved errors. Original paths and outputs preserved; not a new execution.

11. cell 056 output 02

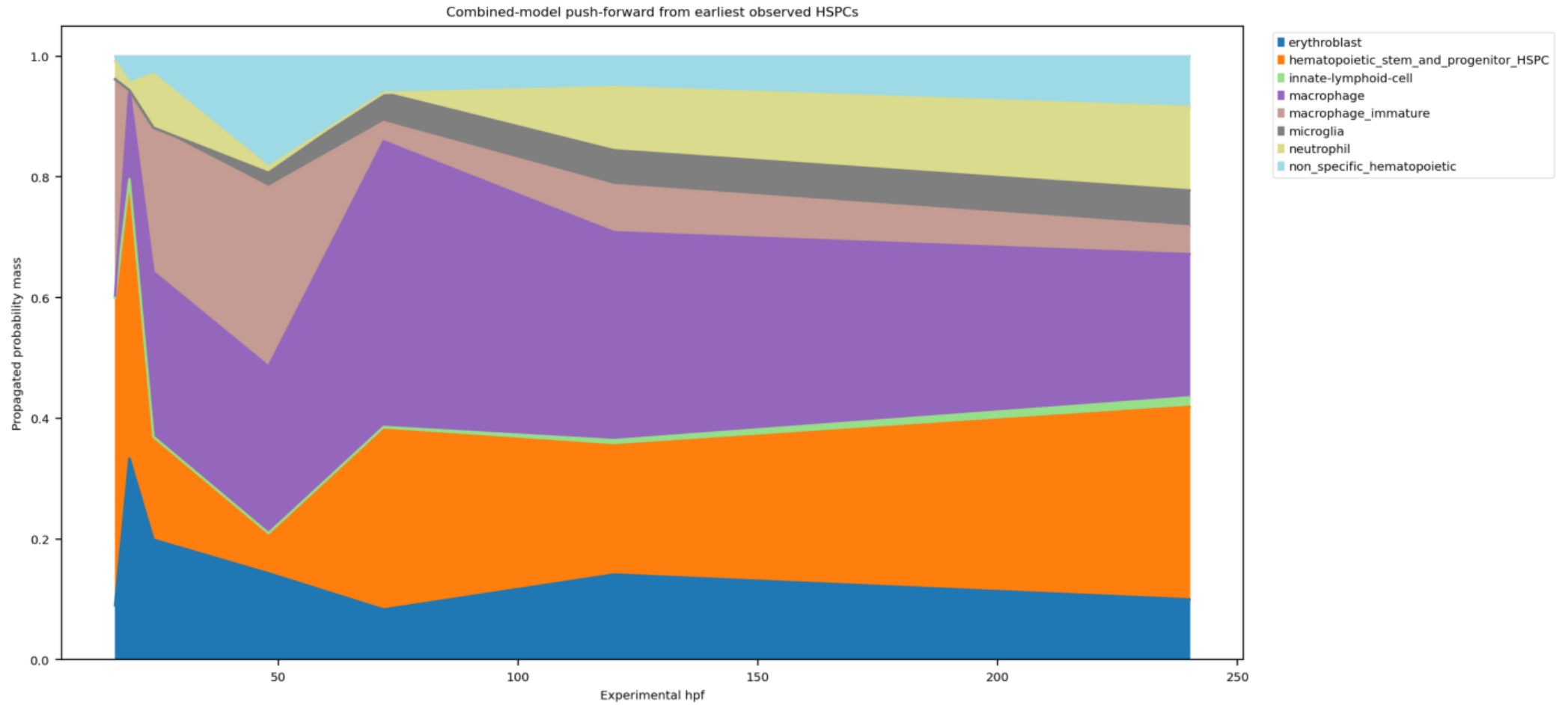
HISTORICAL / SAVED ERROR



21. Aggregate cell-level couplings to official cell-type transitions. Earlier combined model; saved categorical-median error. Superseded by native M0/M1.

12. cell 058 output 02

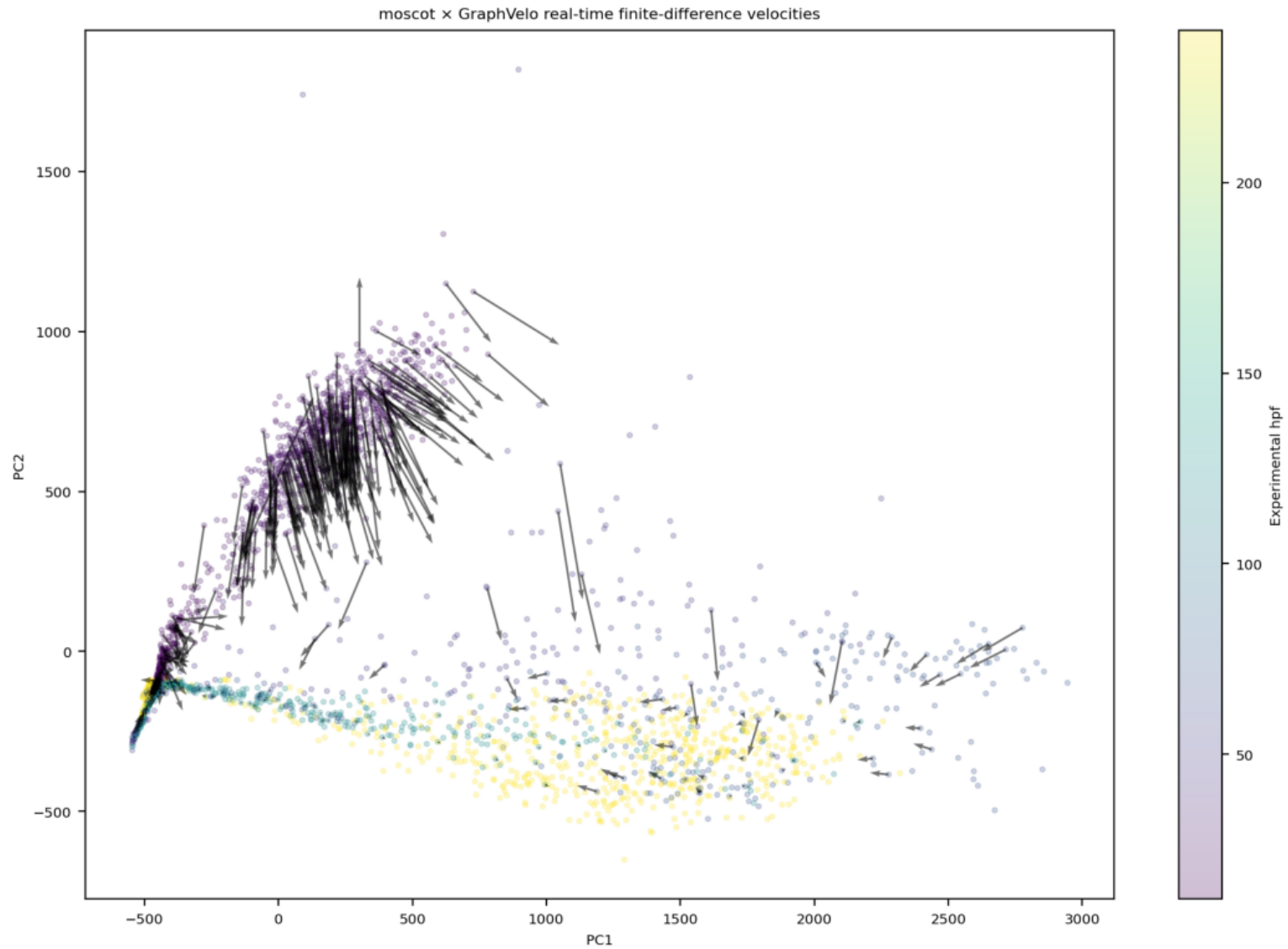
HISTORICAL / SAVED ERROR



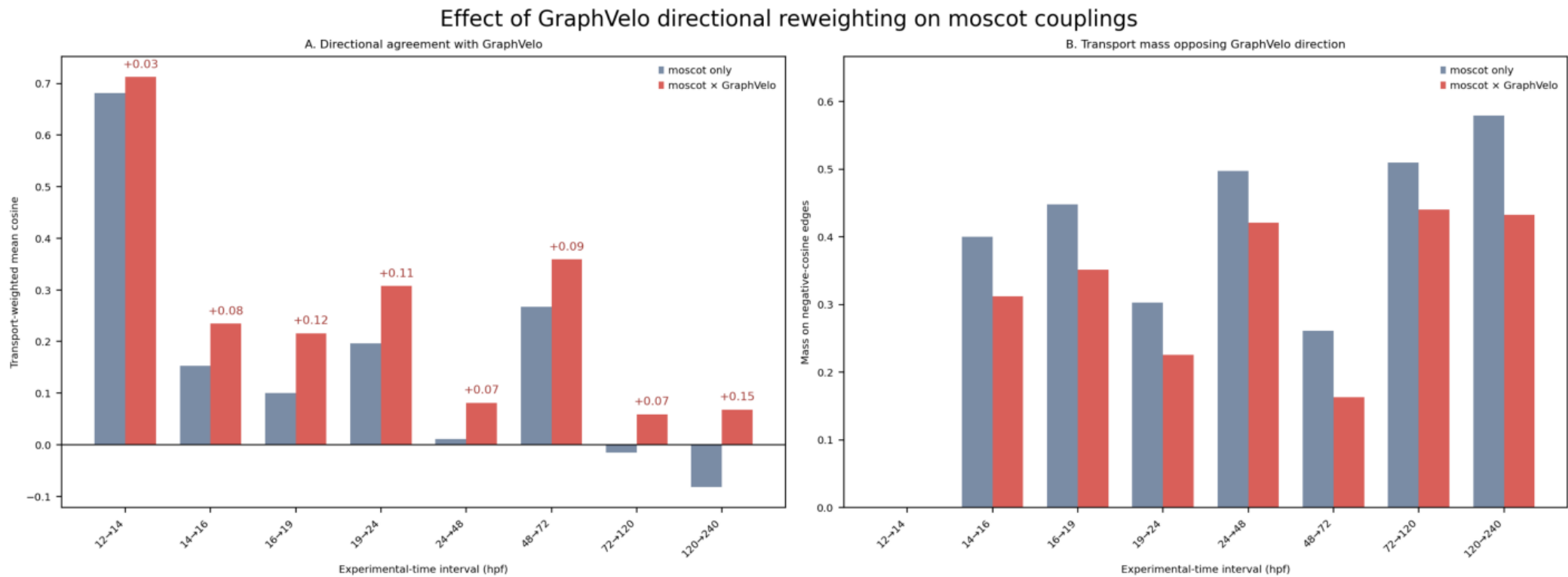
22. HSPC push-forward fate distributions through multiple time points. Earlier combined model; saved categorical-median error. Superseded by native M0/M1.

13. cell 060 output 03

HISTORICAL / SAVED ERROR

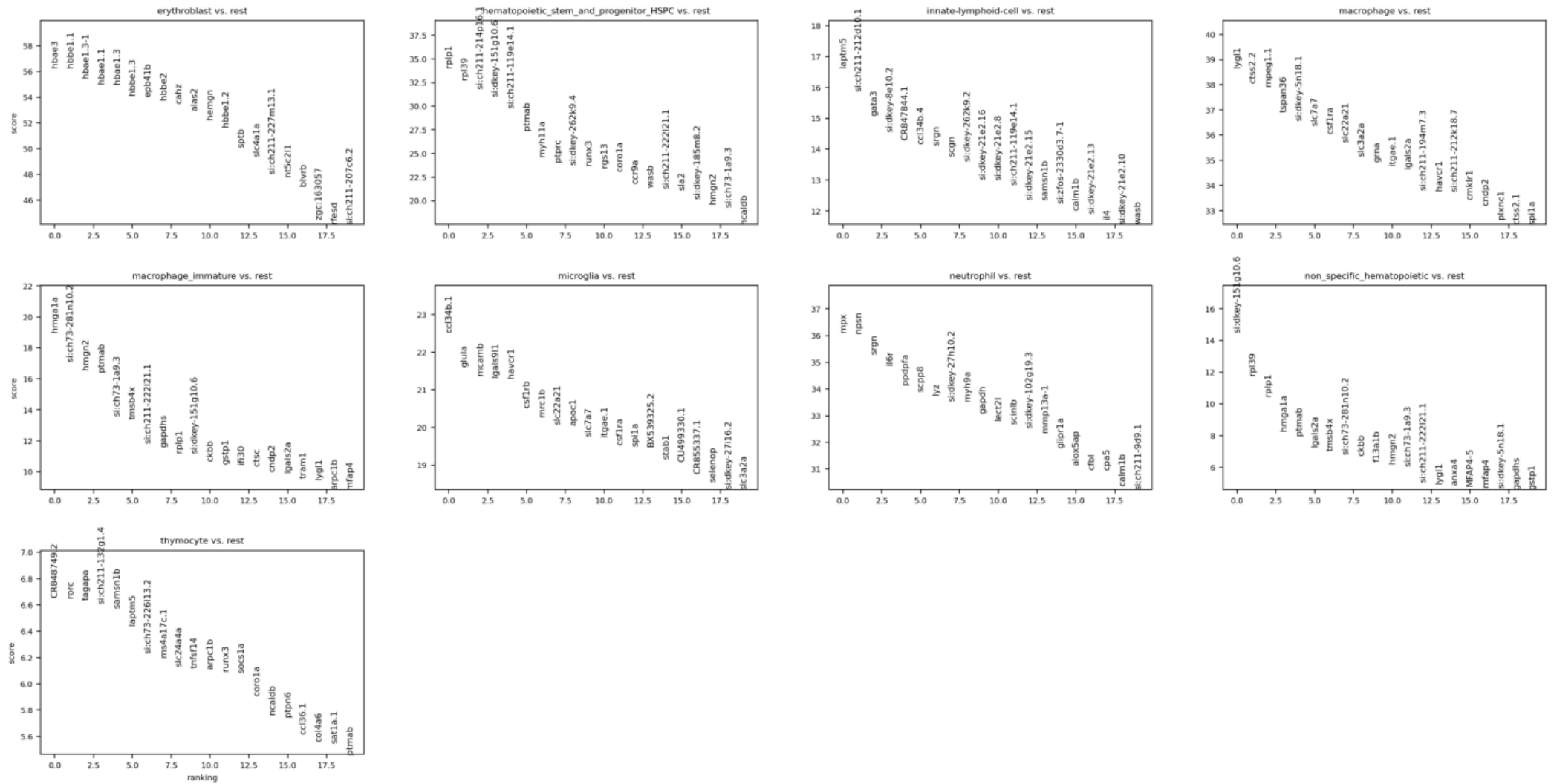


23. Convert combined couplings into real-time PCA velocities. Earlier combined model; saved categorical-median error. Superseded by native M0/M1.



Interpretation checklist before reporting. Earlier combined model; saved categorical-median error. Superseded by native M0/M1.

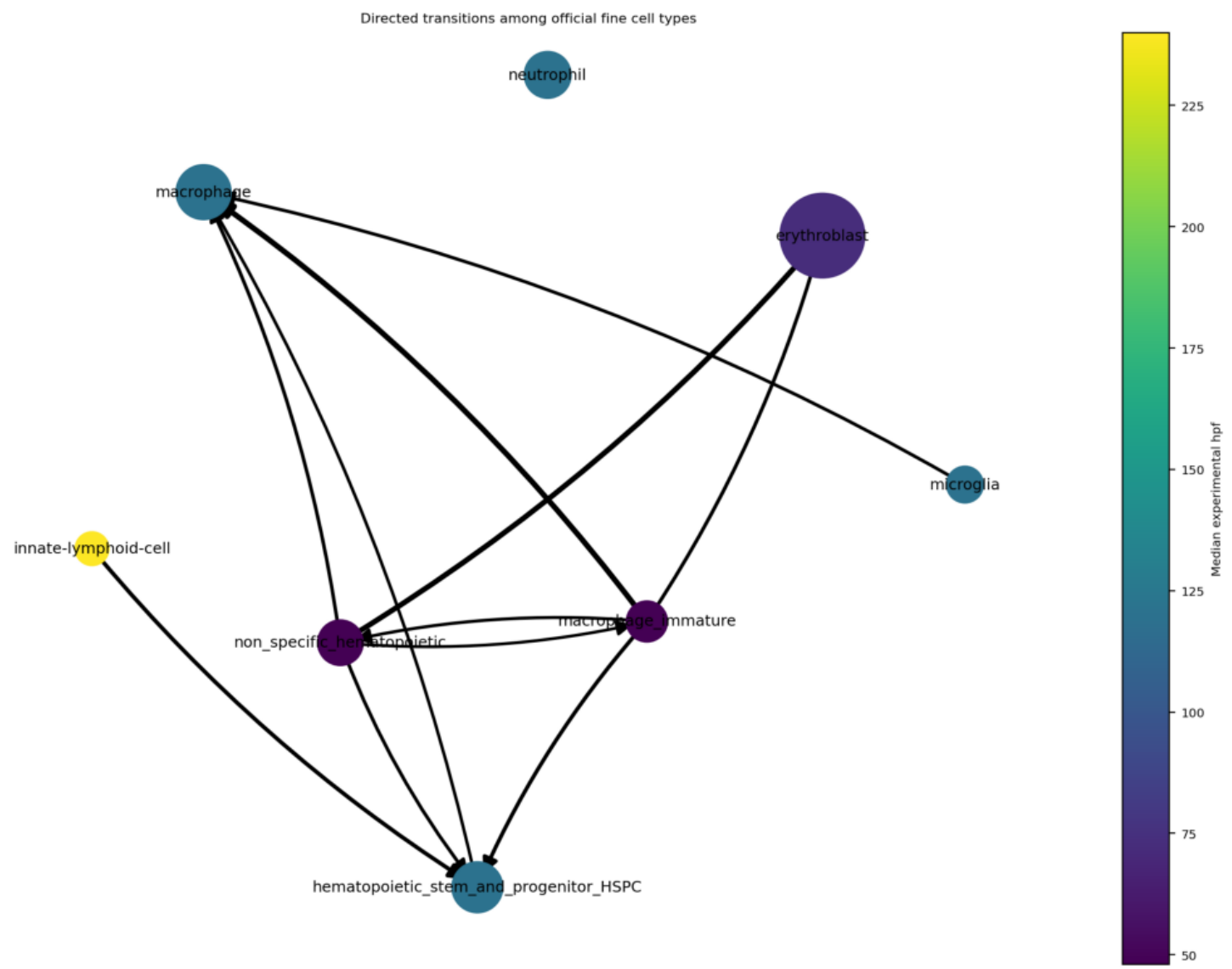
HISTORICAL / PARTIAL RUN



6. Data-driven candidate-marker discovery without relabeling. Earlier GraphVelo stage; some nonempty cells have no execution count.

16. cell 040 output 01

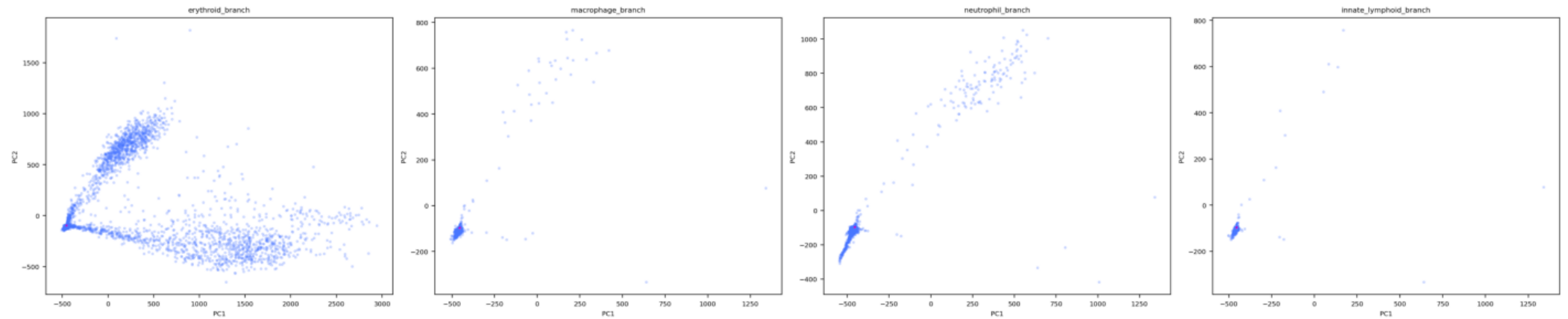
HISTORICAL / PARTIAL RUN



15. Main result: directed weighted official-cell-type network. Earlier GraphVelo stage; some nonempty cells have no execution count.

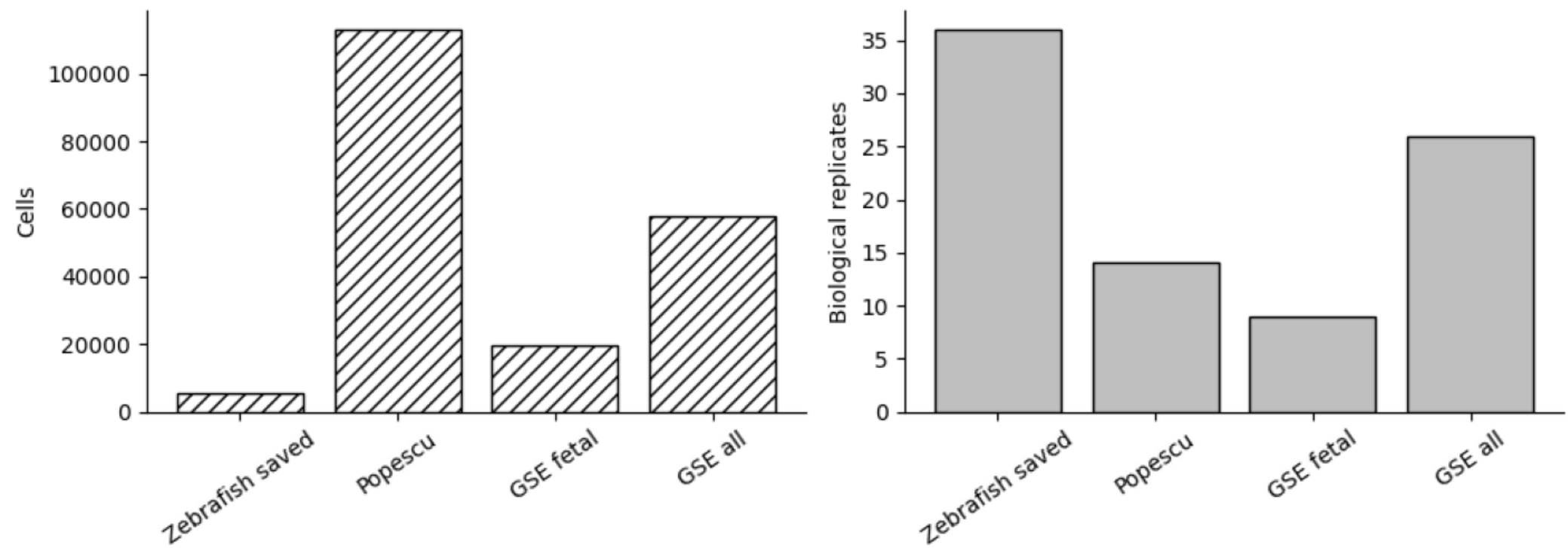
17. cell 047 output 01

HISTORICAL / PARTIAL RUN



16. Optional ODE trajectories in PCA space. Earlier GraphVelo stage; some nonempty cells have no execution count.

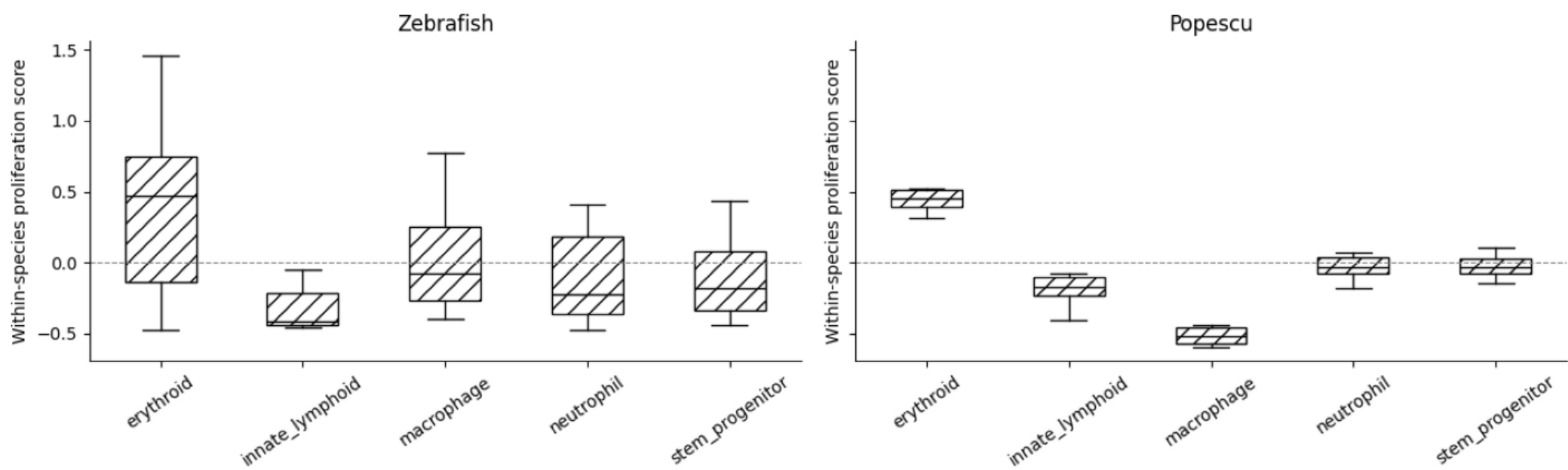
Datasets used in cross-species validation



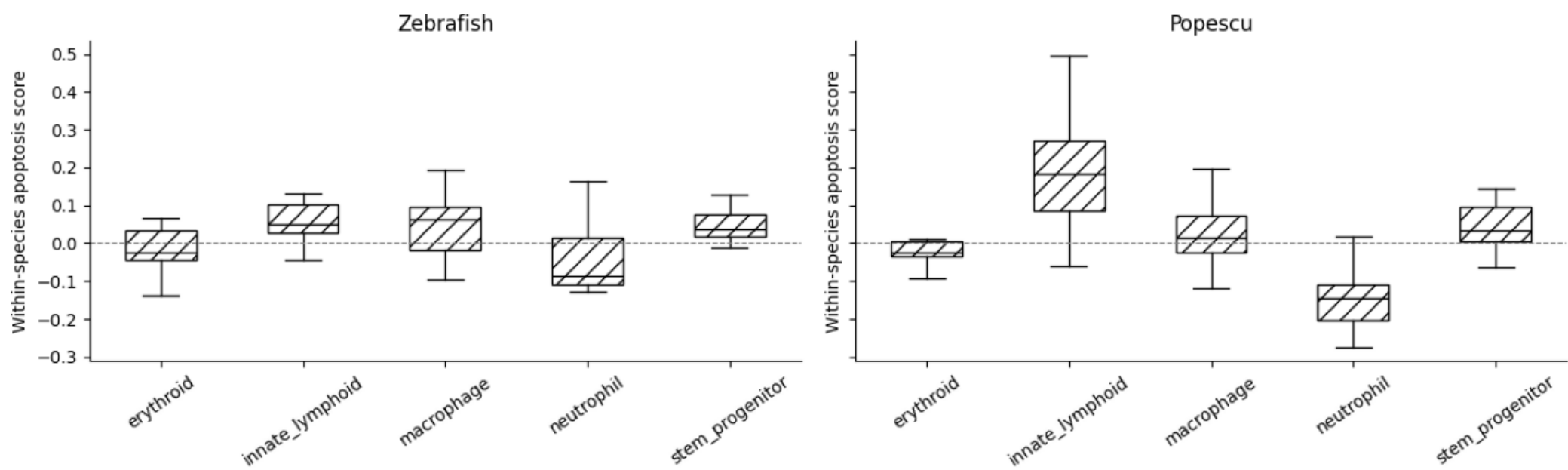
12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.

19. cell 027 output 01

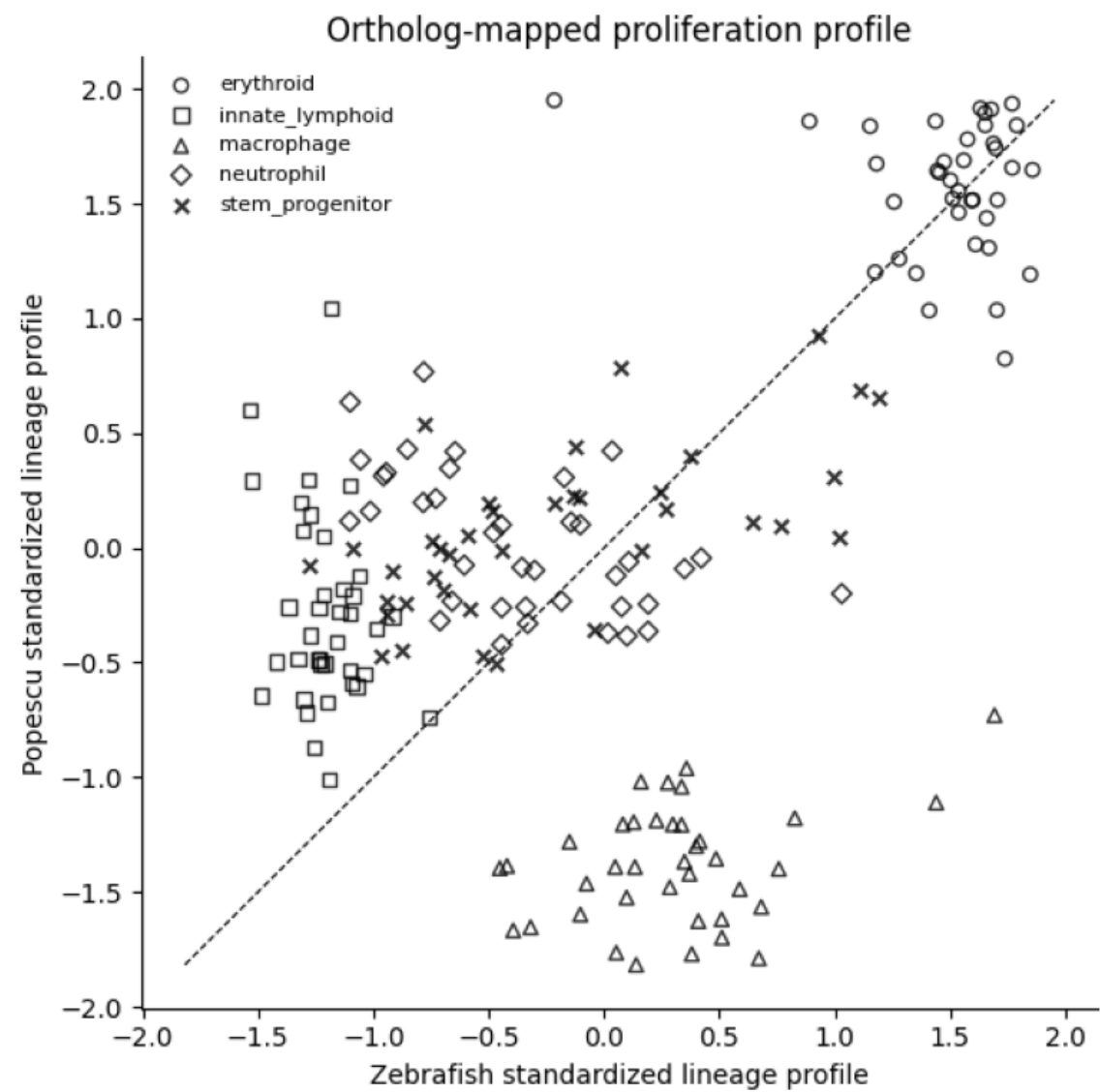
SUPERSEDED HUMAN VALIDATION



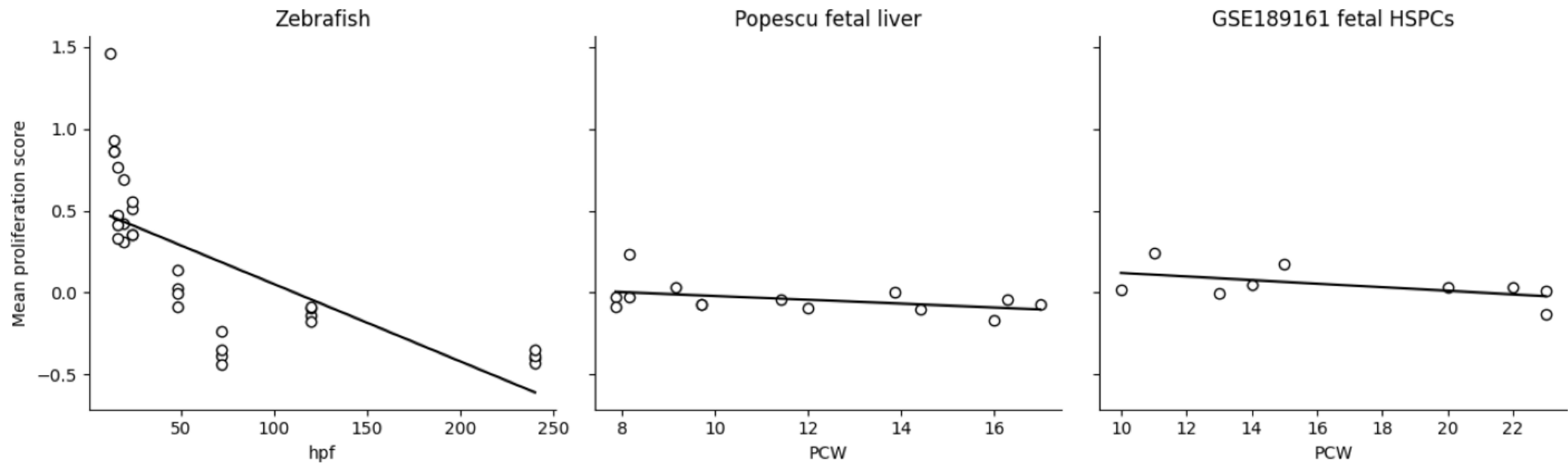
12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.



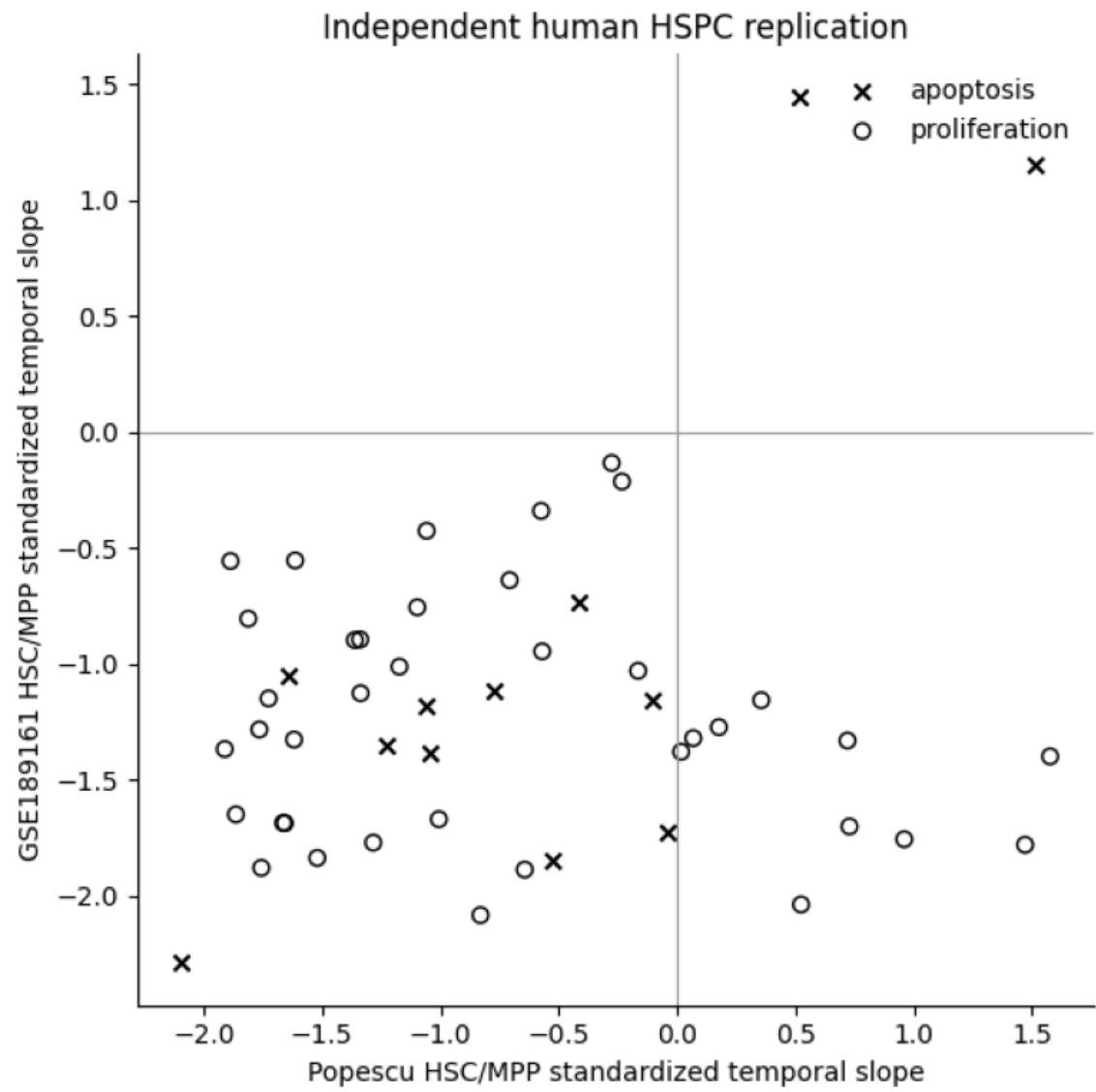
12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.



12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.



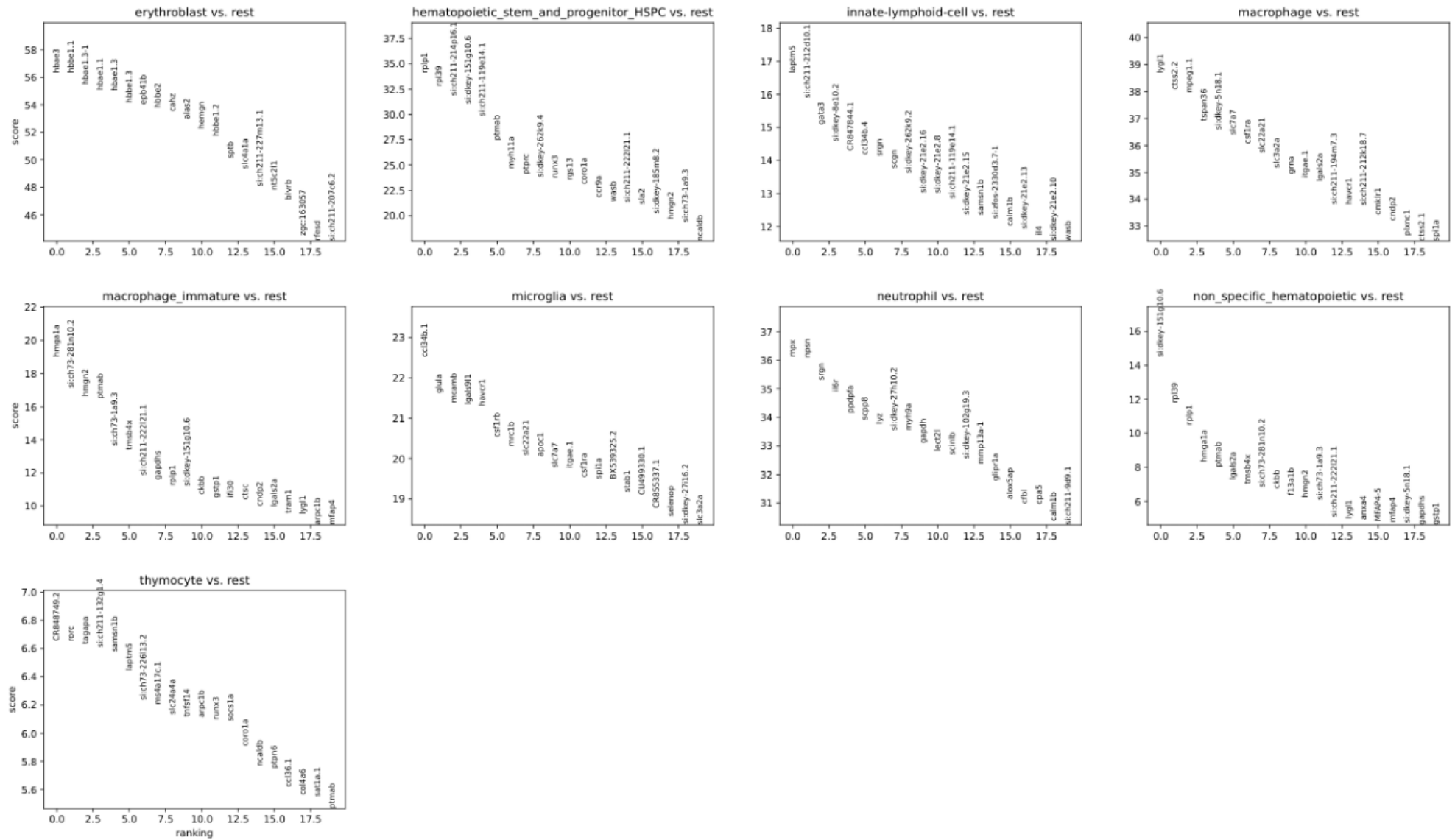
12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.



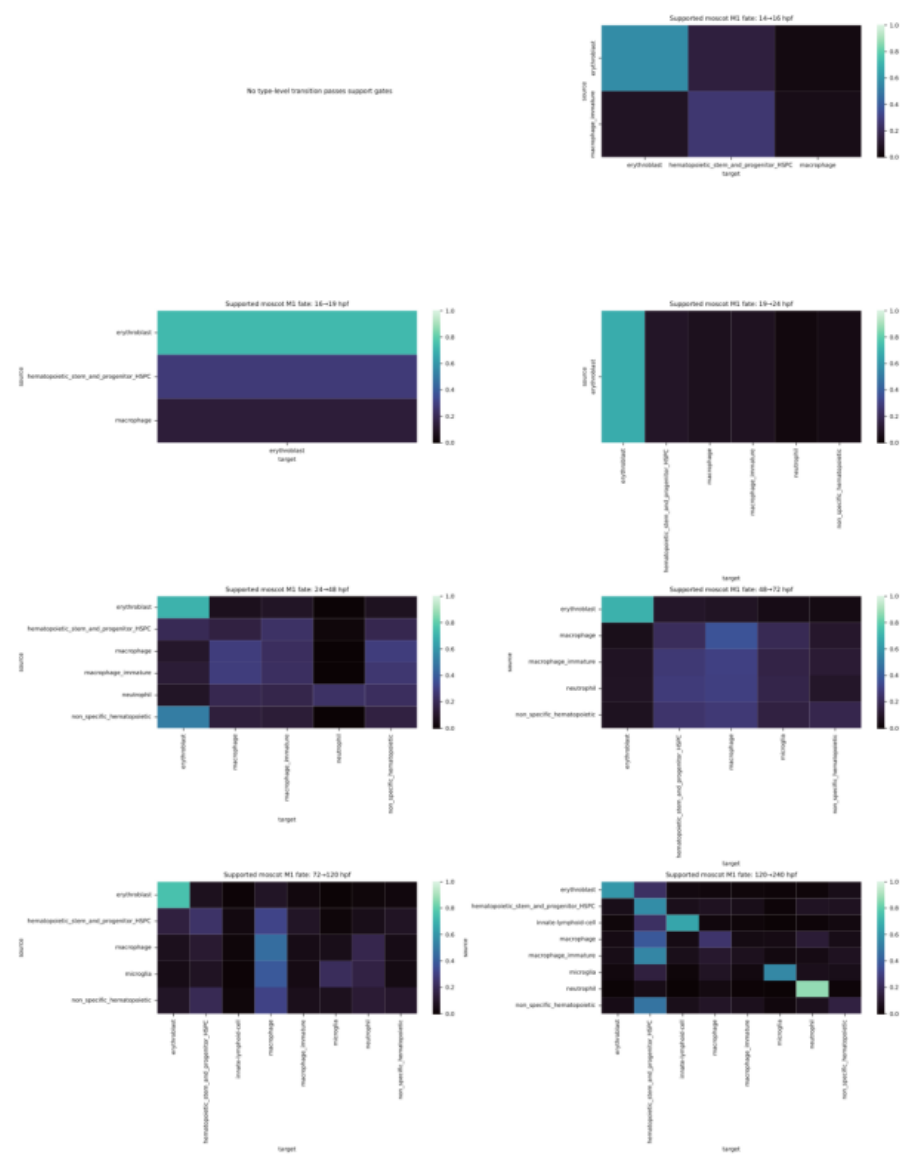
12. Black-and-white figures. Executed older marker-panel analysis, not the current expanded-panel result.

24. exploratory cell level candidate markers / page 1

CURRENT / READ WITH CORRECTED AUDITS



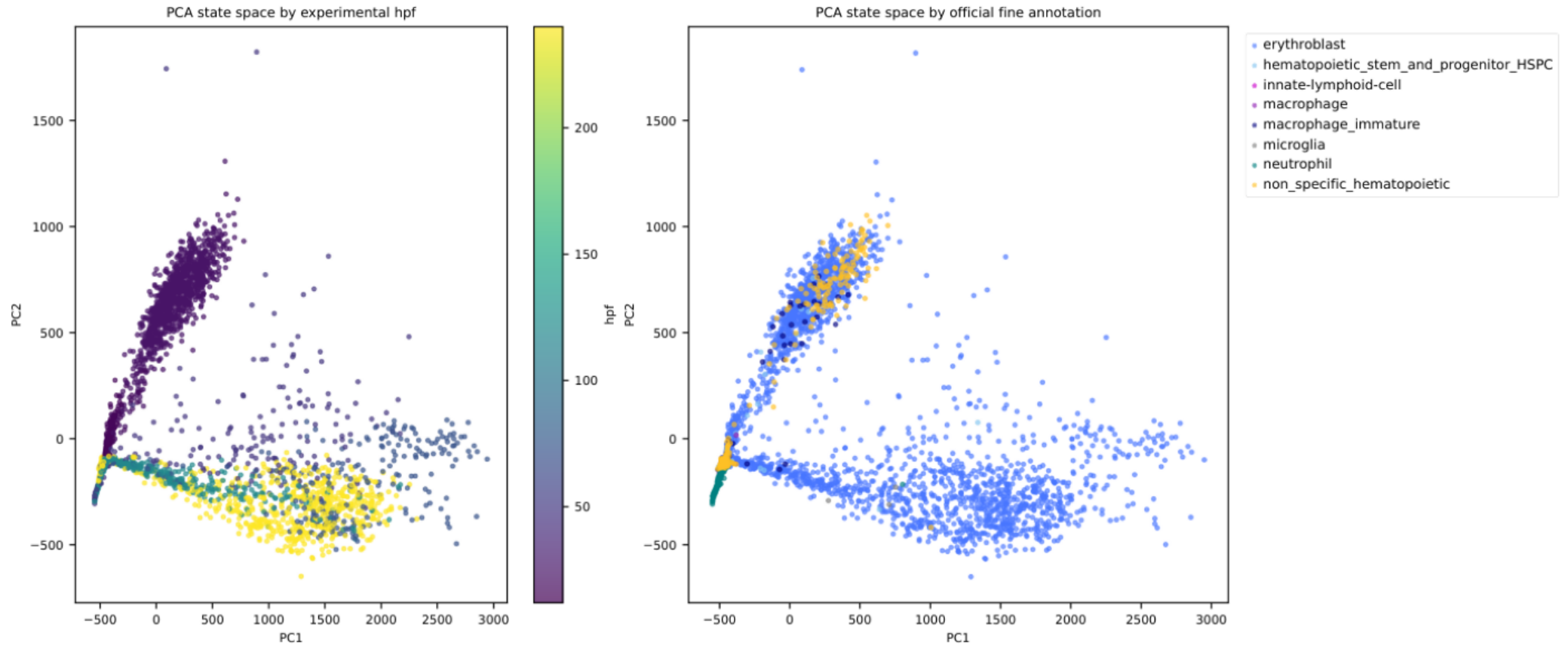
Native M0/M1; cell-level marker plots are exploratory. Use corrected directional evidence and support gates for interpretation.



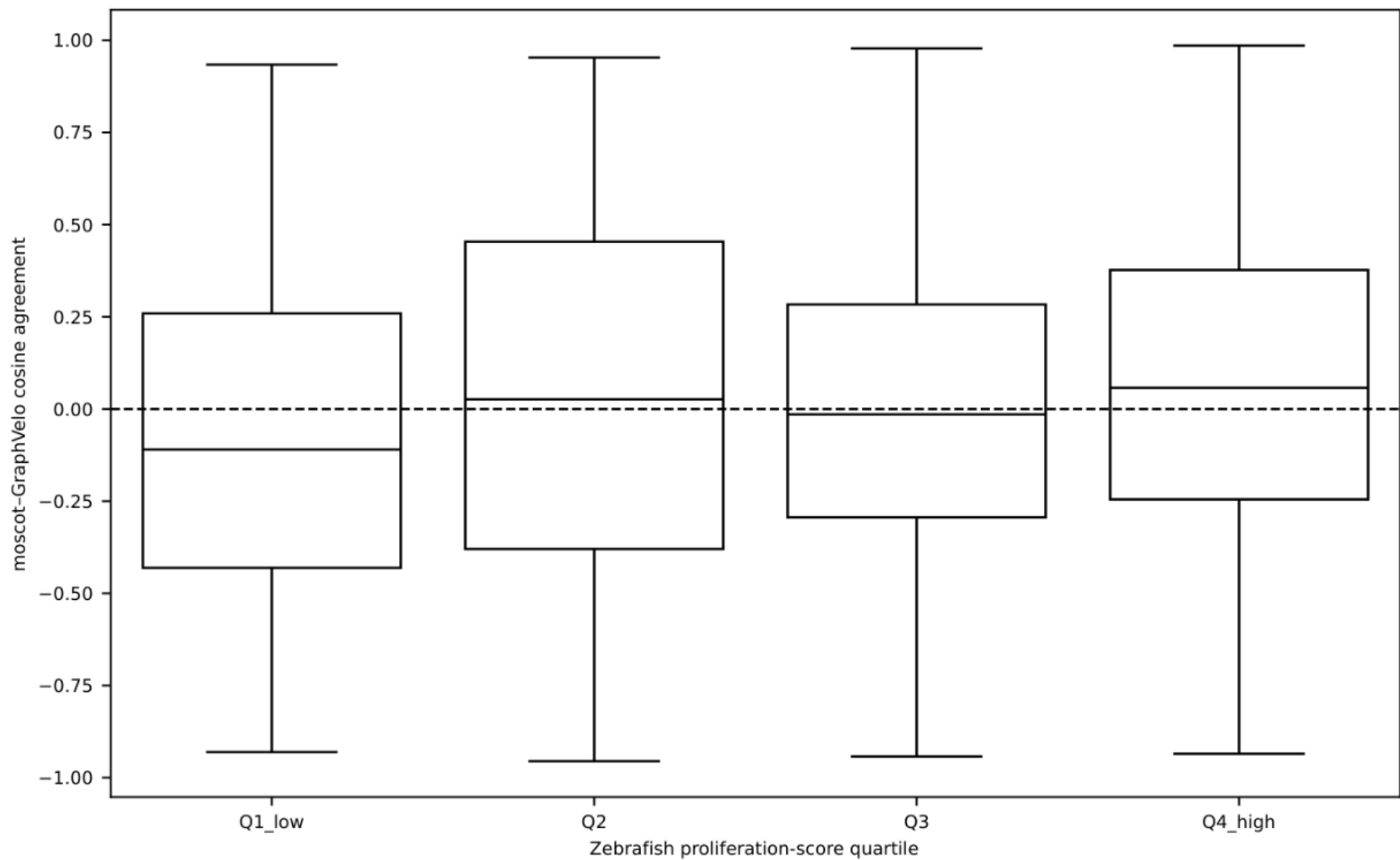
Native M0/M1; cell-level marker plots are exploratory. Use corrected directional evidence and support gates for interpretation.

26. official fine type pca state space / page 1

CURRENT / READ WITH CORRECTED AUDITS

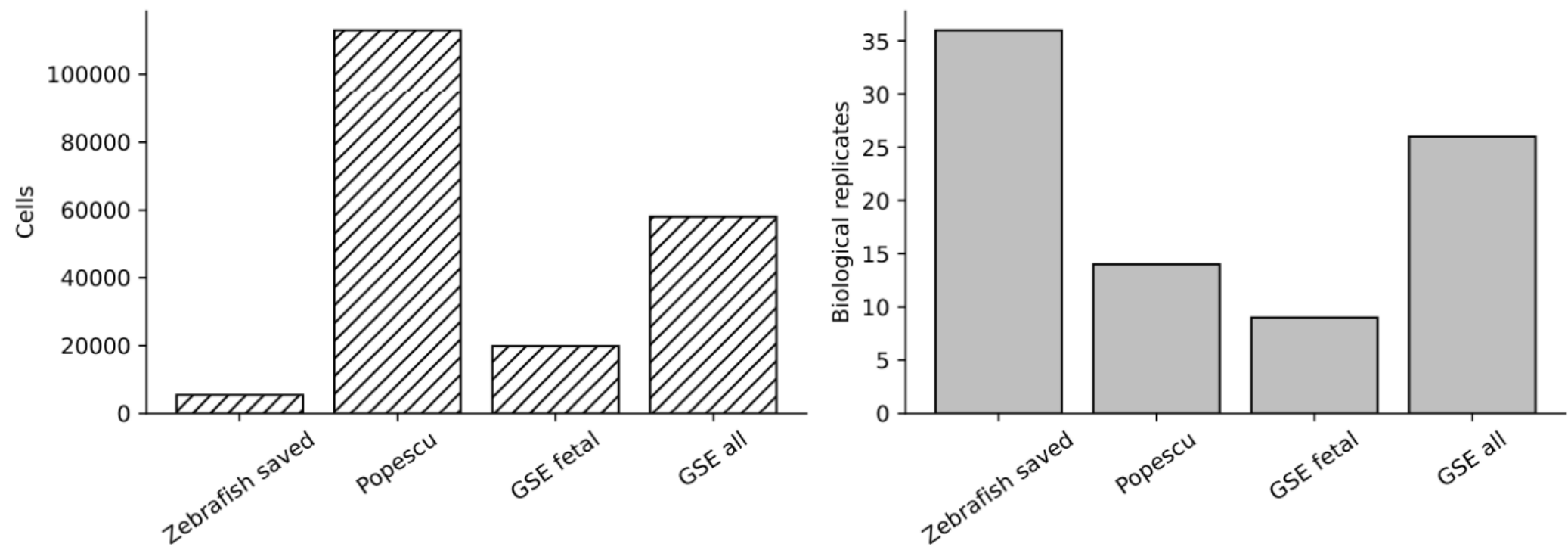


Native M0/M1; cell-level marker plots are exploratory. Use corrected directional evidence and support gates for interpretation.

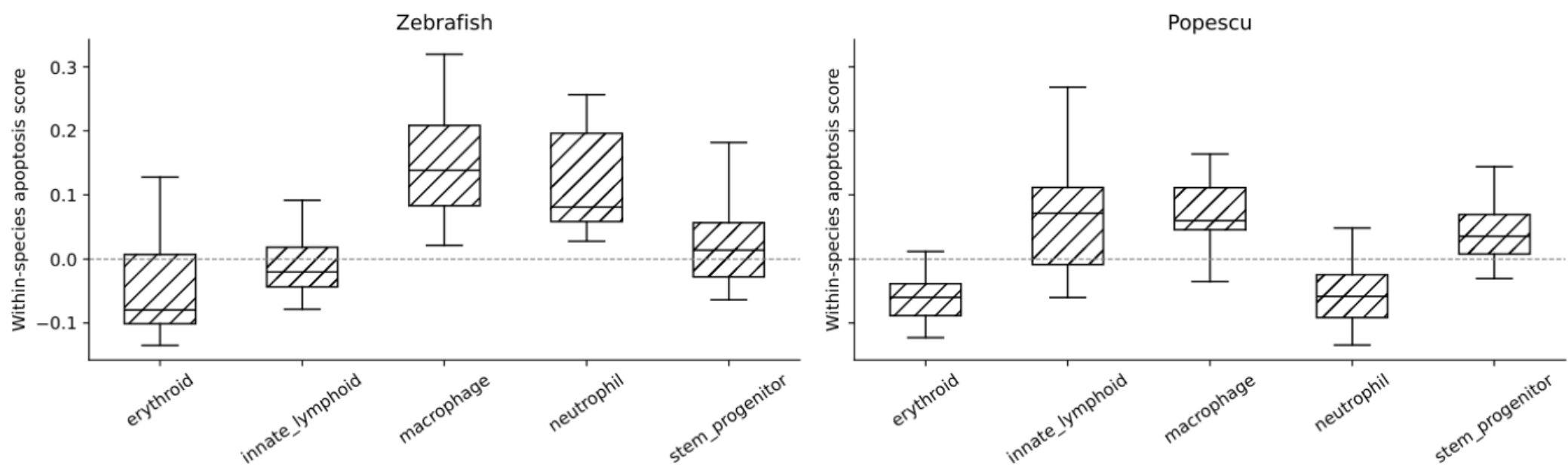


Native M0/M1; cell-level marker plots are exploratory. Use corrected directional evidence and support gates for interpretation.

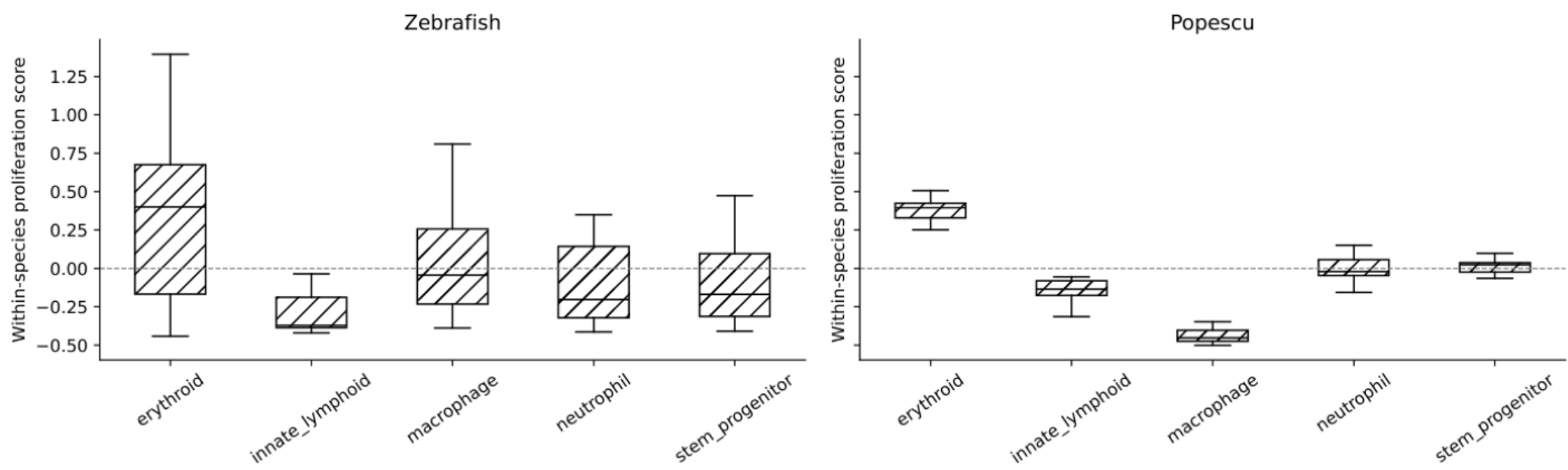
Datasets used in cross-species validation



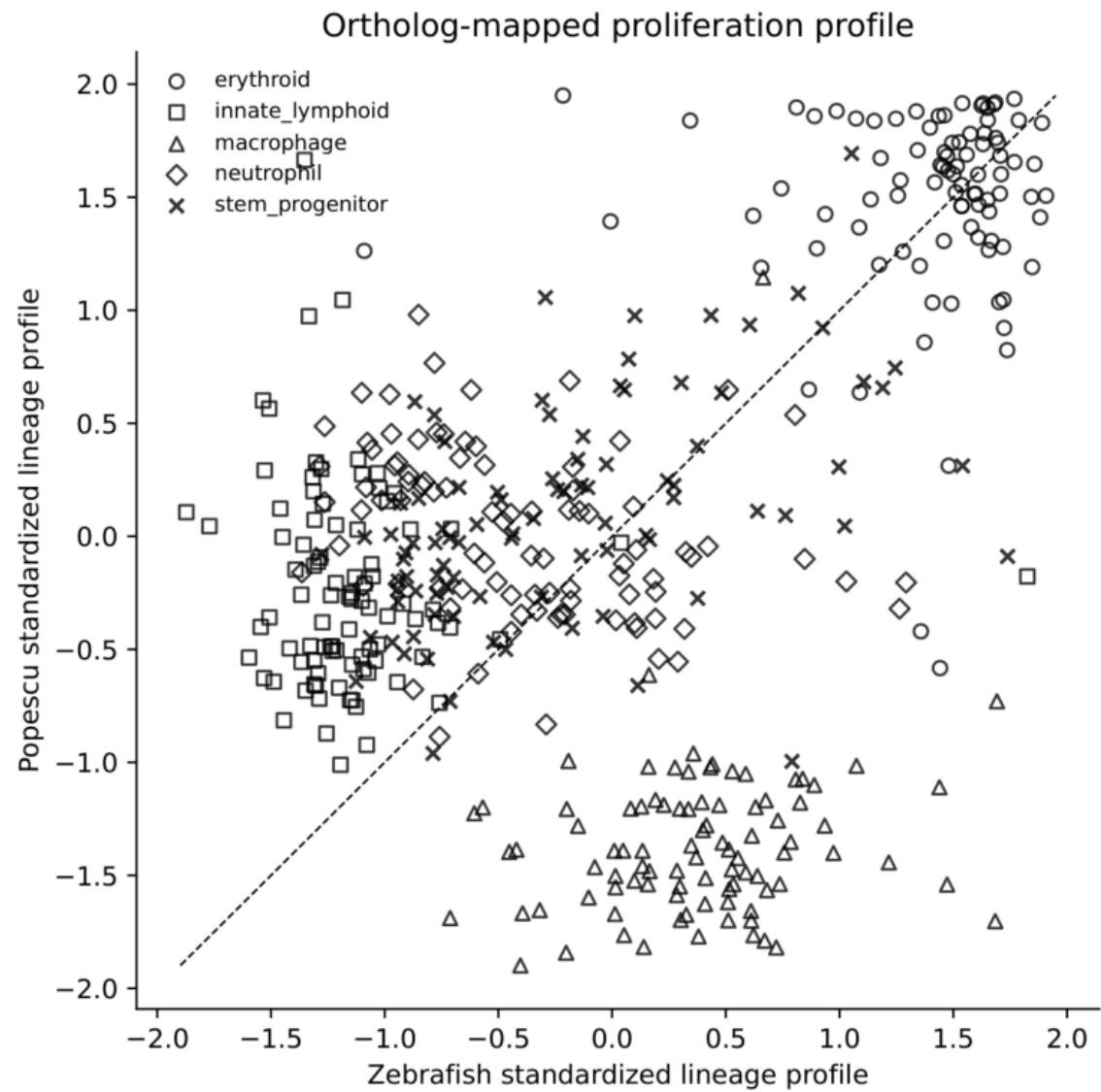
Human comparisons test expression programs, not fate, velocity or absolute growth.



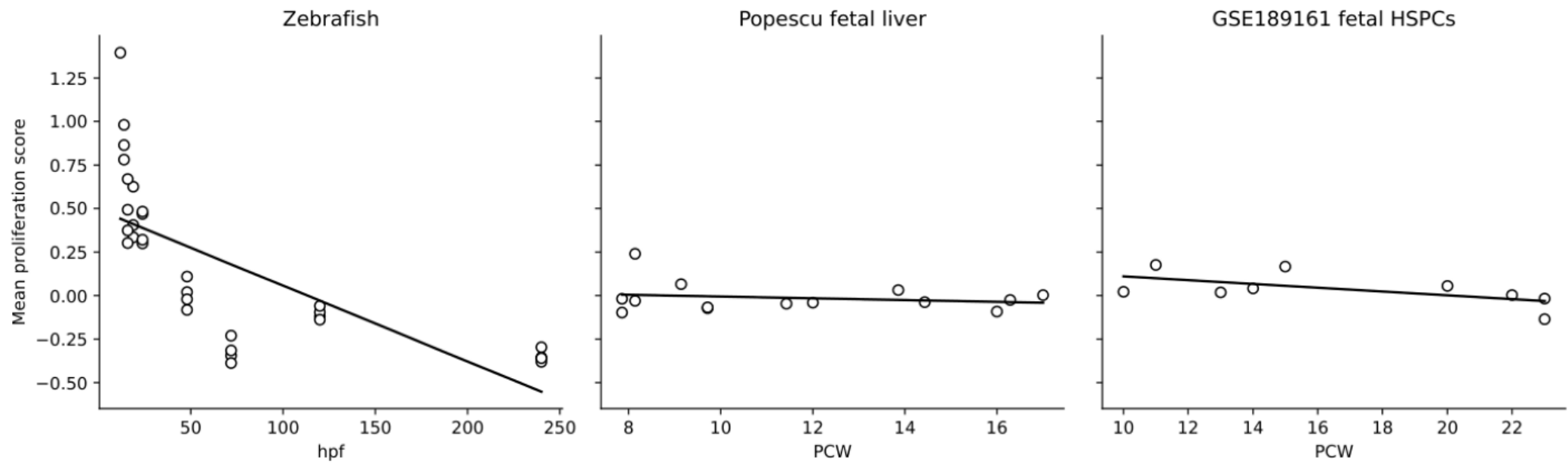
Human comparisons test expression programs, not fate, velocity or absolute growth.



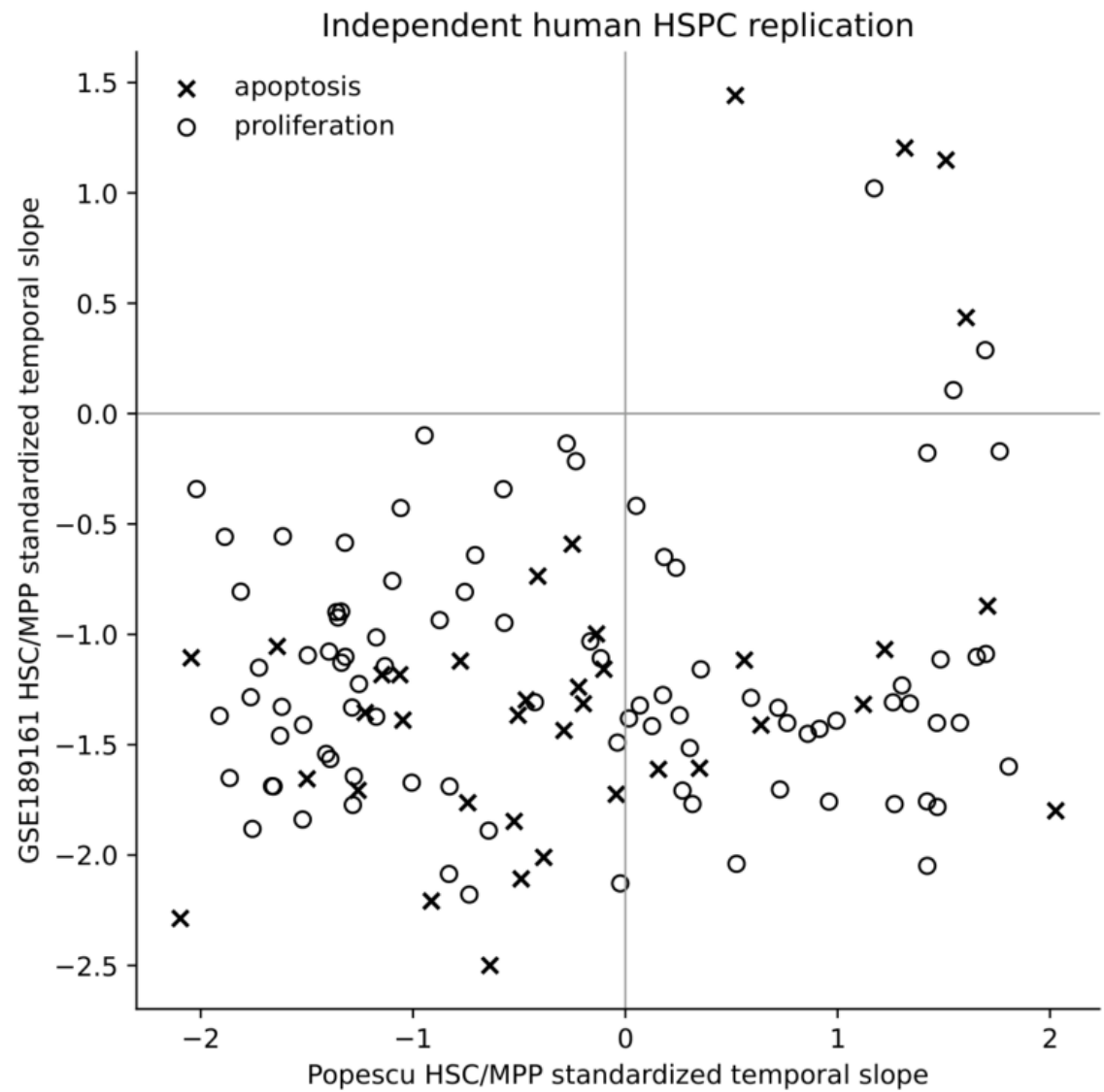
Human comparisons test expression programs, not fate, velocity or absolute growth.



Human comparisons test expression programs, not fate, velocity or absolute growth.

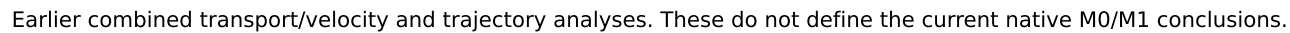


Human comparisons test expression programs, not fate, velocity or absolute growth.



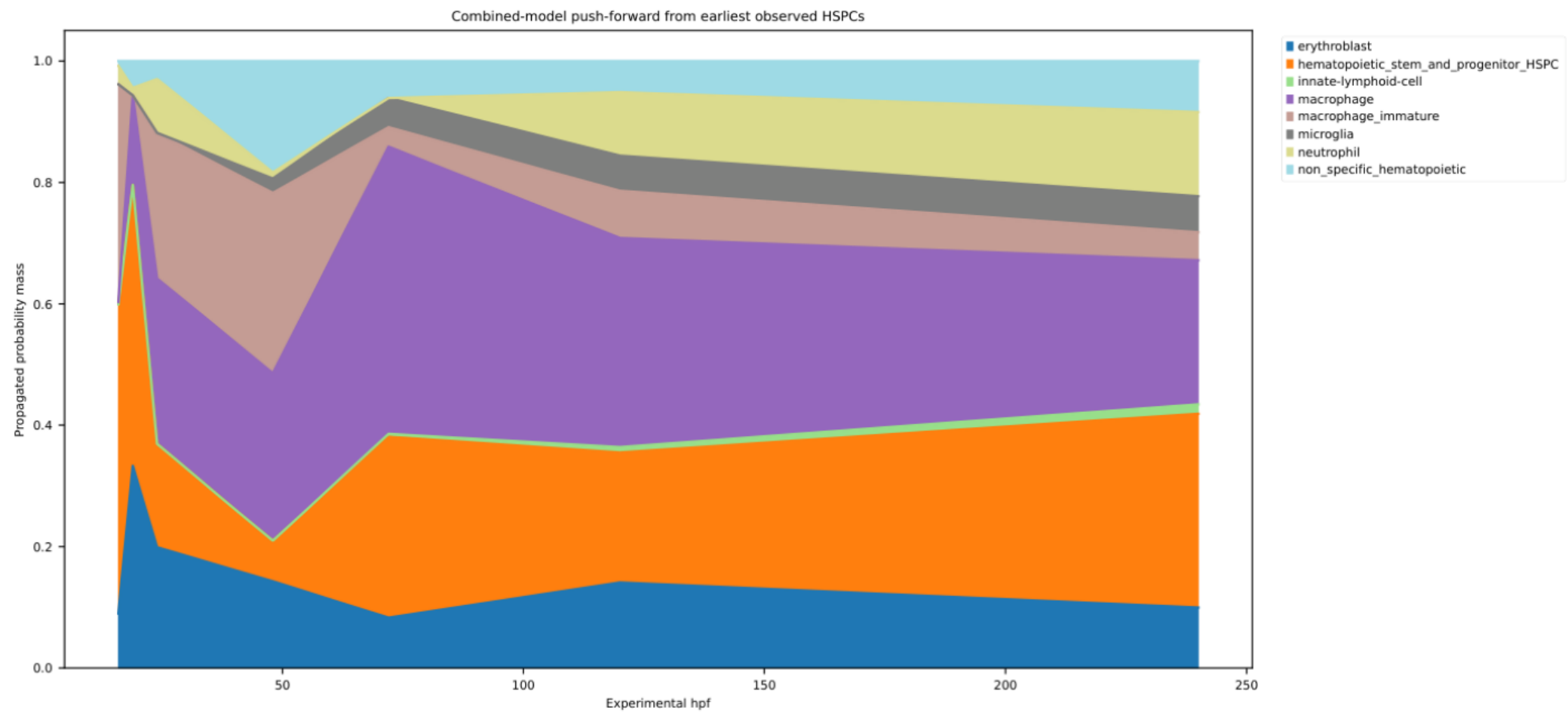
Human comparisons test expression programs, not fate, velocity or absolute growth.

HISTORICAL / SUPERSEDED



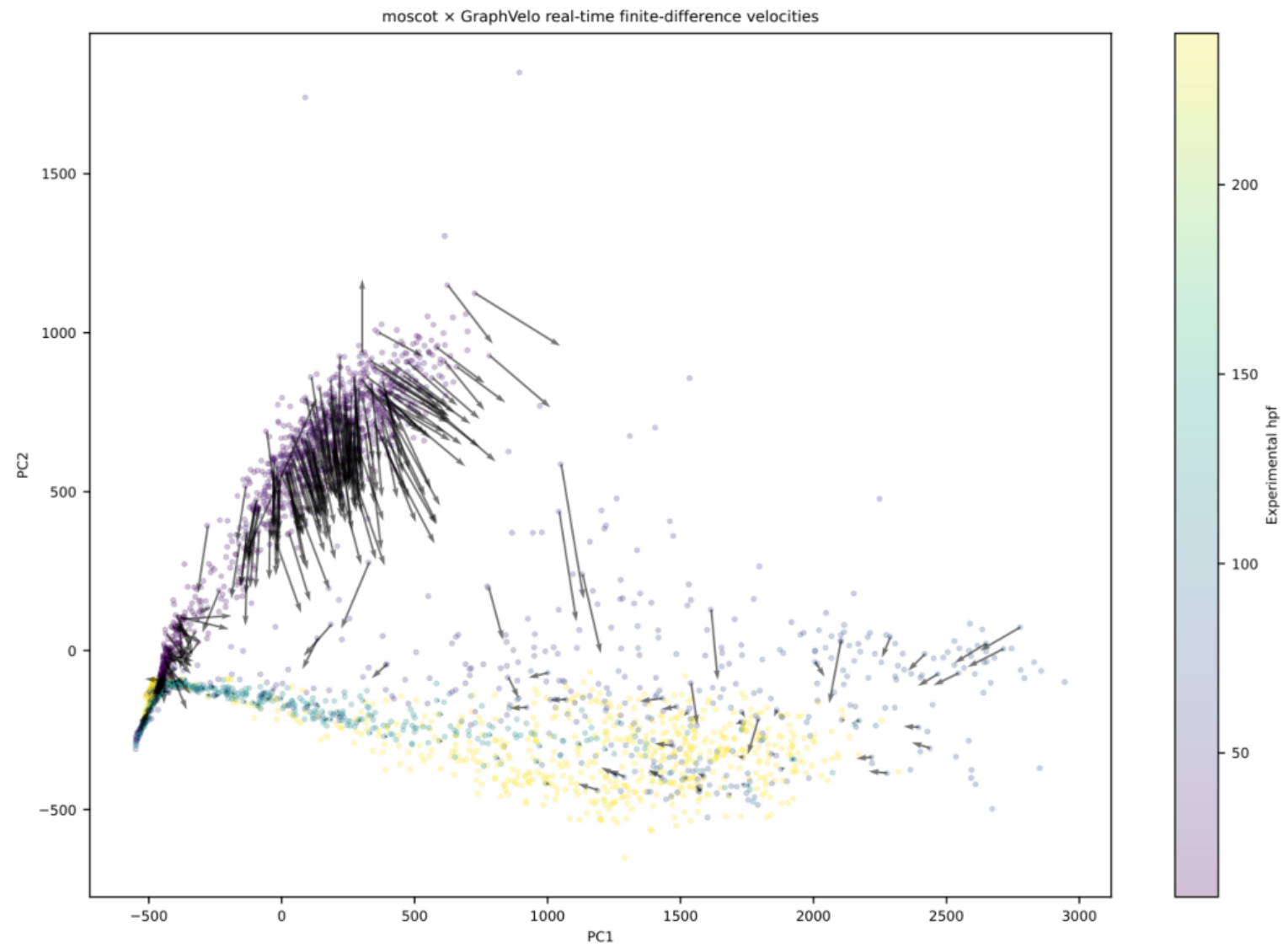
35. hspc combined pushforward fates / page 1

HISTORICAL / SUPERSEDED



Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

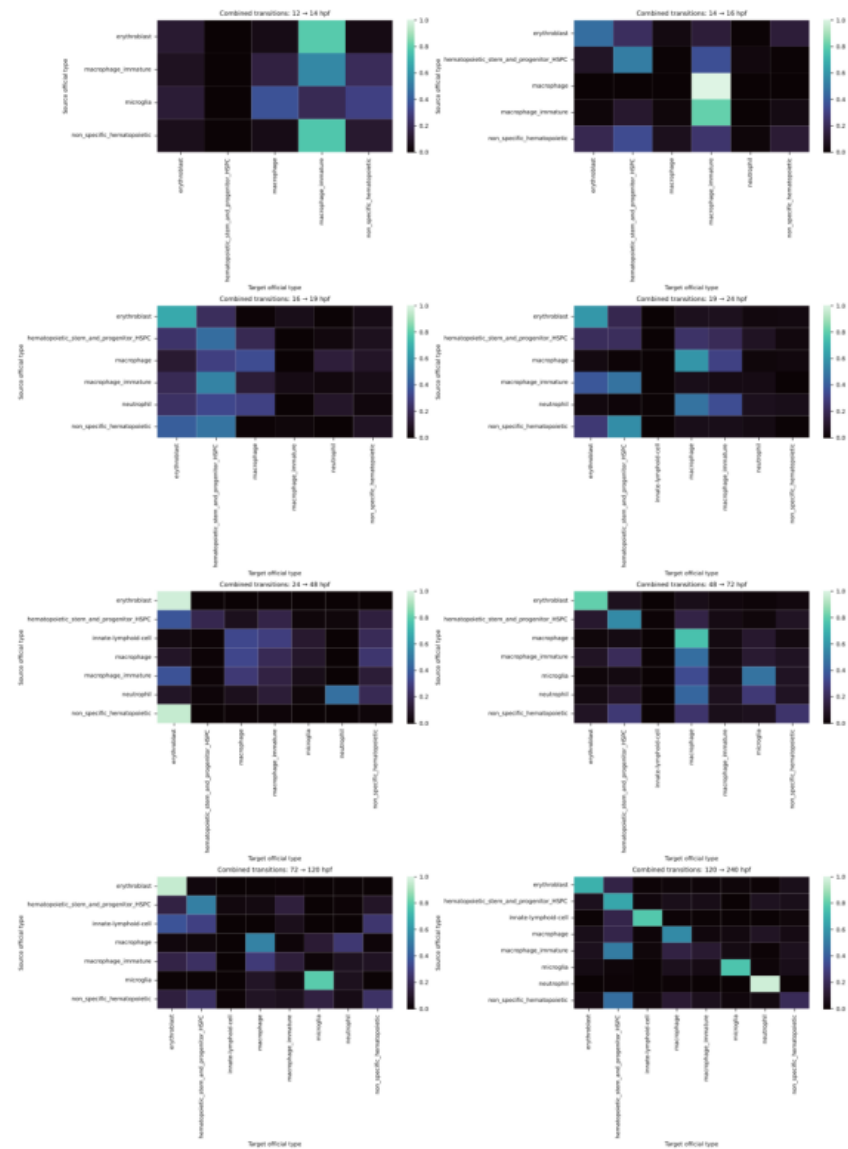
HISTORICAL / SUPERSEDED



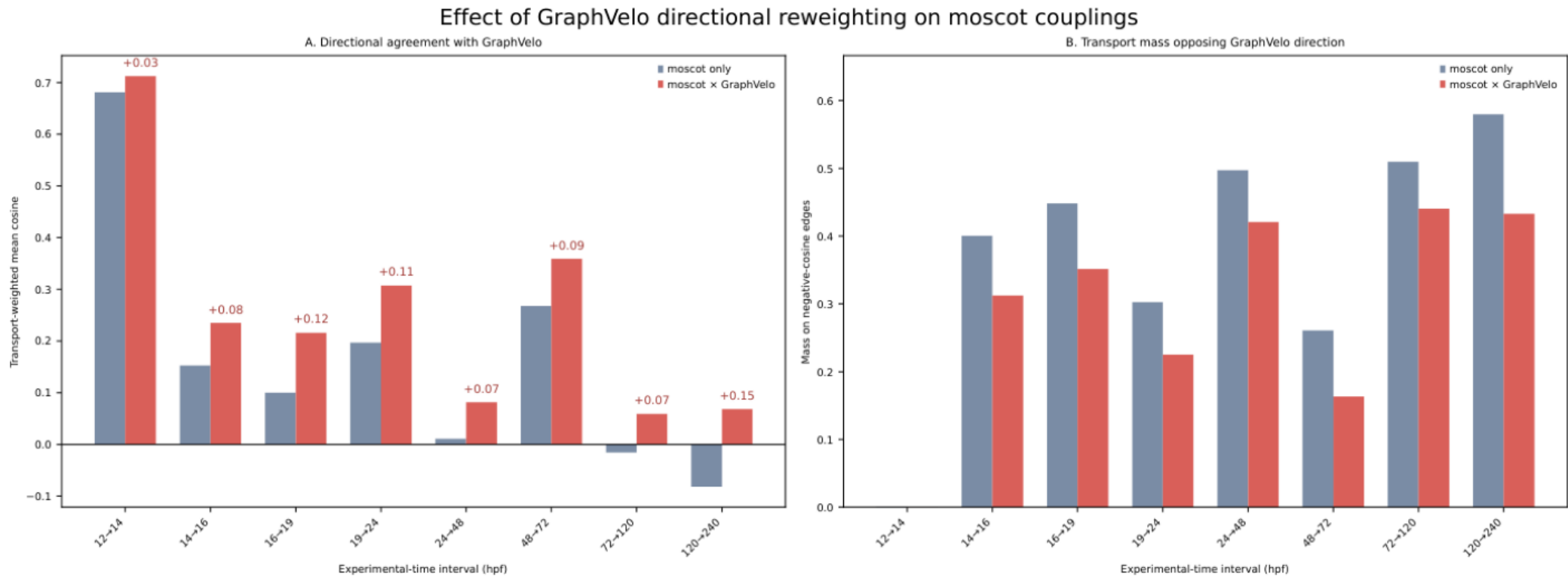
Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

37. moscot graphvelo type transition heatmaps / page 1

HISTORICAL / SUPERSEDED

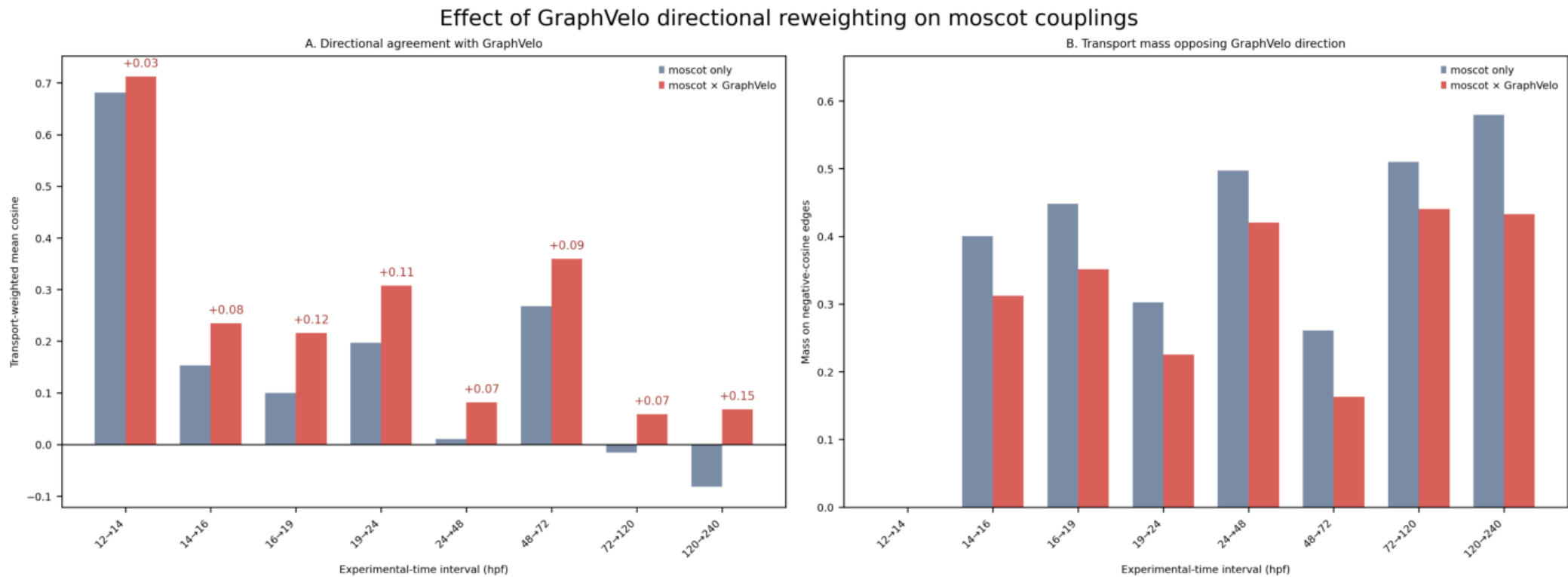


Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.



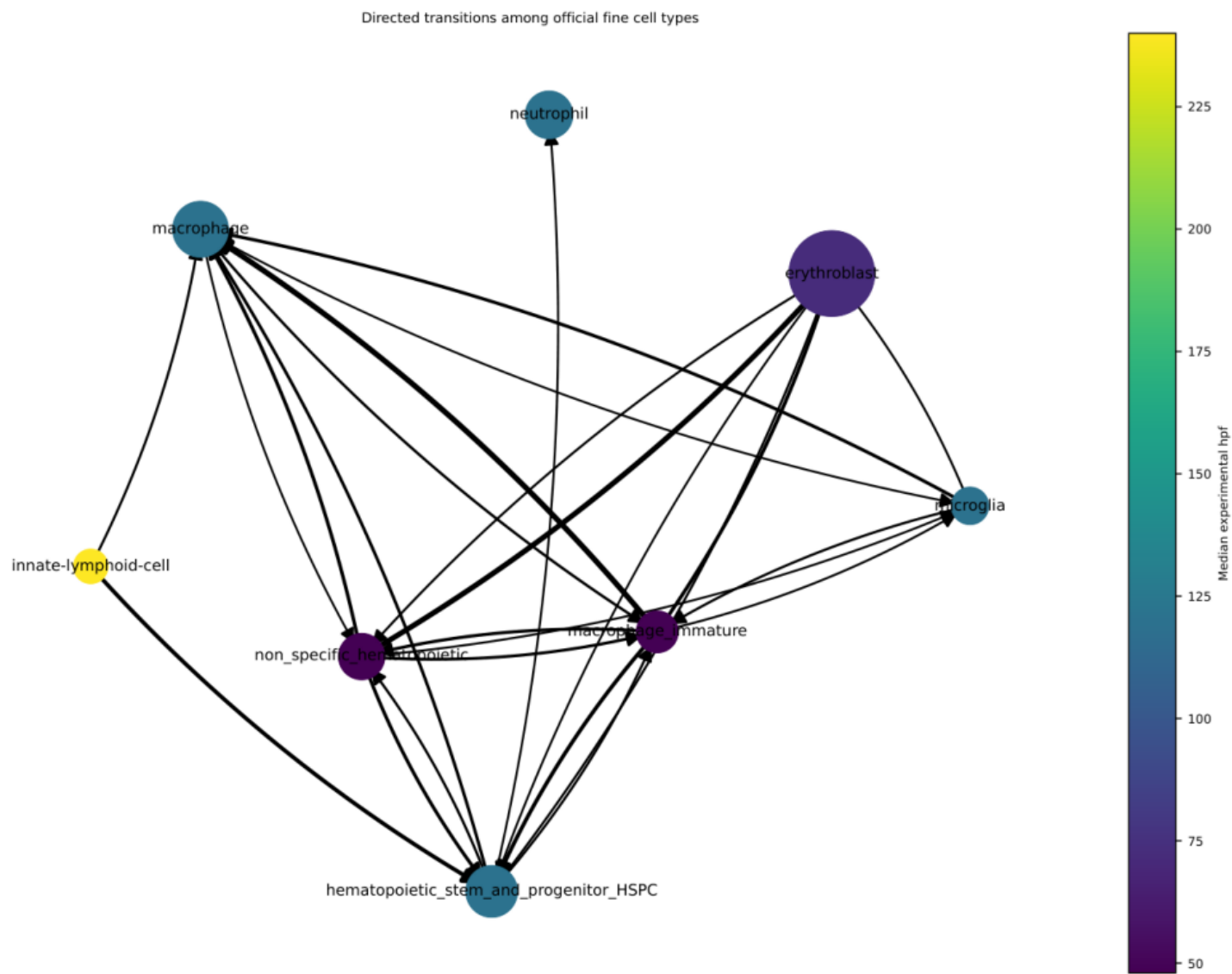
Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

39. moscot vs combined directional diagnostics
HISTORICAL / SUPERSEDED



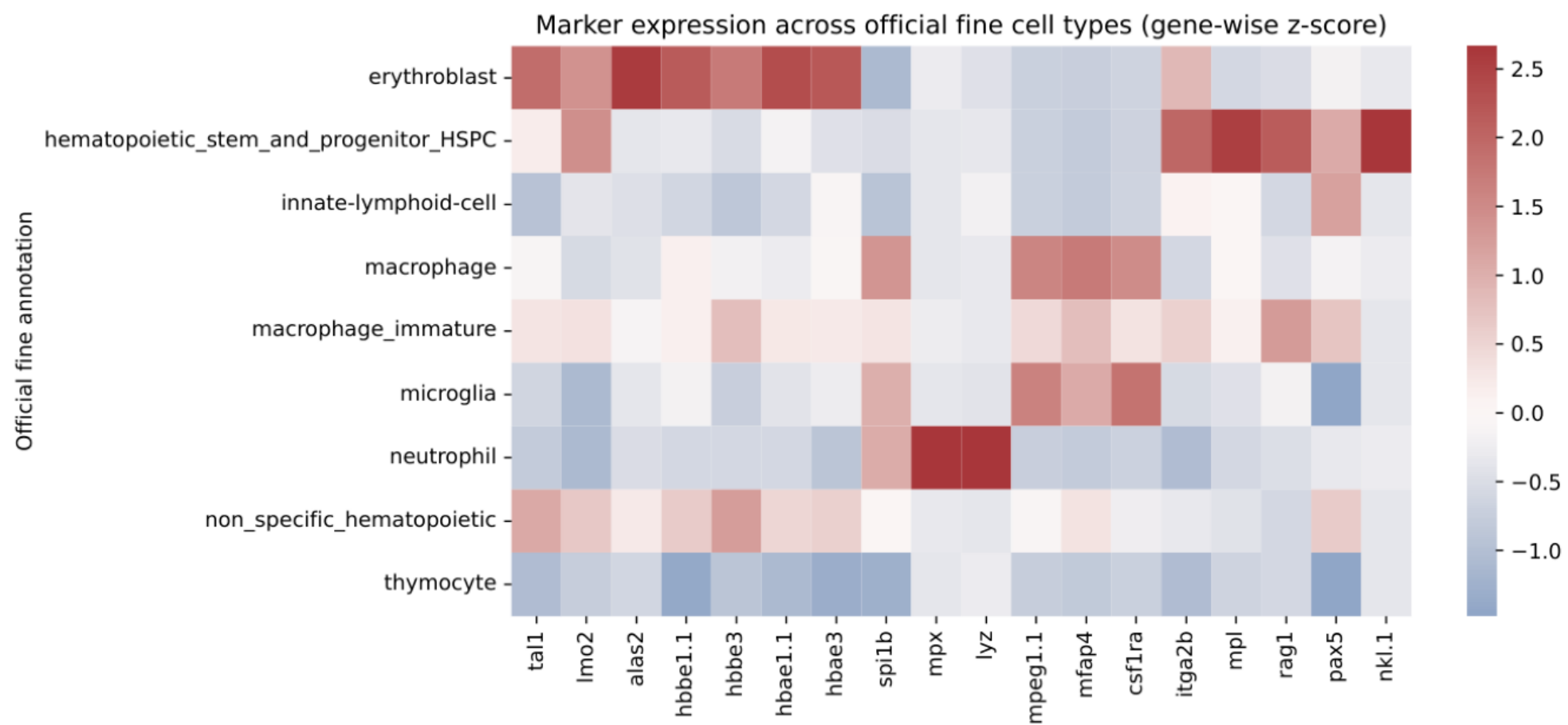
Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

HISTORICAL / SUPERSEDED



Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

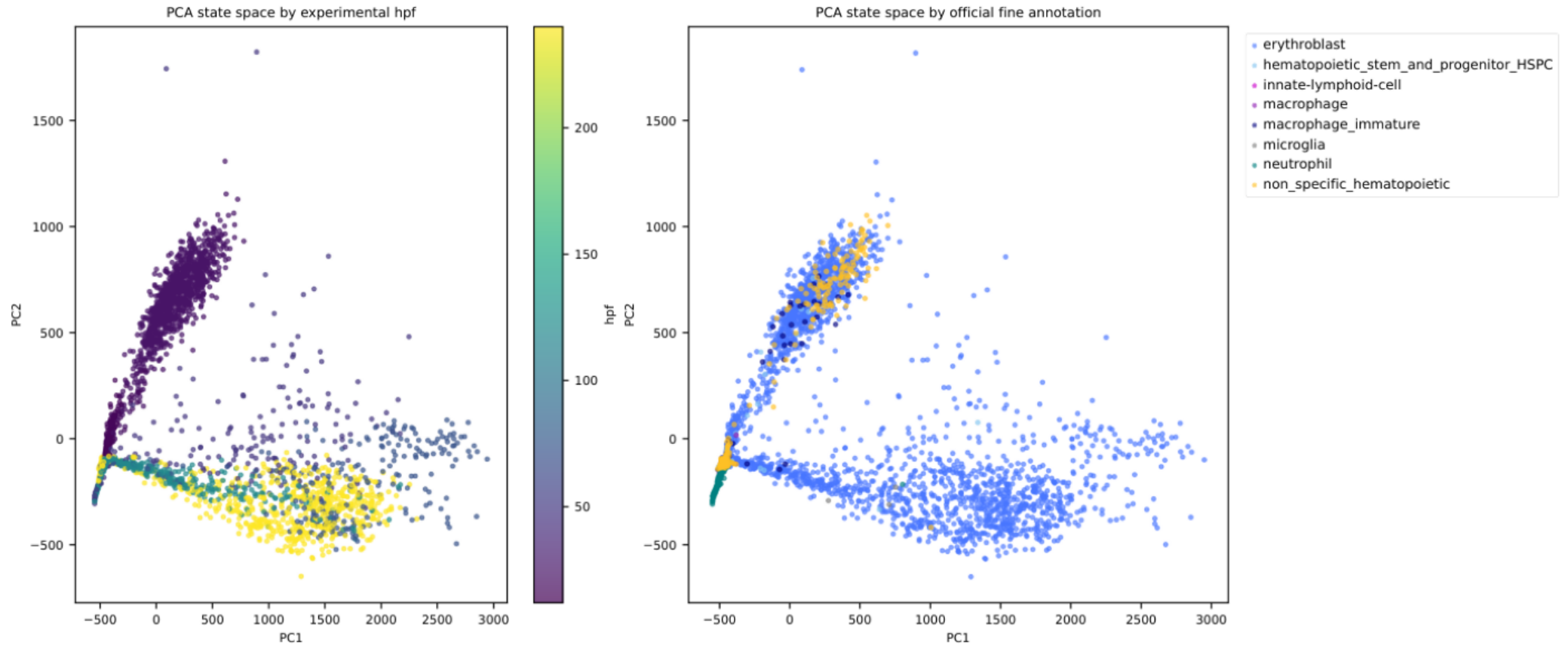
HISTORICAL / SUPERSEDED



Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

42. official fine type pca state space / page 1

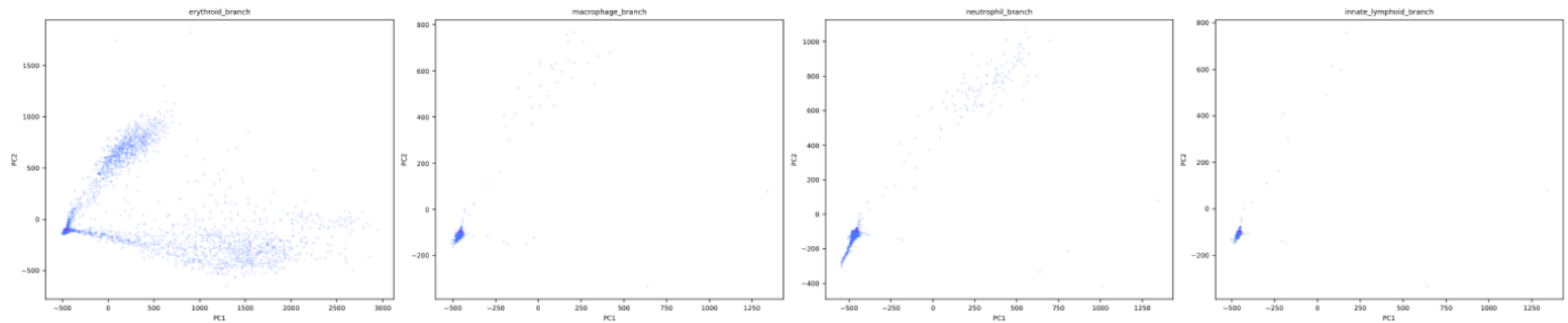
HISTORICAL / SUPERSEDED



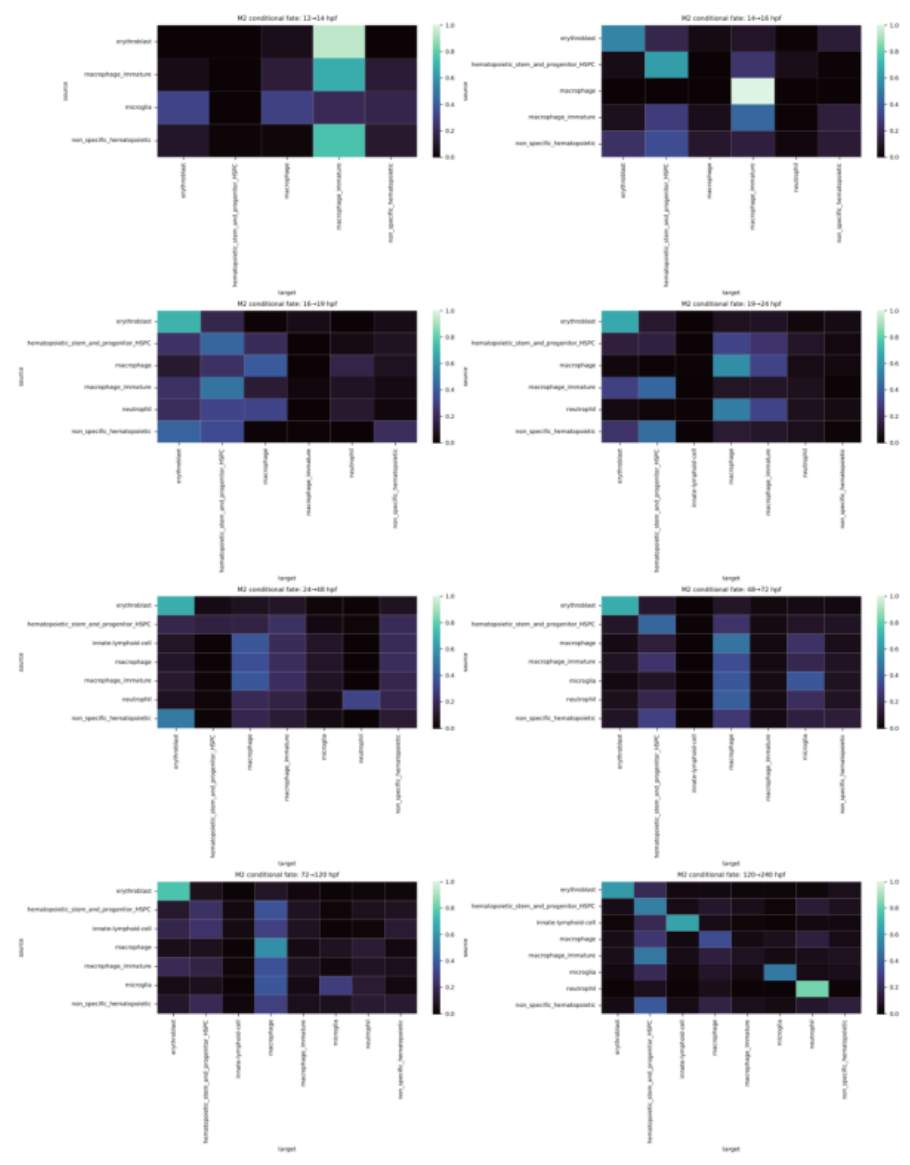
Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

43. official label ode trajectories / page 1

HISTORICAL / SUPERSEDED

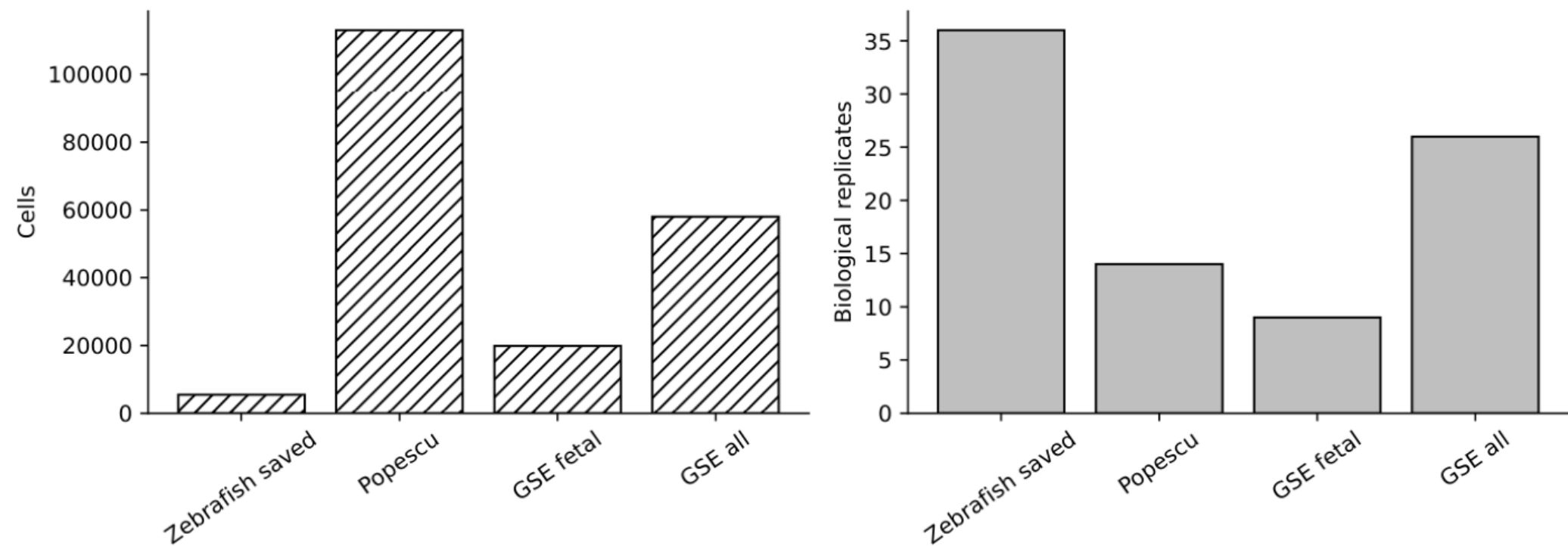


Earlier combined transport/velocity and trajectory analyses. These do not define the current native M0/M1 conclusions.

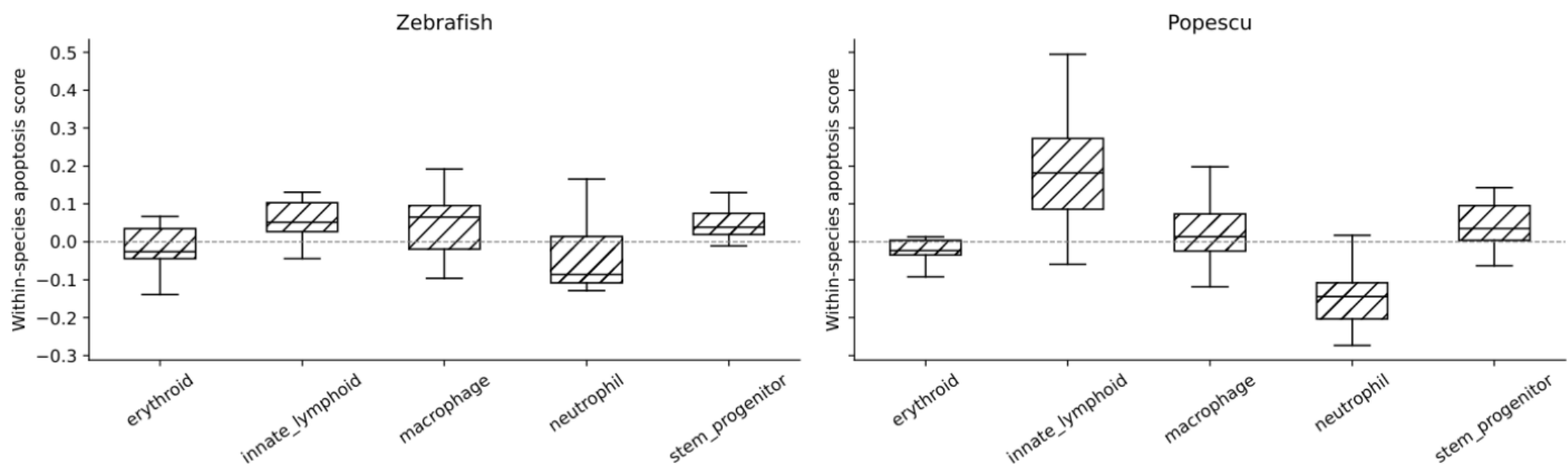


Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

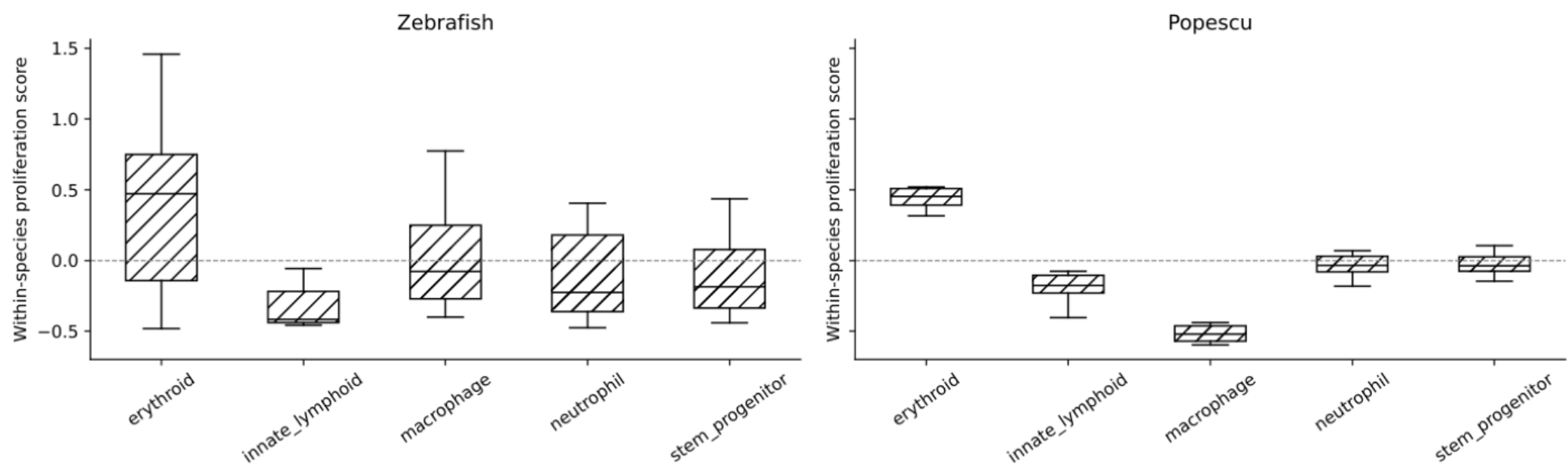
Datasets used in cross-species validation



Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

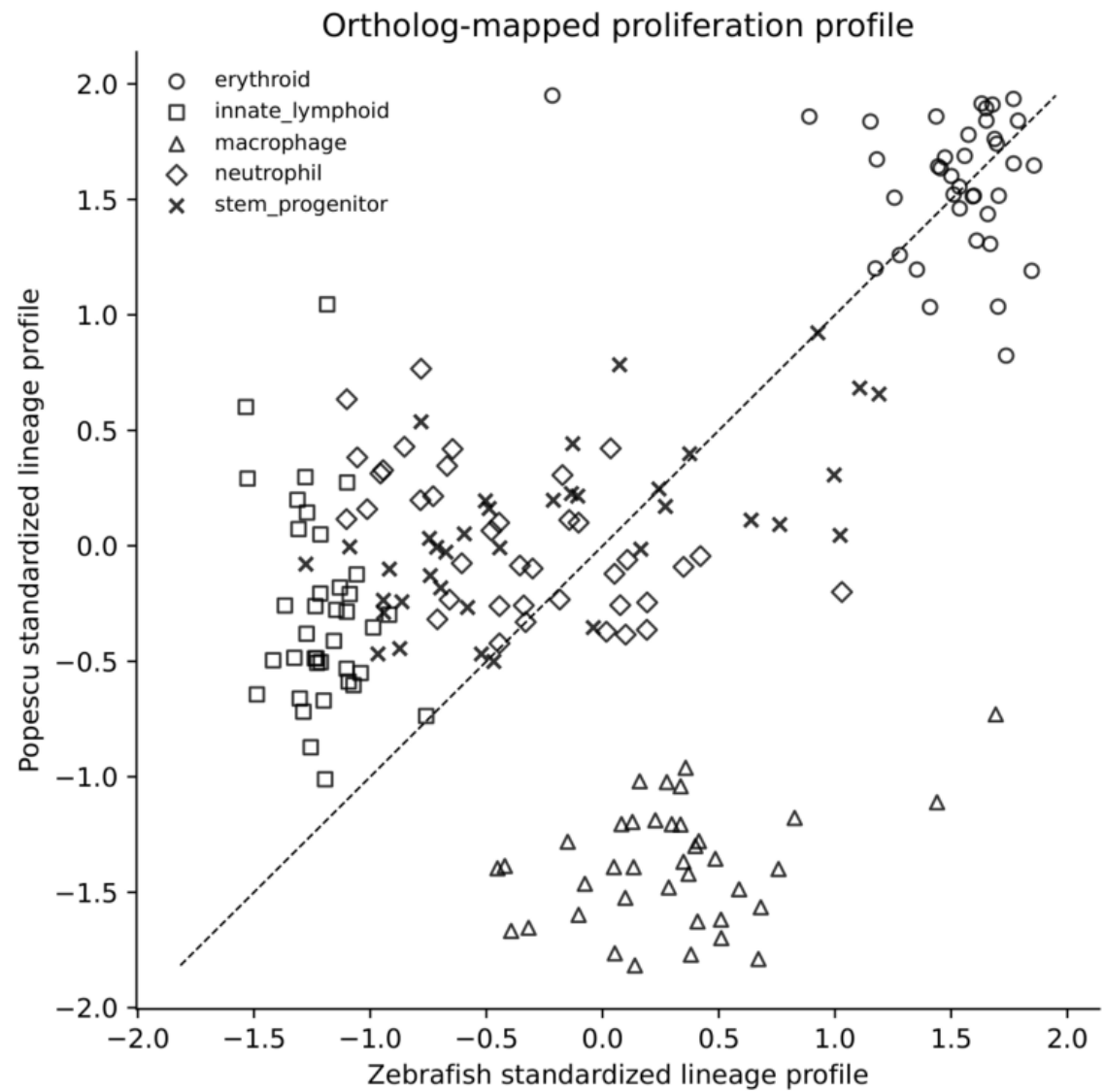


Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

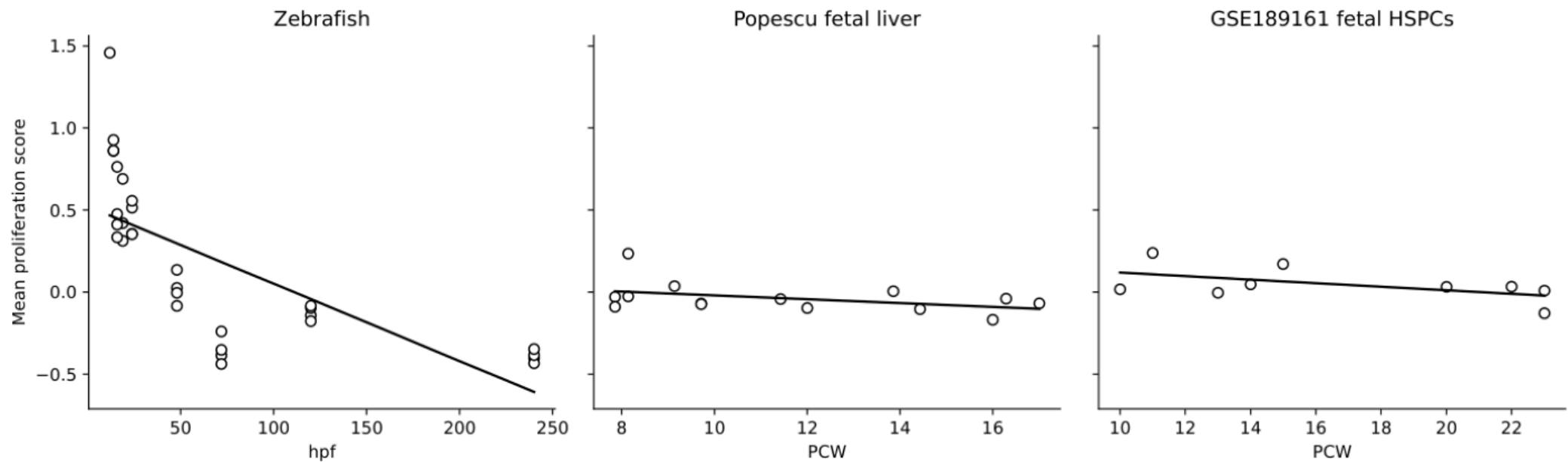


Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

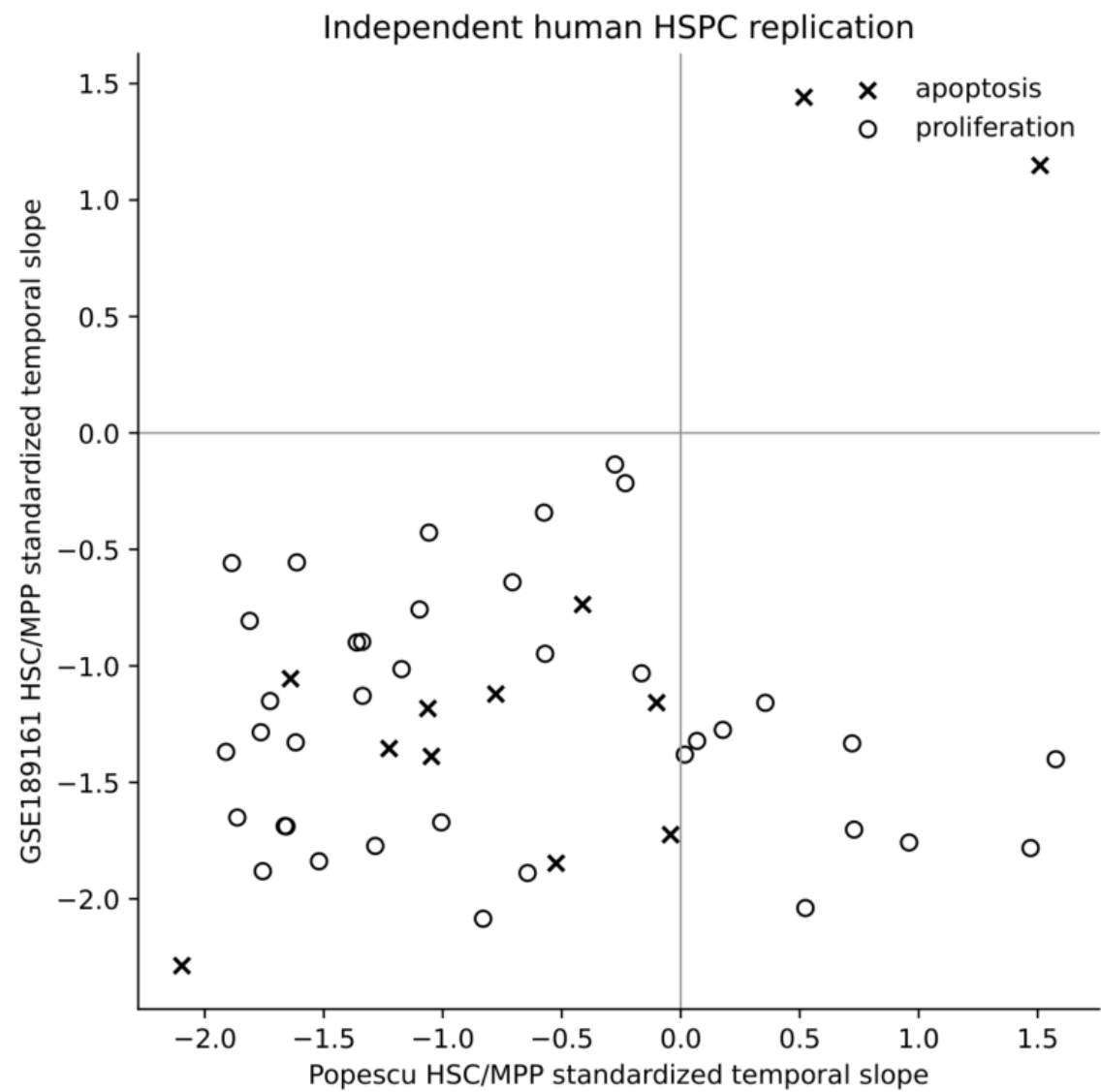
ARCHIVED / SUPERSEDED



Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

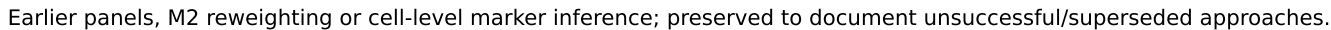


Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.



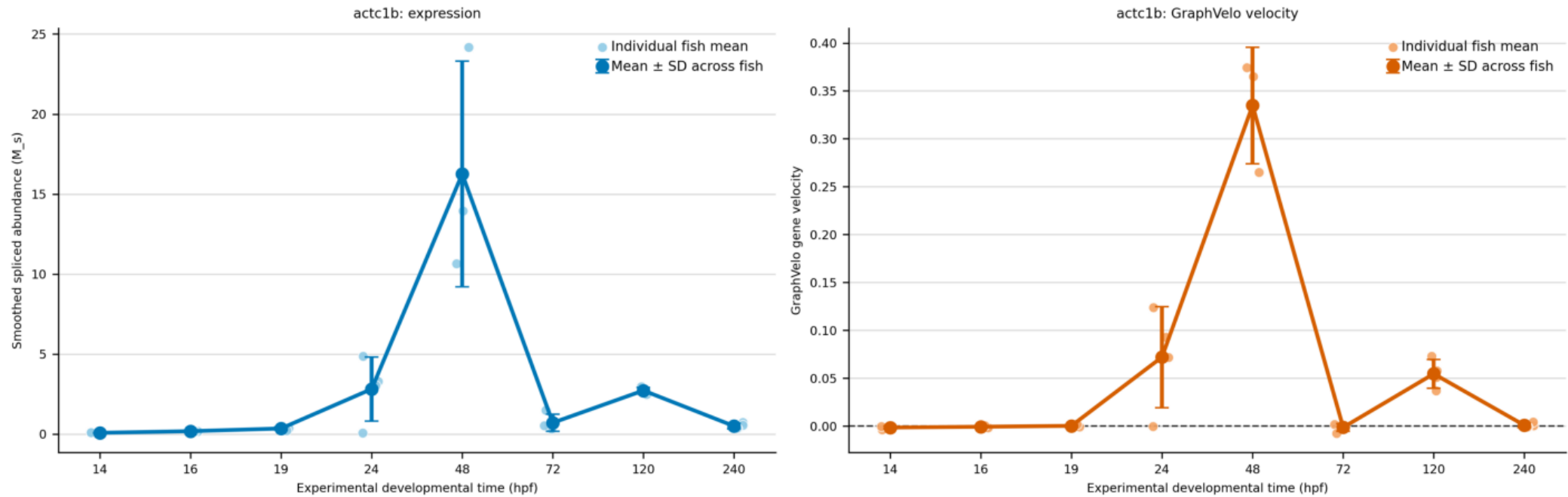
Earlier panels, M2 reweighting or cell-level marker inference; preserved to document unsuccessful/superseded approaches.

ARCHIVED / SUPERSEDED



52. actc1b pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

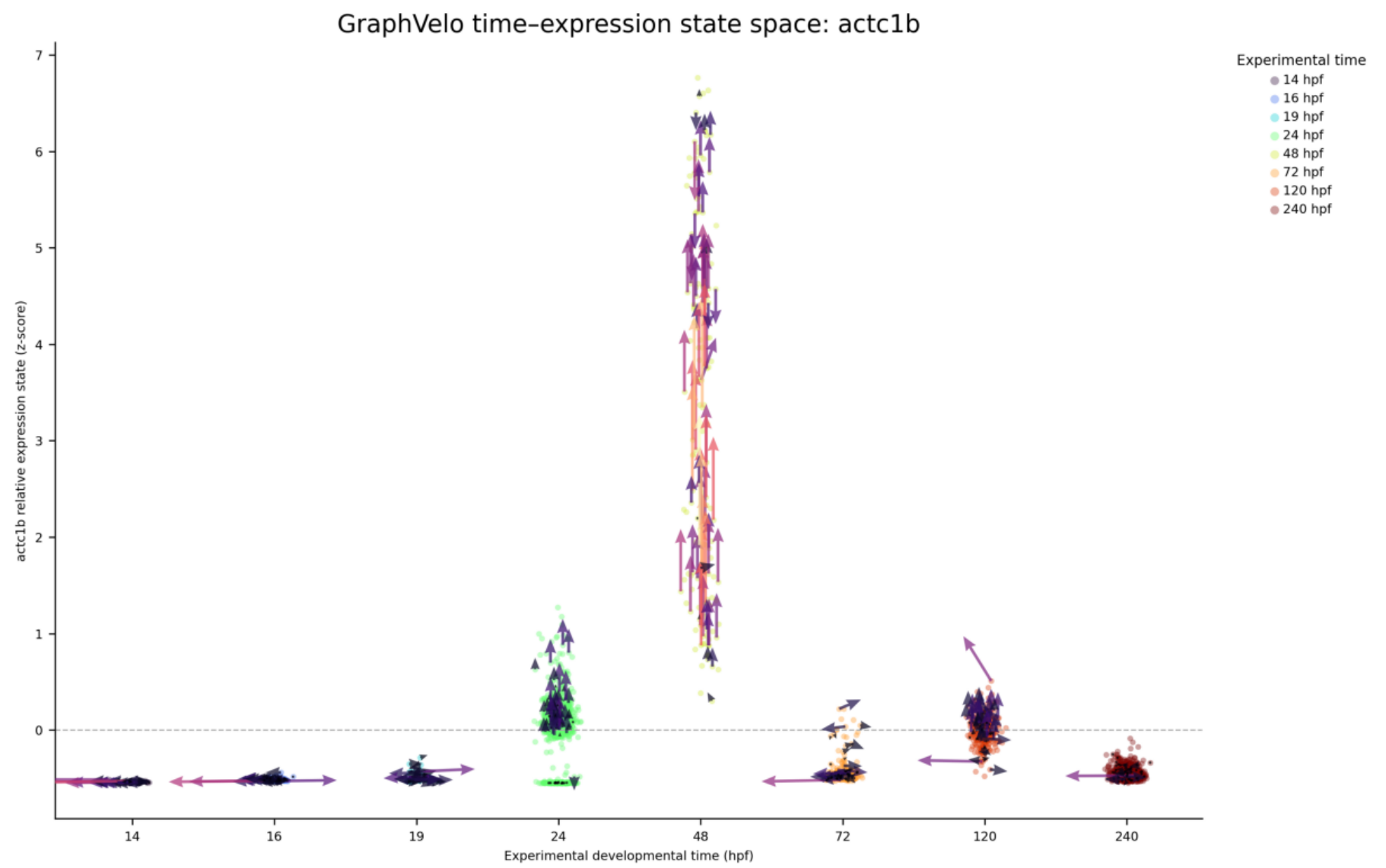
Cluster-23 pilot real-time trajectory: actc1b
GraphVelo-projected gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

53. actc1b time expression manifold

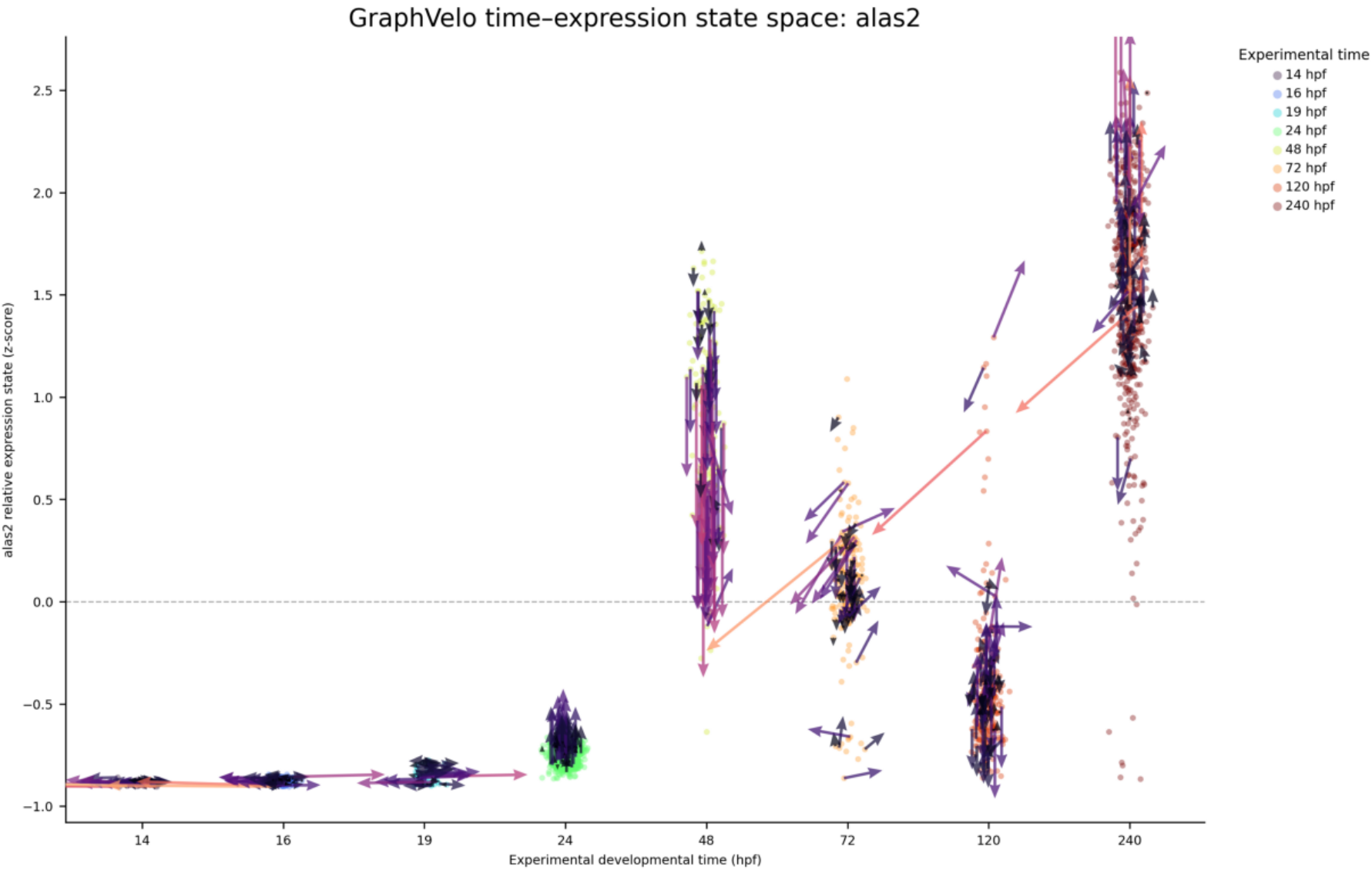
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

54. alas2 time expression manifold

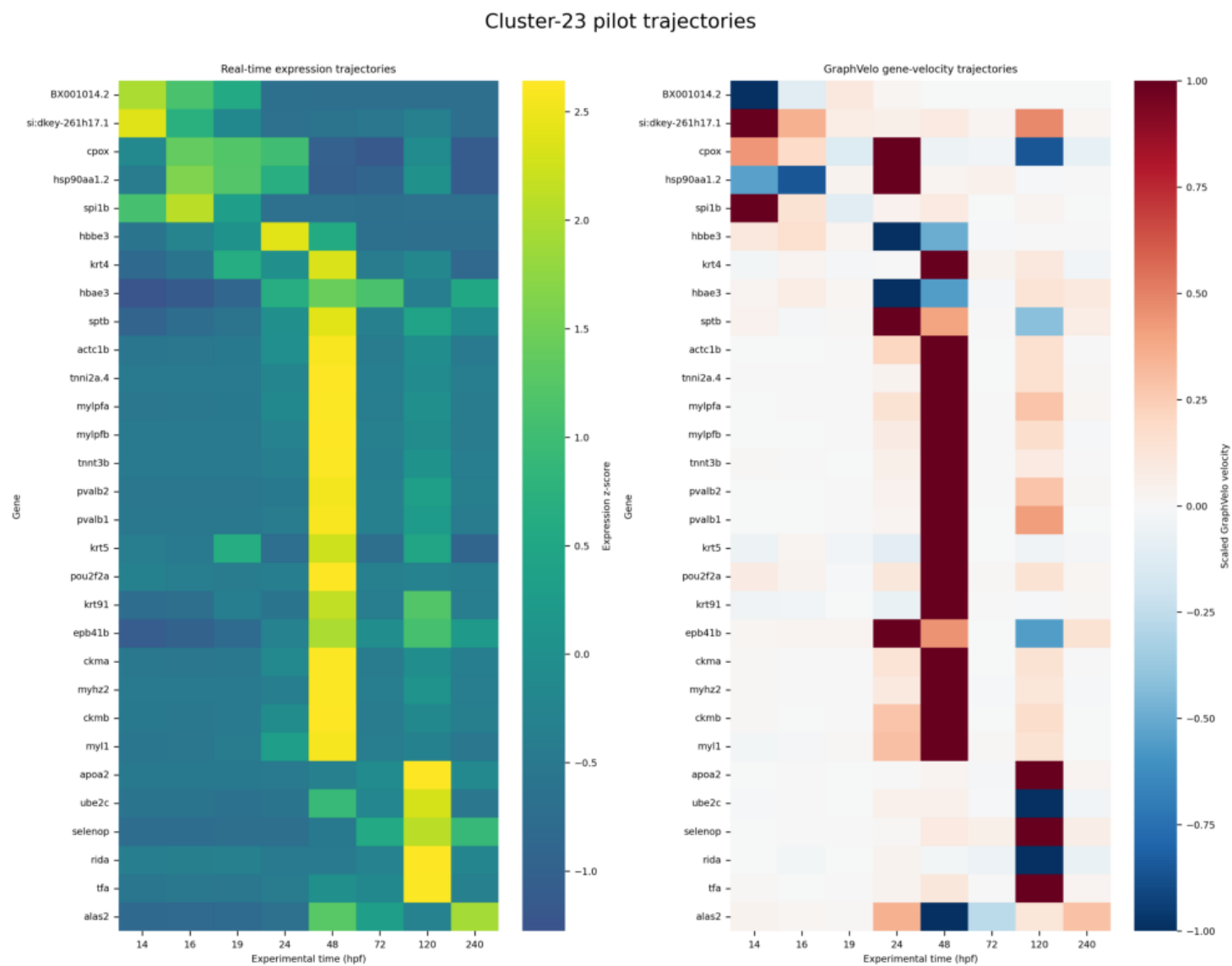
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

55. cluster23 multigene pilot trajectory

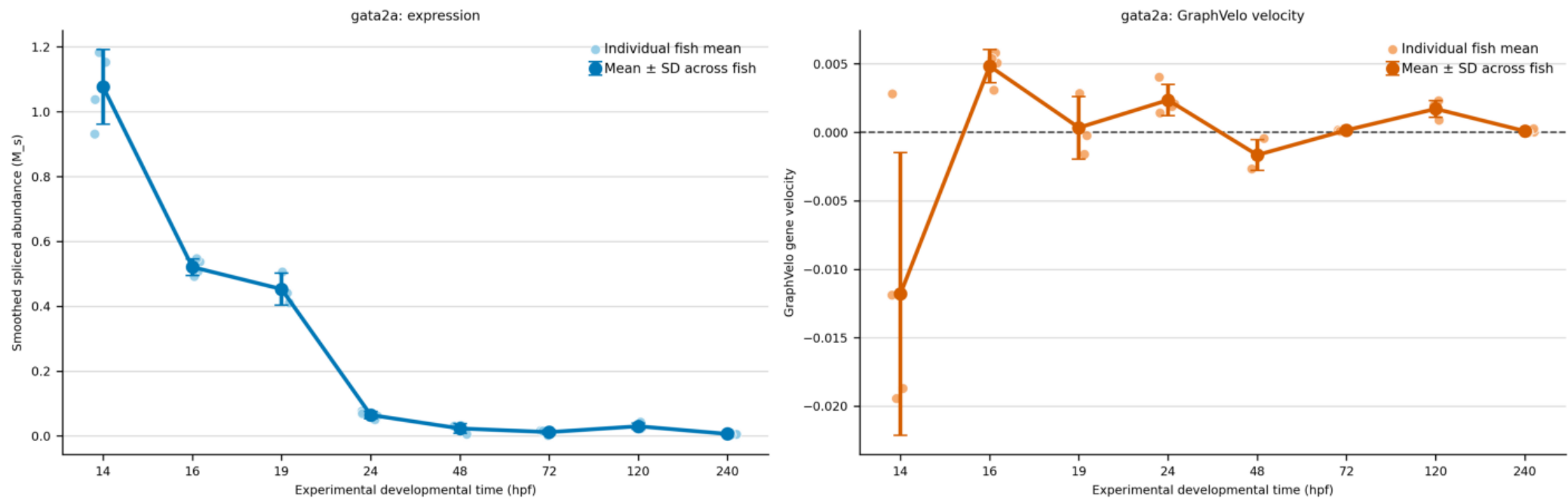
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

56. gata2a pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

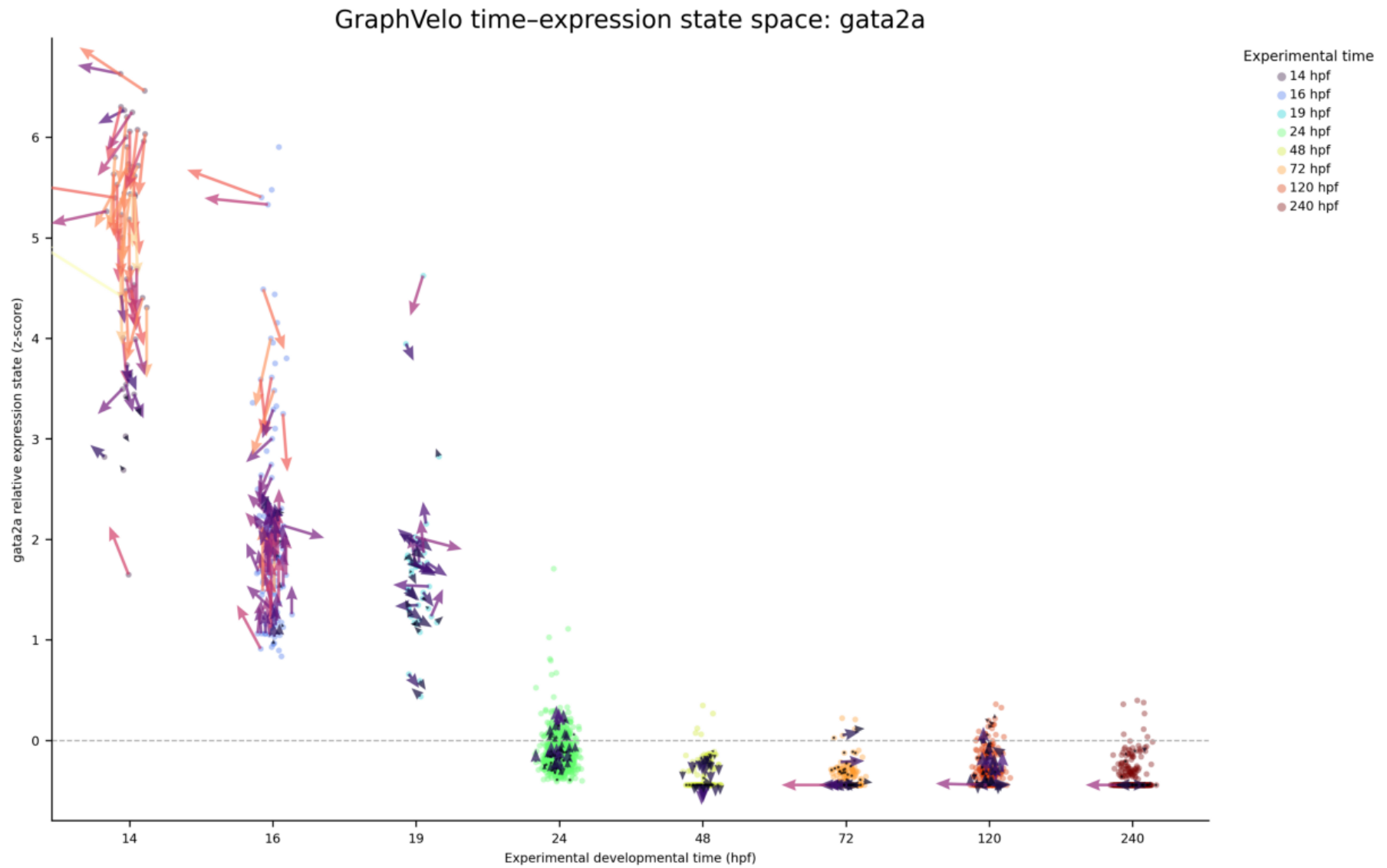
Cluster-23 pilot real-time trajectory: gata2a
GraphVelo training gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

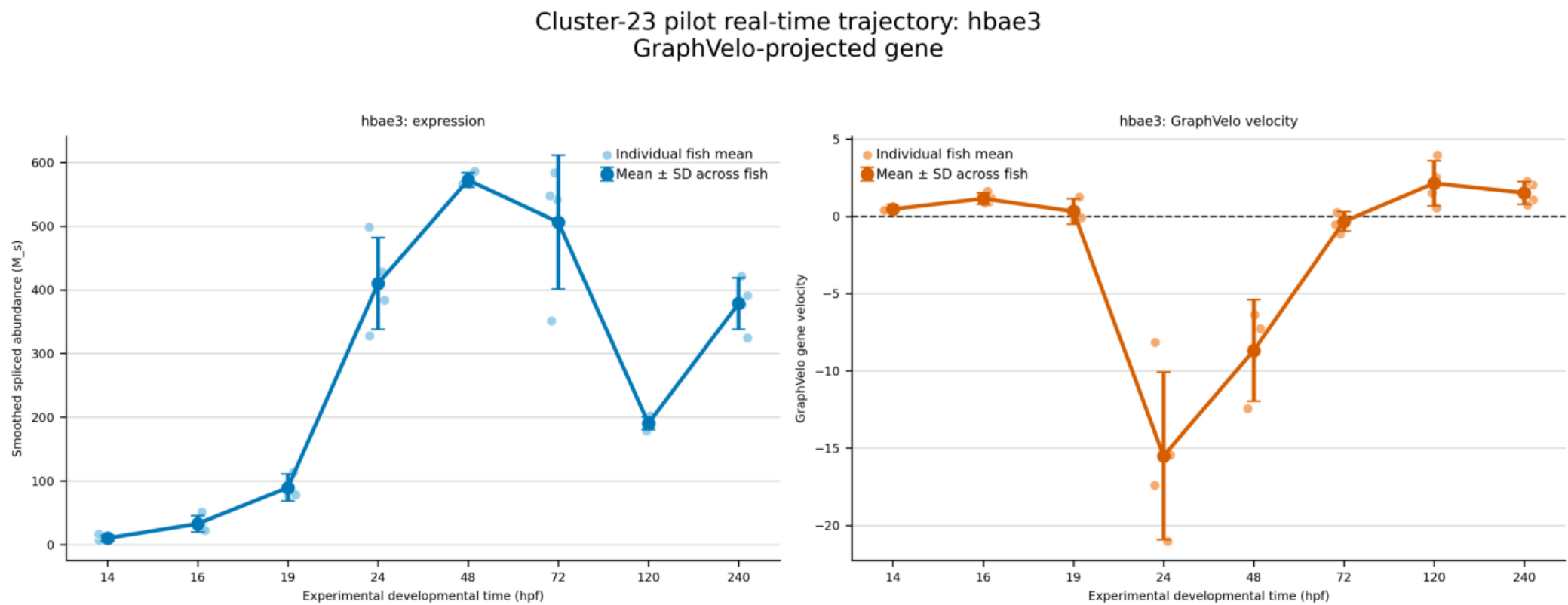
57. gata2a time expression manifold

EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

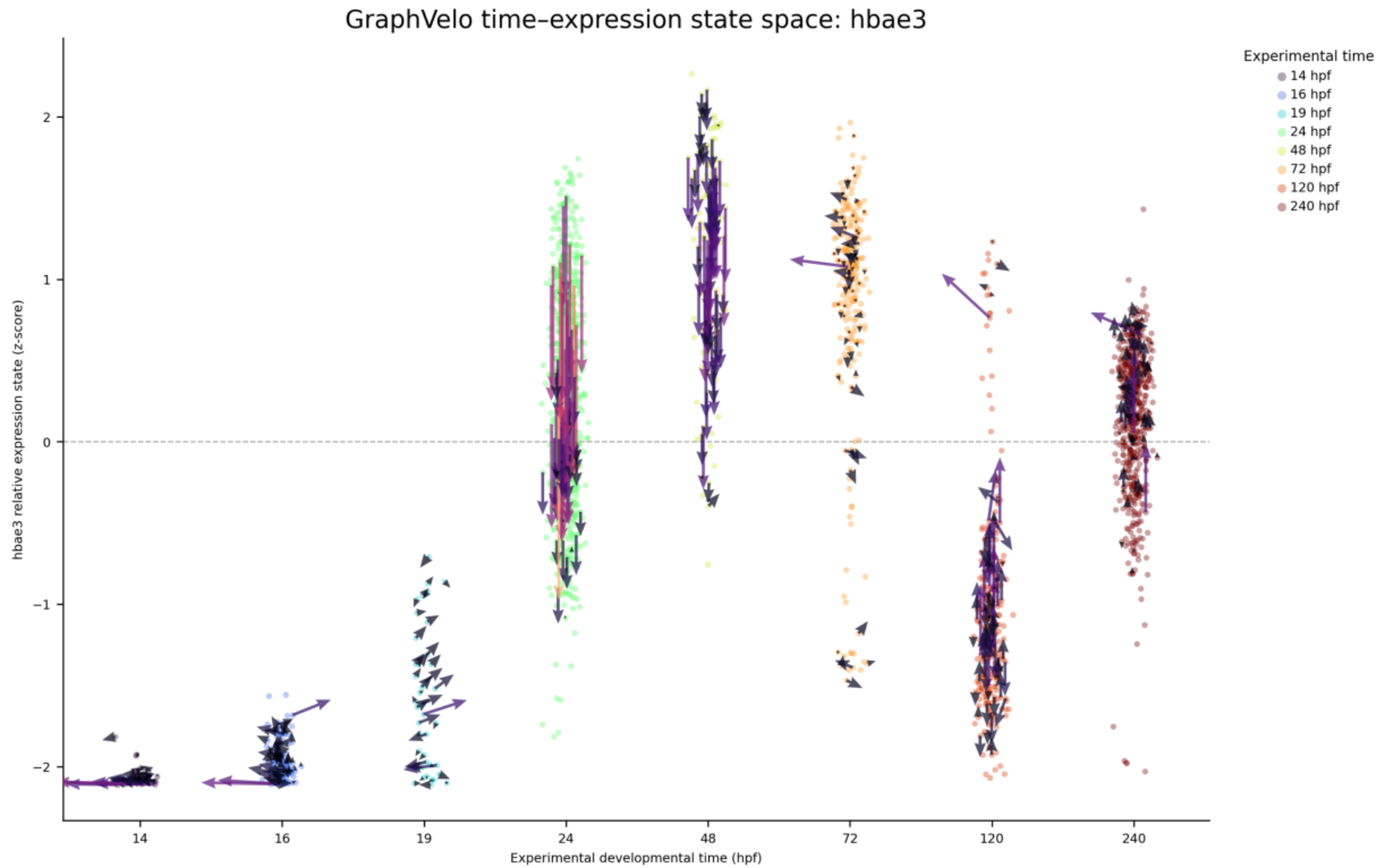
58. hbae3 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

59. hbae3 time expression manifold

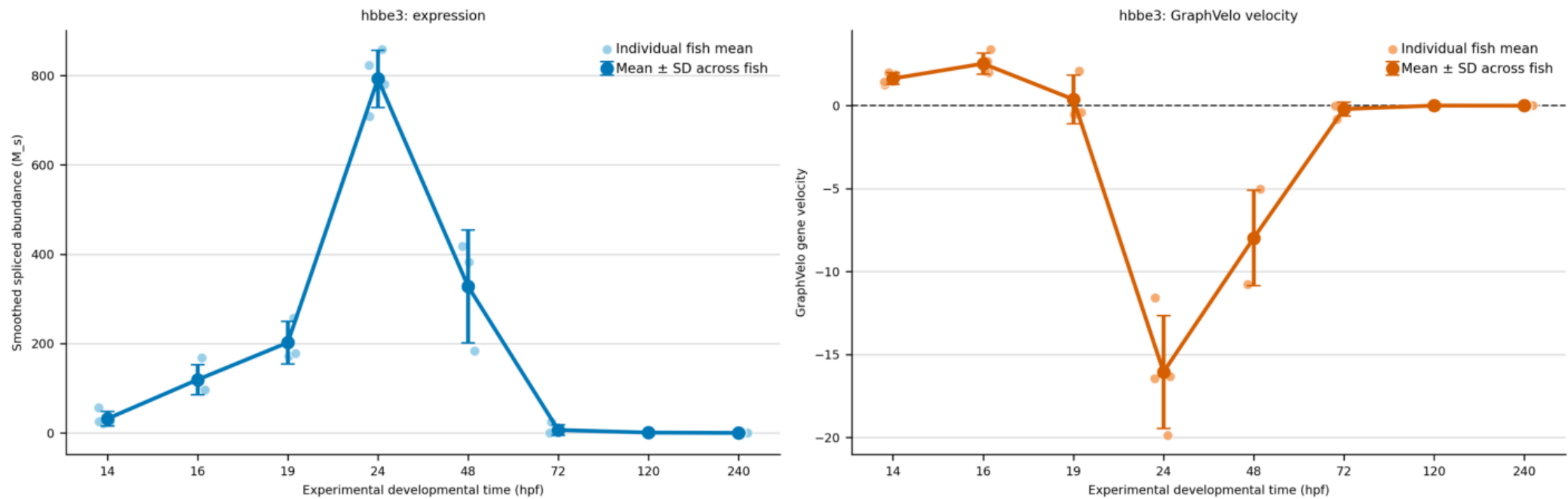
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

60. hbbe3 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

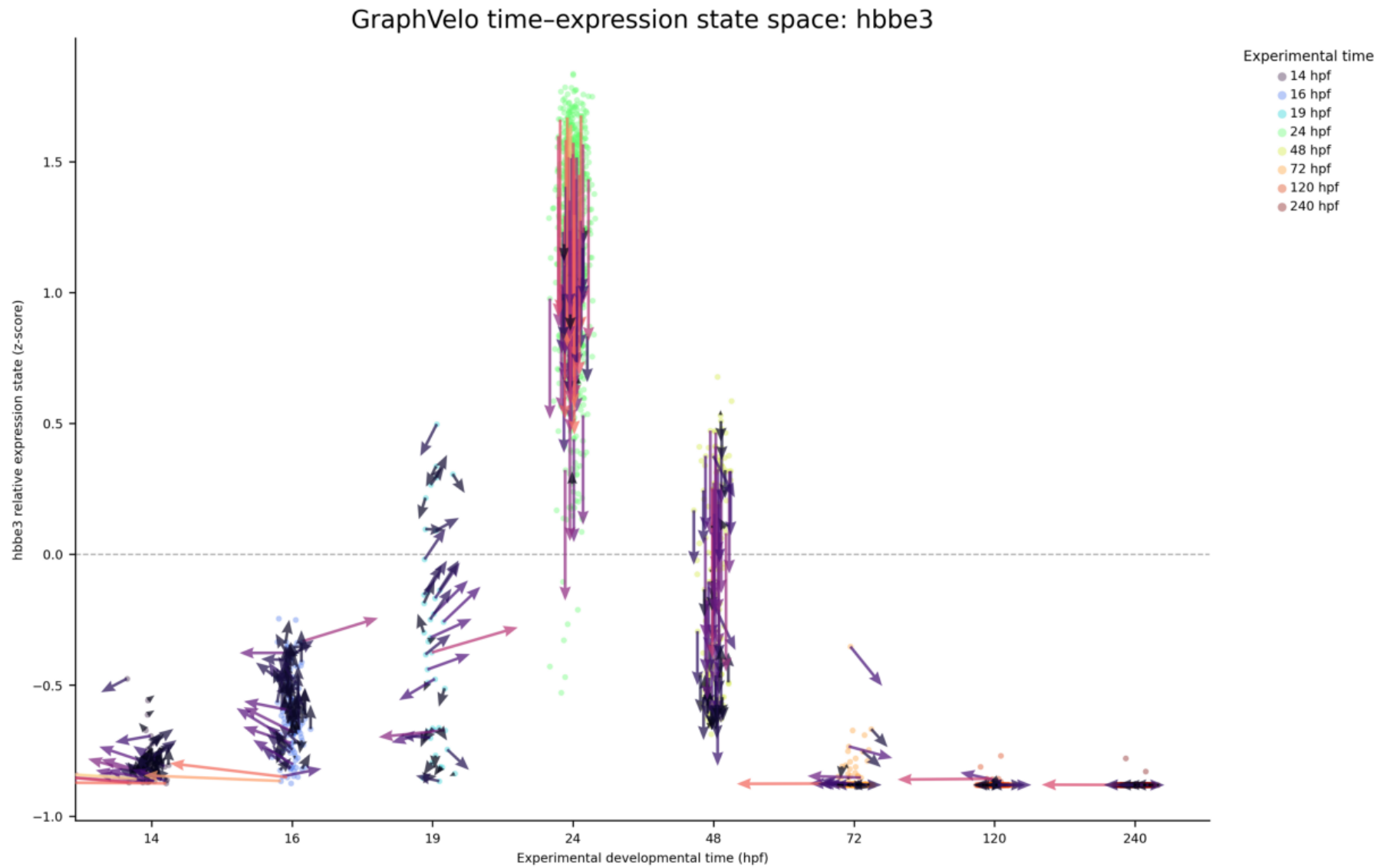
Cluster-23 pilot real-time trajectory: hbbe3
GraphVelo training gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

61. hbbe3 time expression manifold

EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

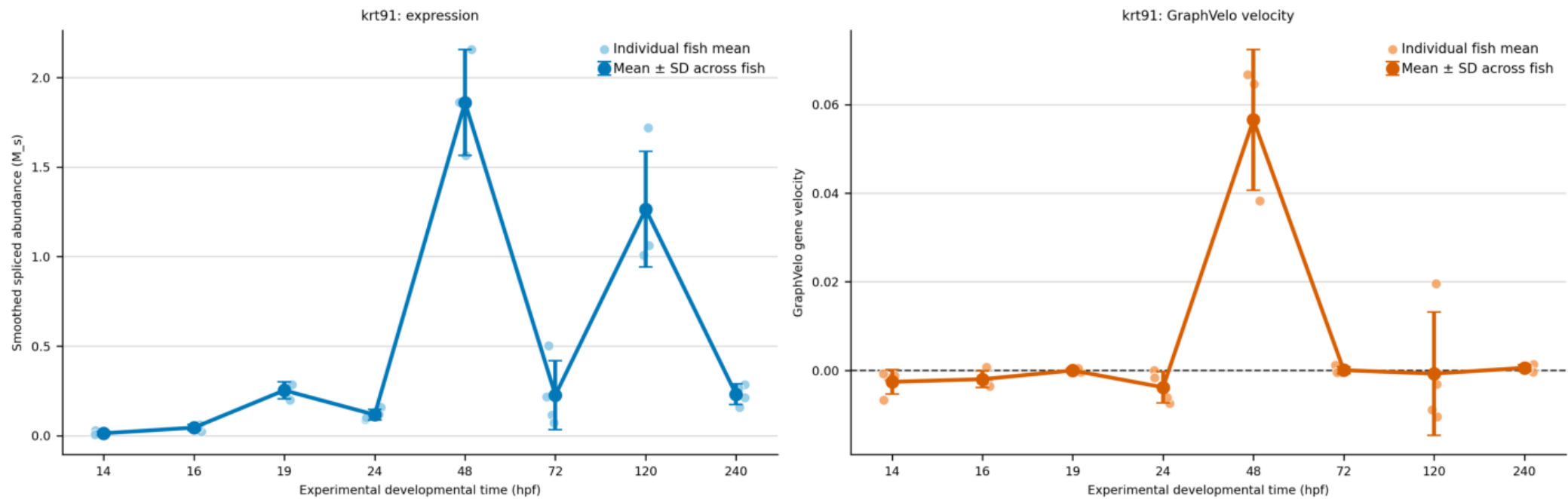
62. krt4 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

63. krt91 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

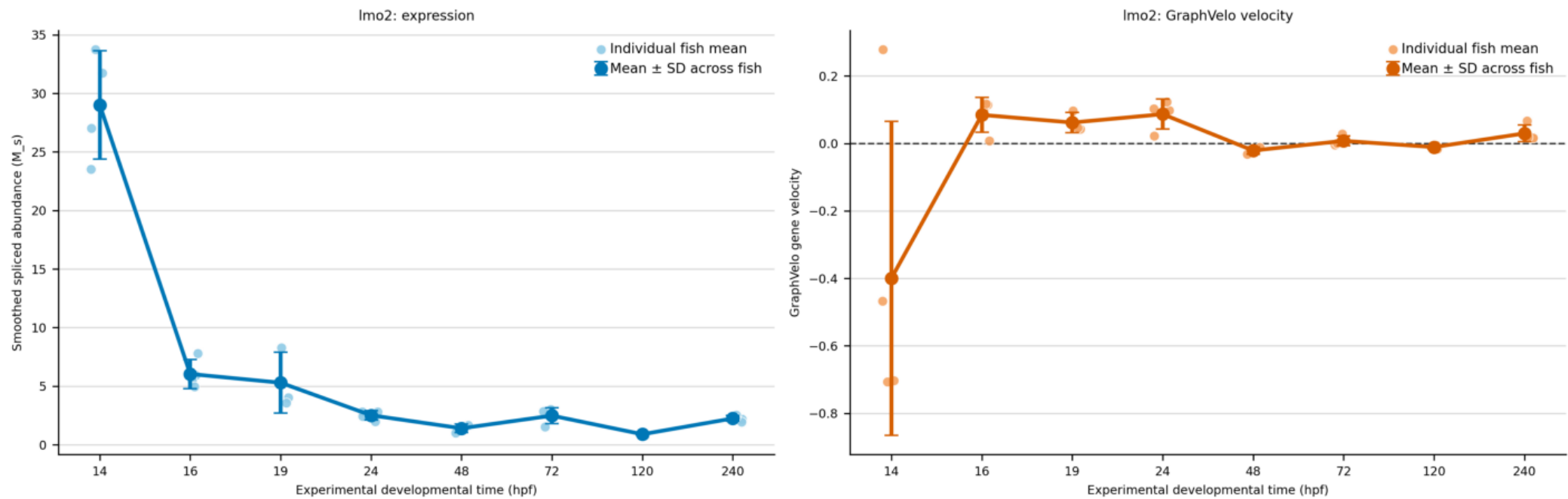
Cluster-23 pilot real-time trajectory: krt91
GraphVelo-projected gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

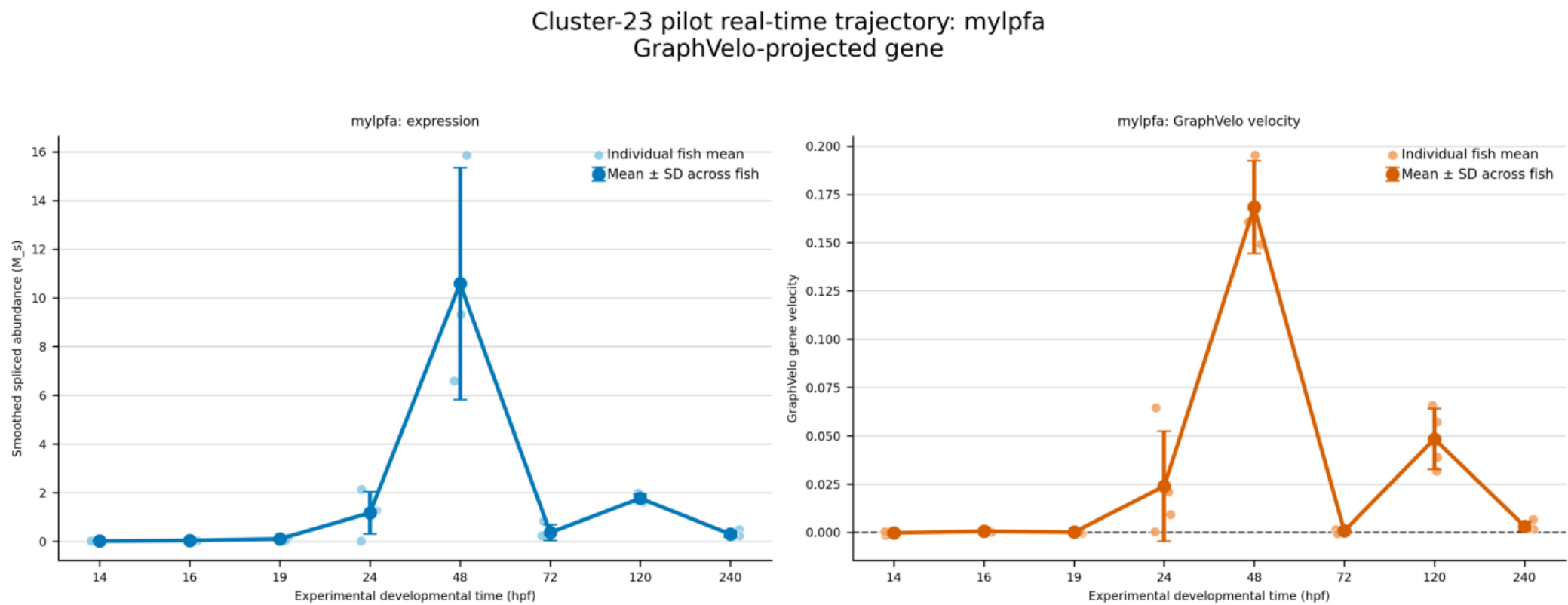
64. Imo2 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

Cluster-23 pilot real-time trajectory: Imo2
GraphVelo training gene



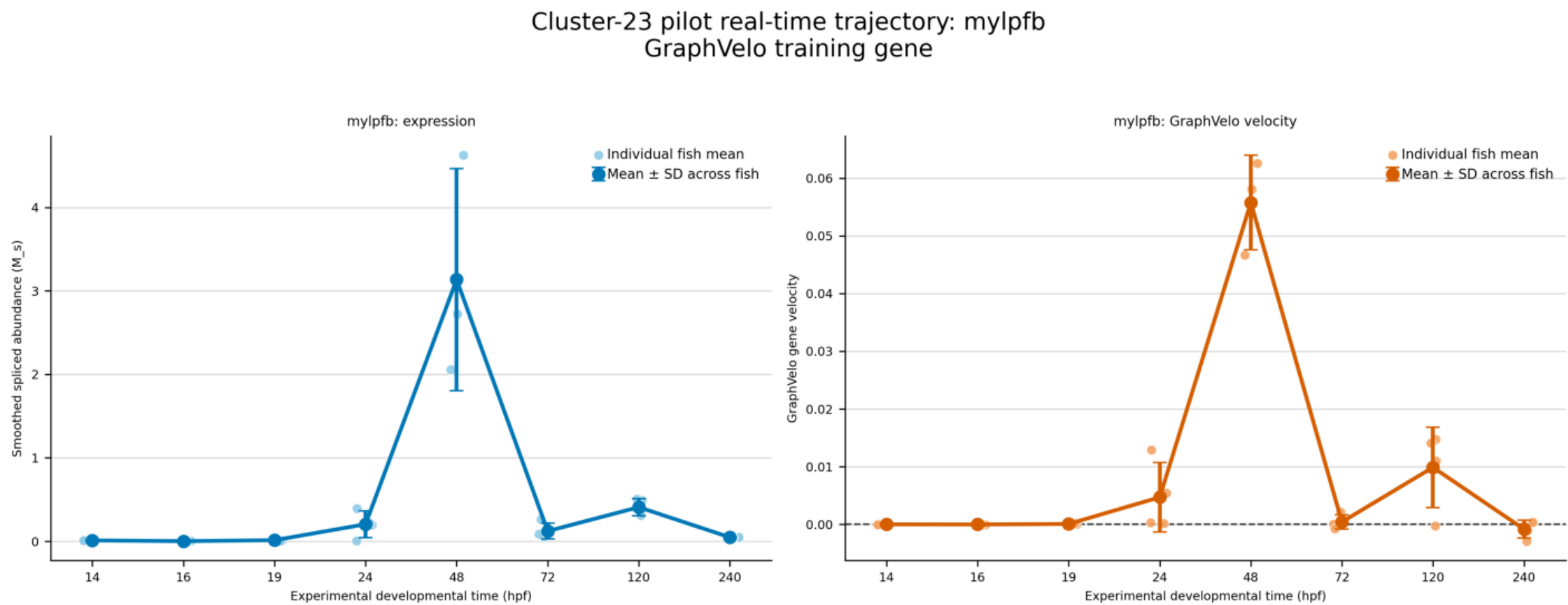
Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

65. mylpfa pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

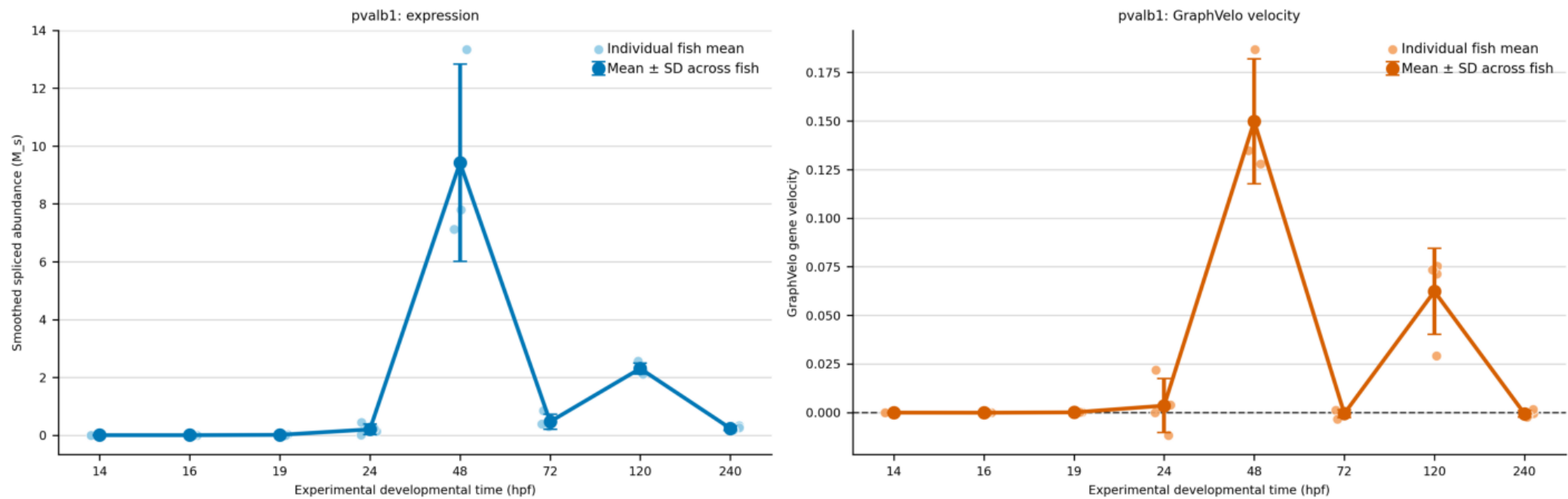
66. mylpfb pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

67. pvalb1 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

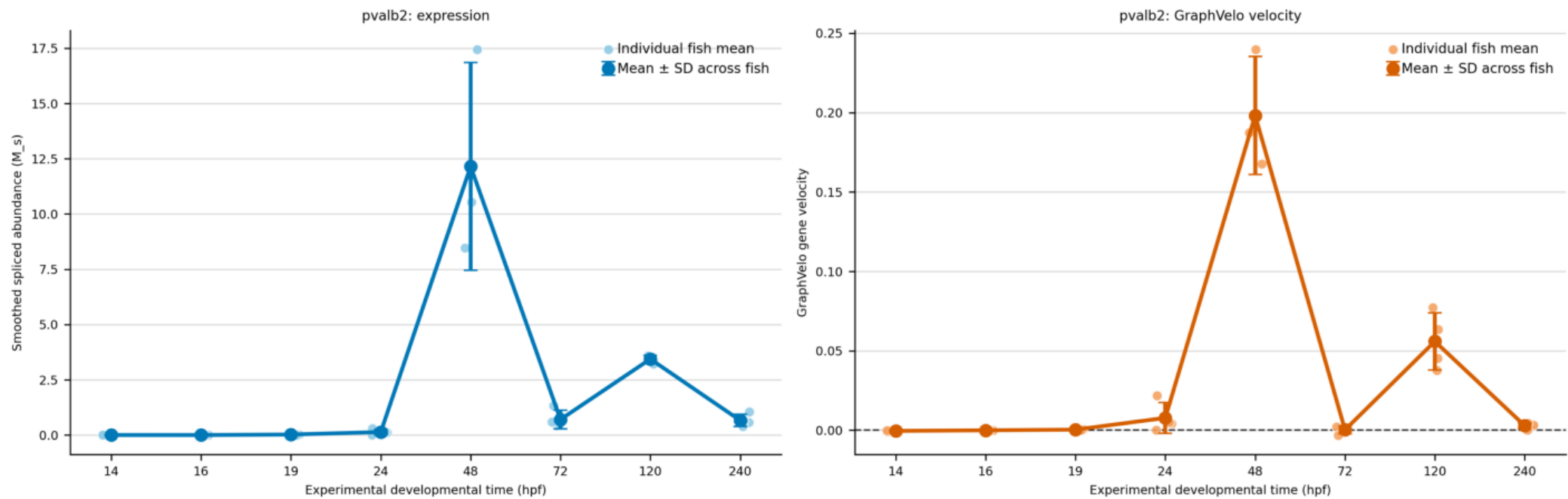
Cluster-23 pilot real-time trajectory: pvalb1
GraphVelo-projected gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

68. pvalb2 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

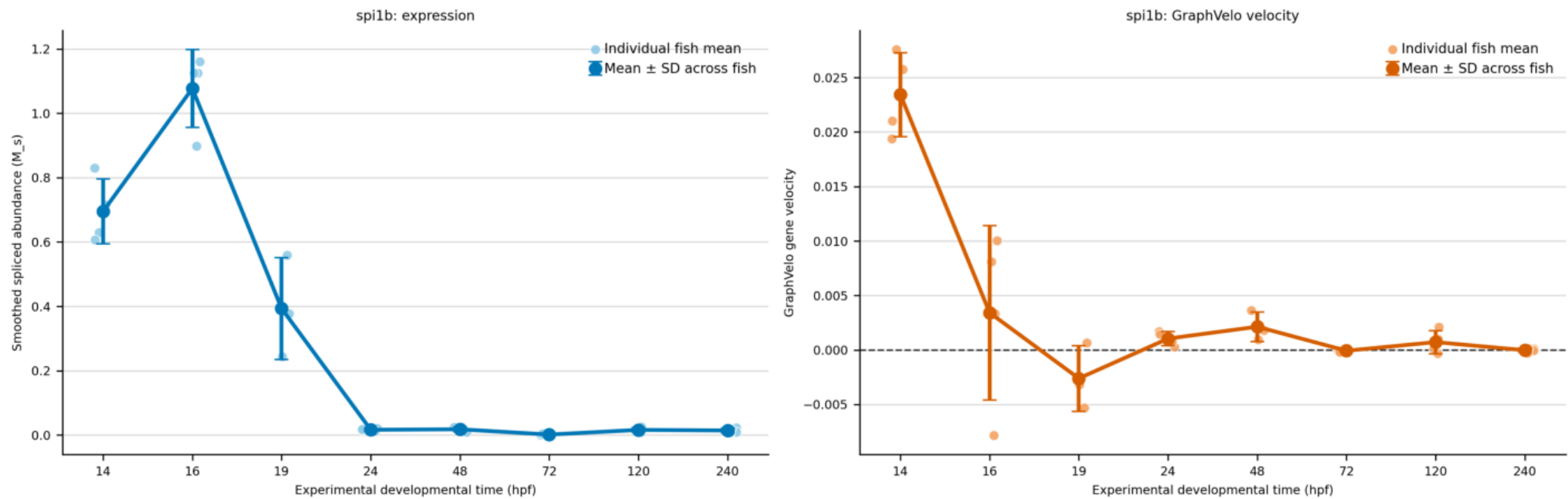
Cluster-23 pilot real-time trajectory: pvalb2
GraphVelo-projected gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

69. spi1b pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

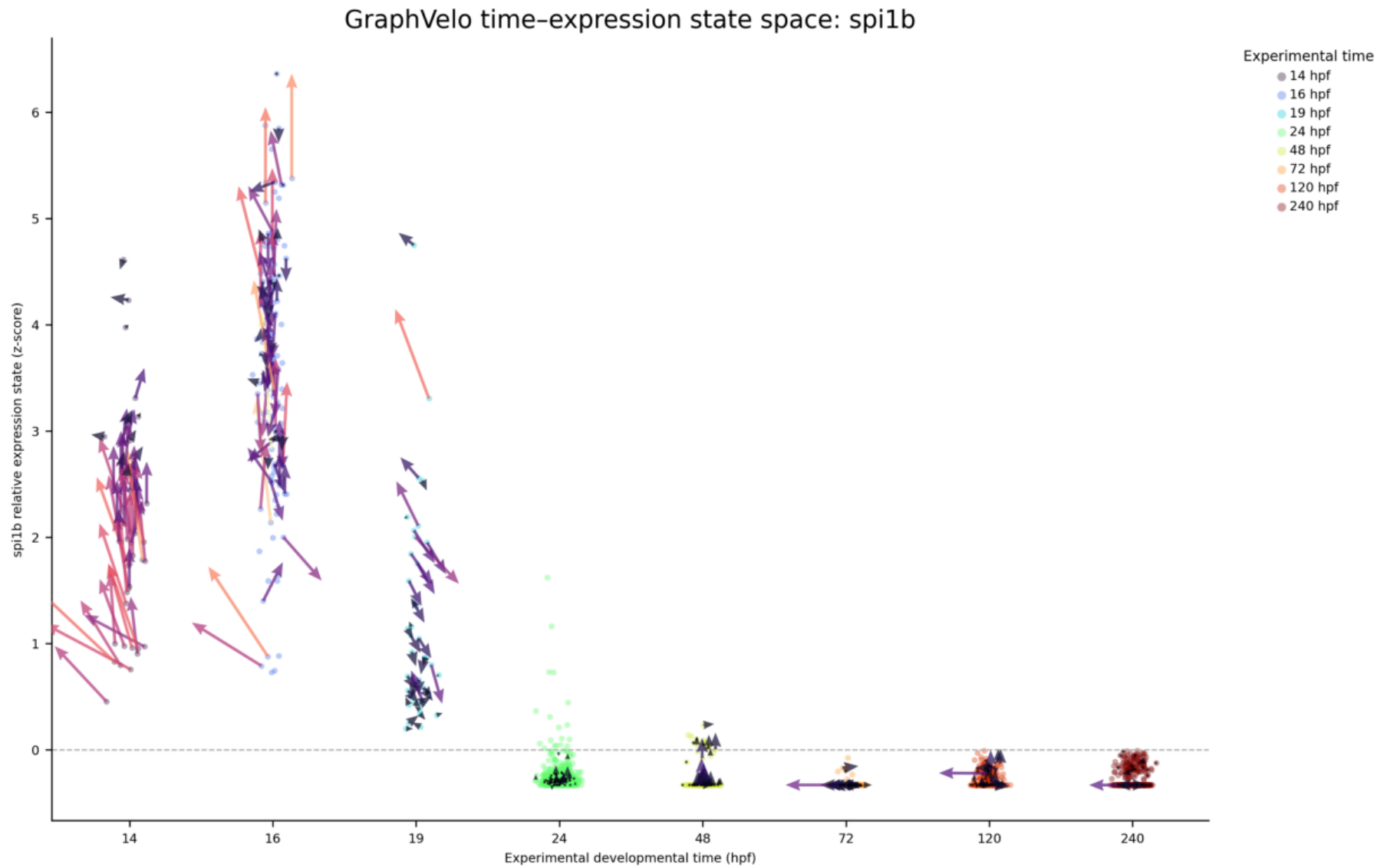
Cluster-23 pilot real-time trajectory: spi1b
GraphVelo training gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

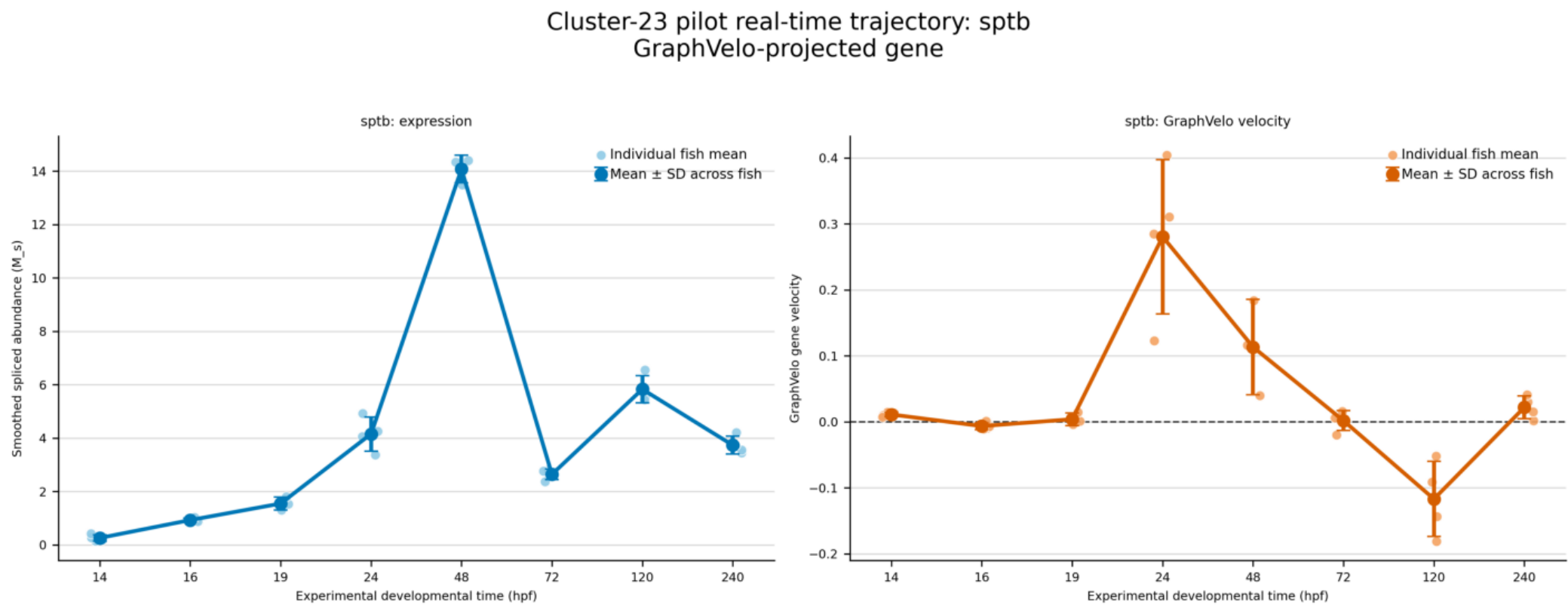
70. spi1b time expression manifold

EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

71. sptb pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

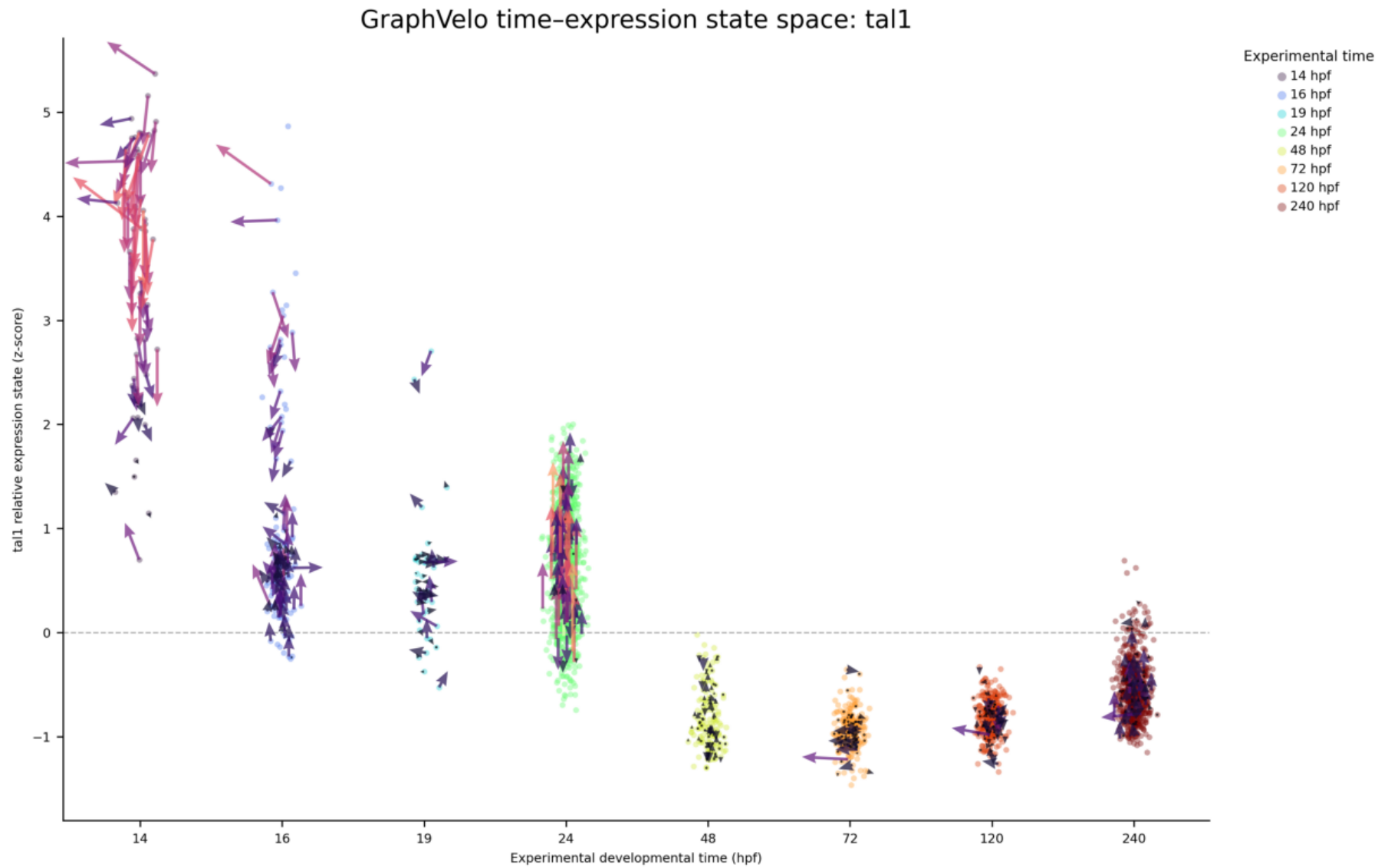
72. tal1 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

73. tal1 time expression manifold

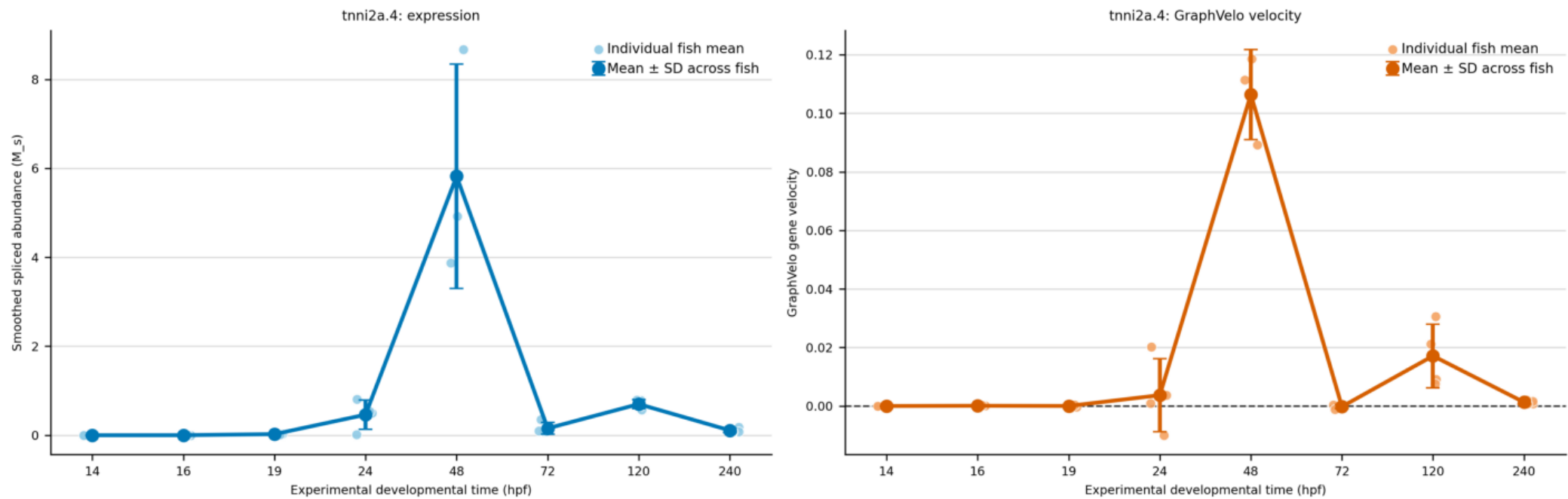
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

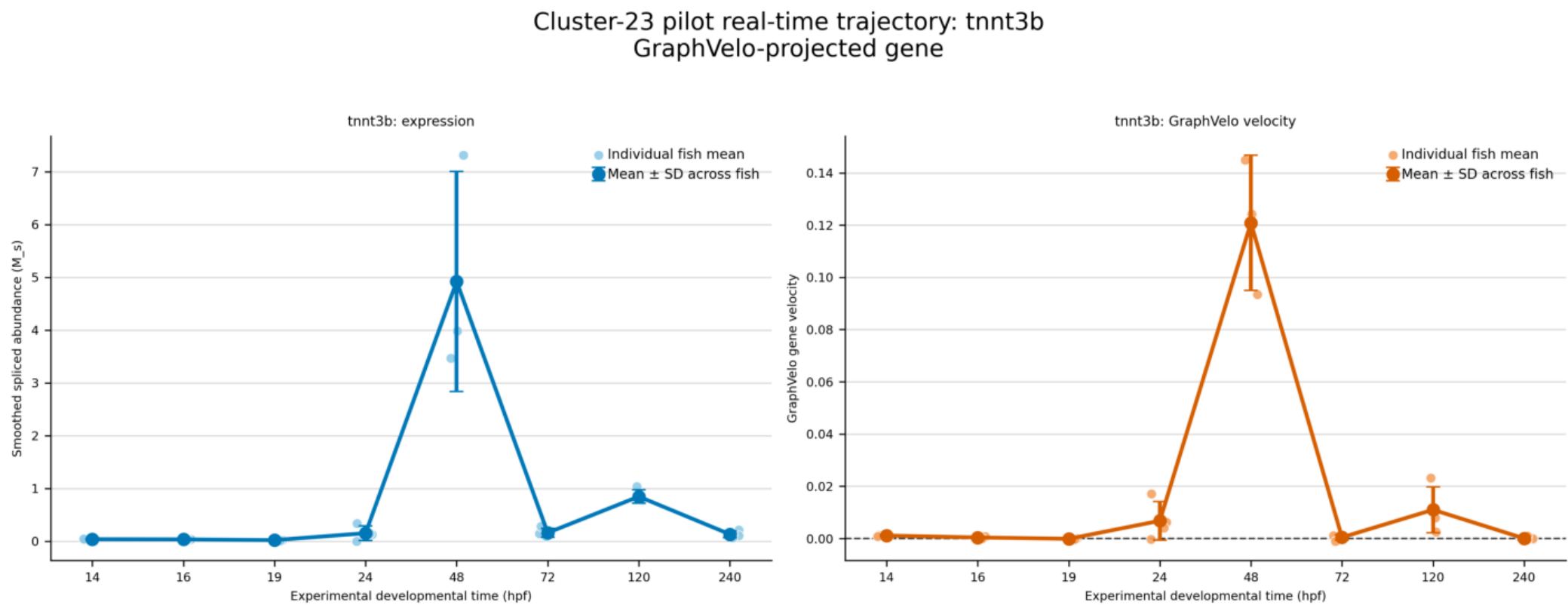
74. tnni2a.4 pilot trajectory
EXPLORATORY CLUSTER-23 PILOT

Cluster-23 pilot real-time trajectory: tnni2a.4
GraphVelo training gene



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

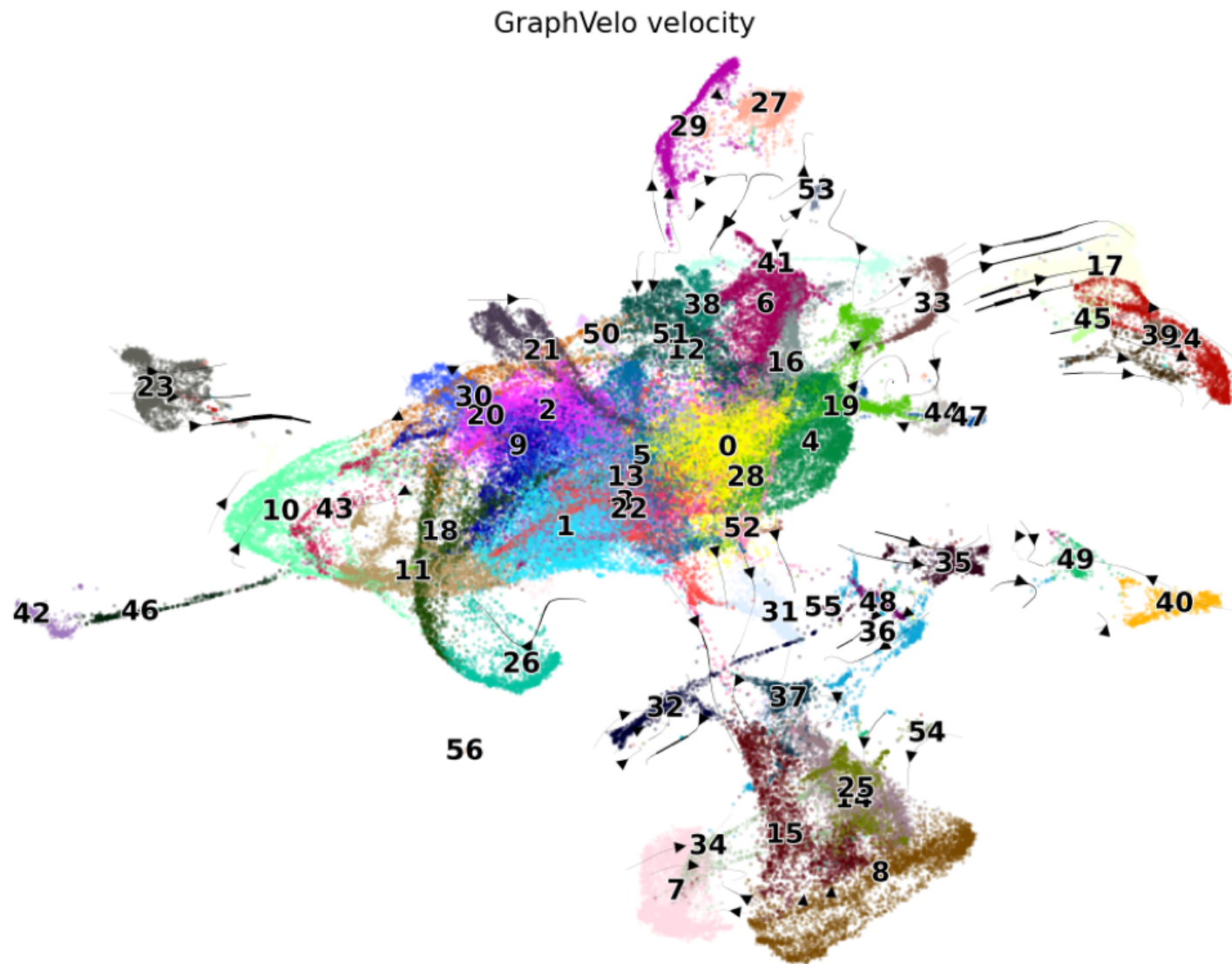
75. tnnt3b pilot trajectory
EXPLORATORY CLUSTER-23 PILOT



Cluster is not a lineage identity. Mixed blood/non-blood gene trajectories are exploratory and do not validate blood fate.

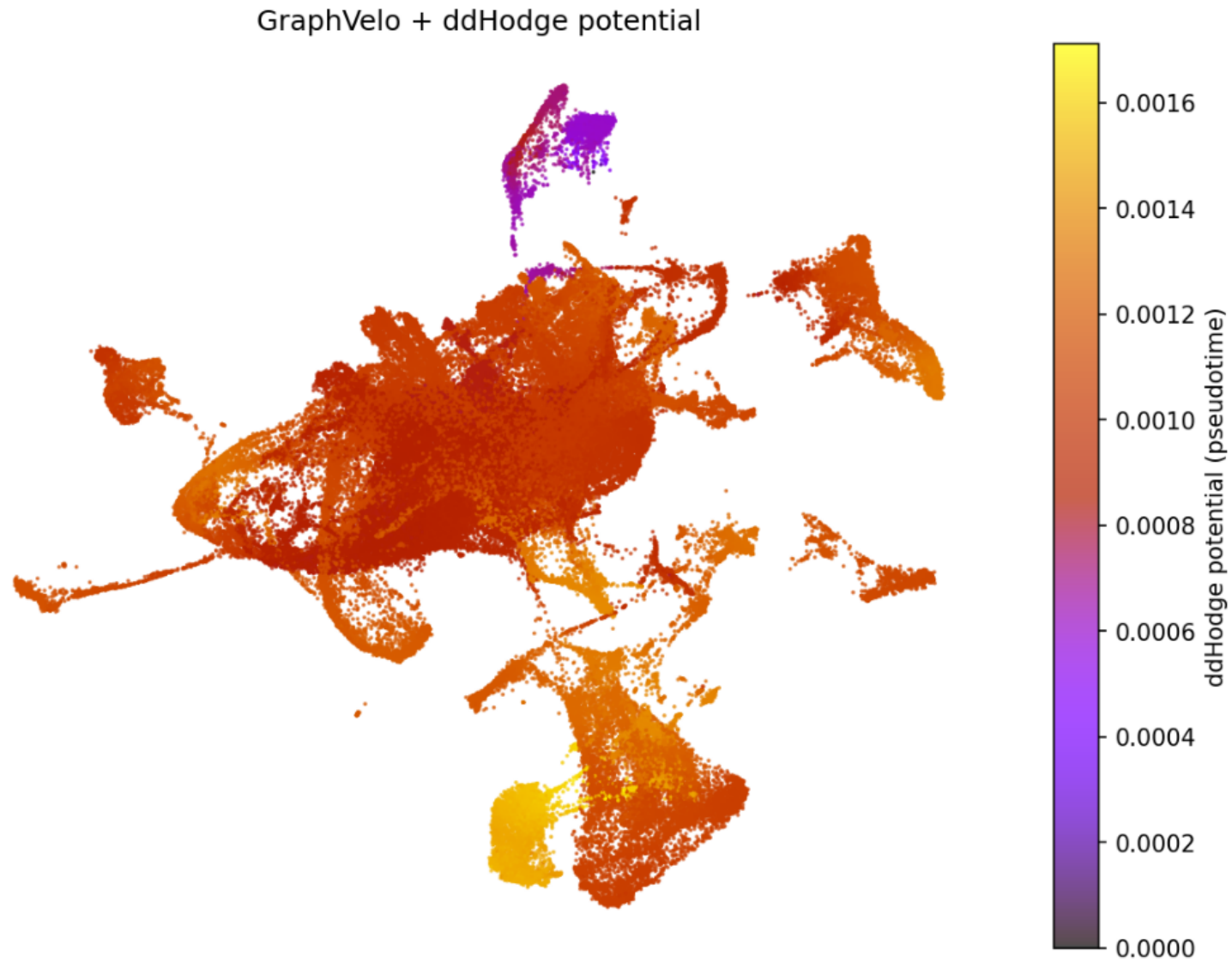
76. fig1 gv stream

EXPLORATORY WHOLE-ATLAS GRAPHVELO

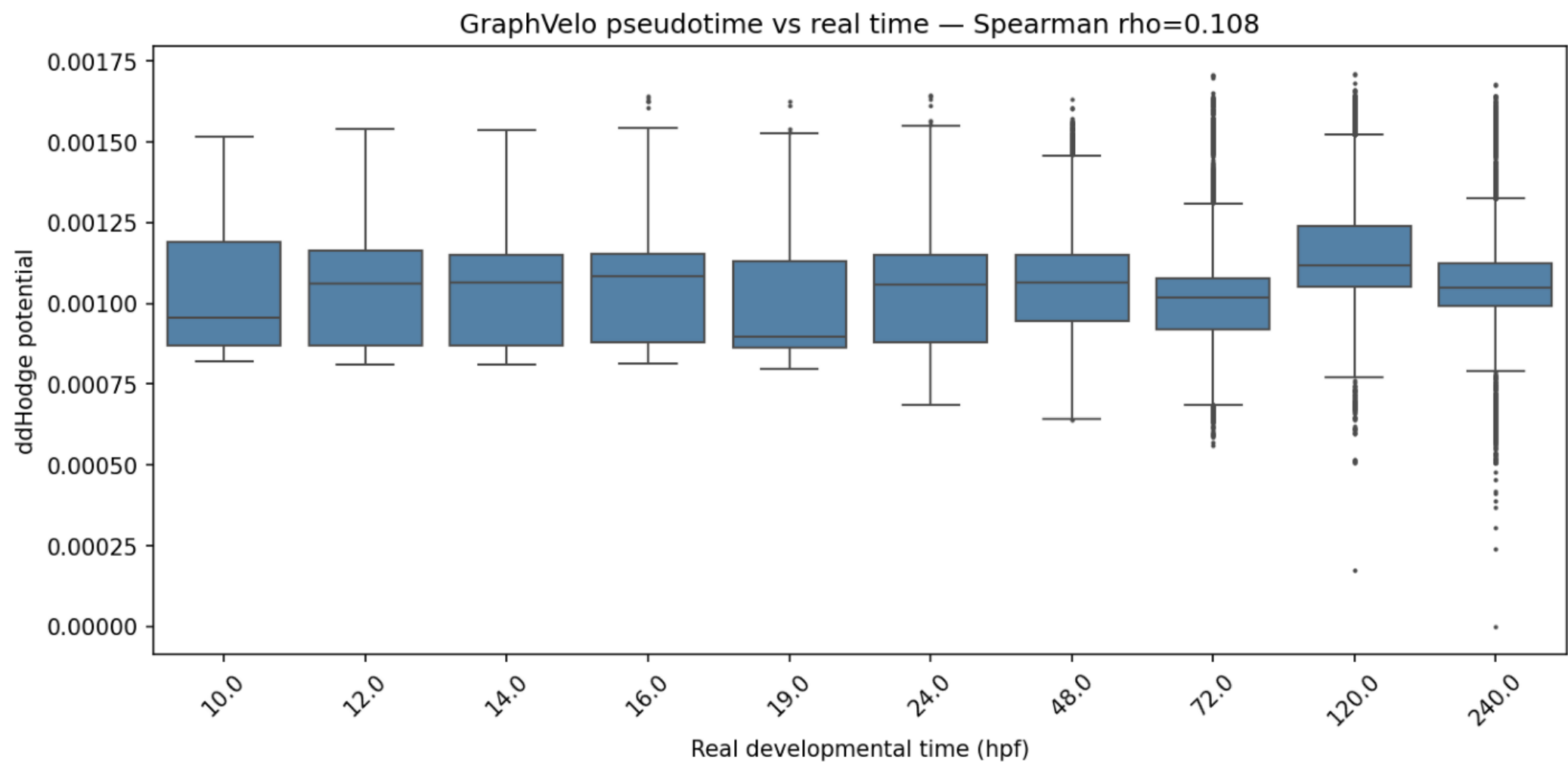


Whole-atlas velocity visualizations; not independent validation of the final blood cohort.

77. fig2 ddhodge potential
EXPLORATORY WHOLE-ATLAS GRAPHVELO



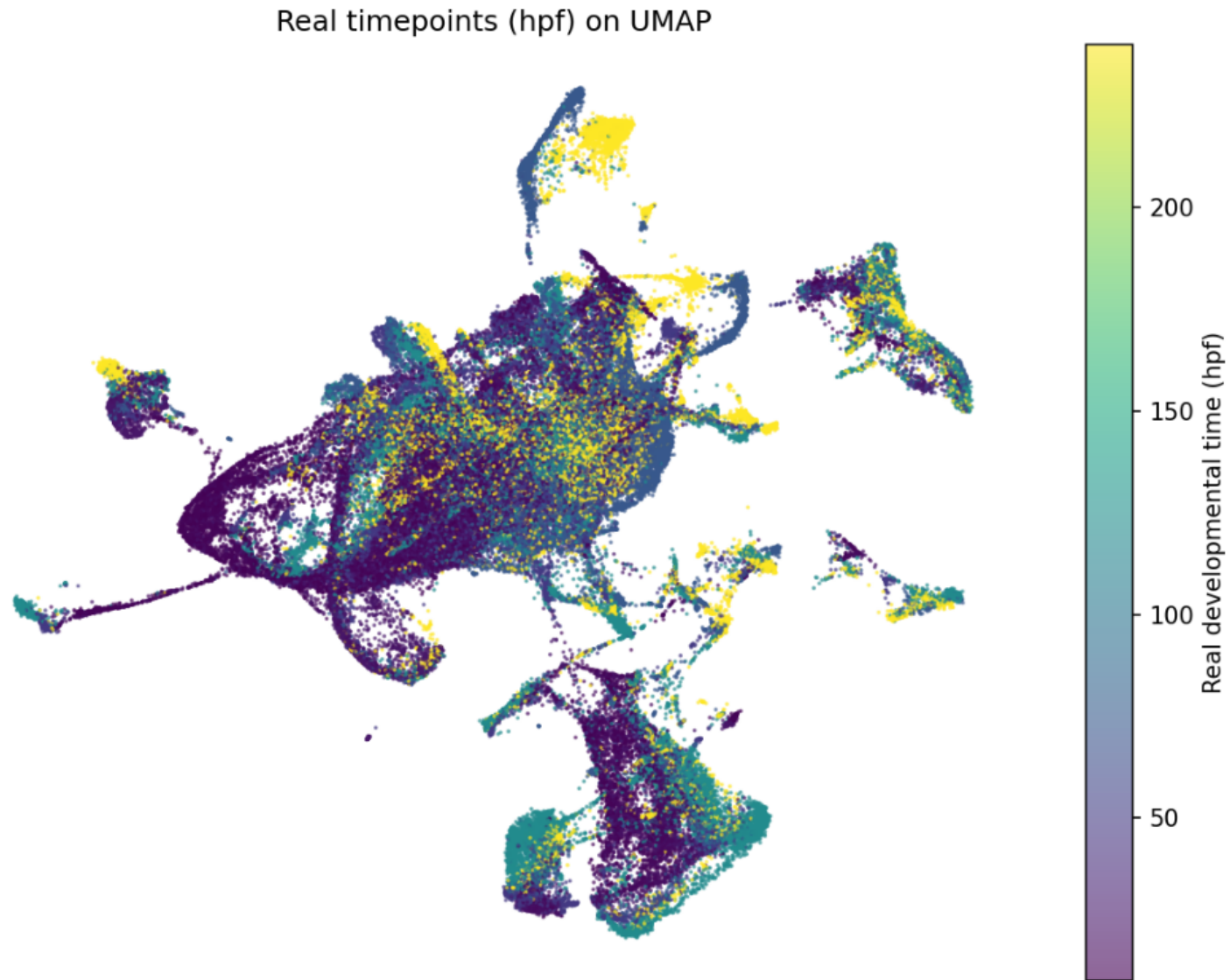
Whole-atlas velocity visualizations; not independent validation of the final blood cohort.



Whole-atlas velocity visualizations; not independent validation of the final blood cohort.

79. fig4 hpf umap

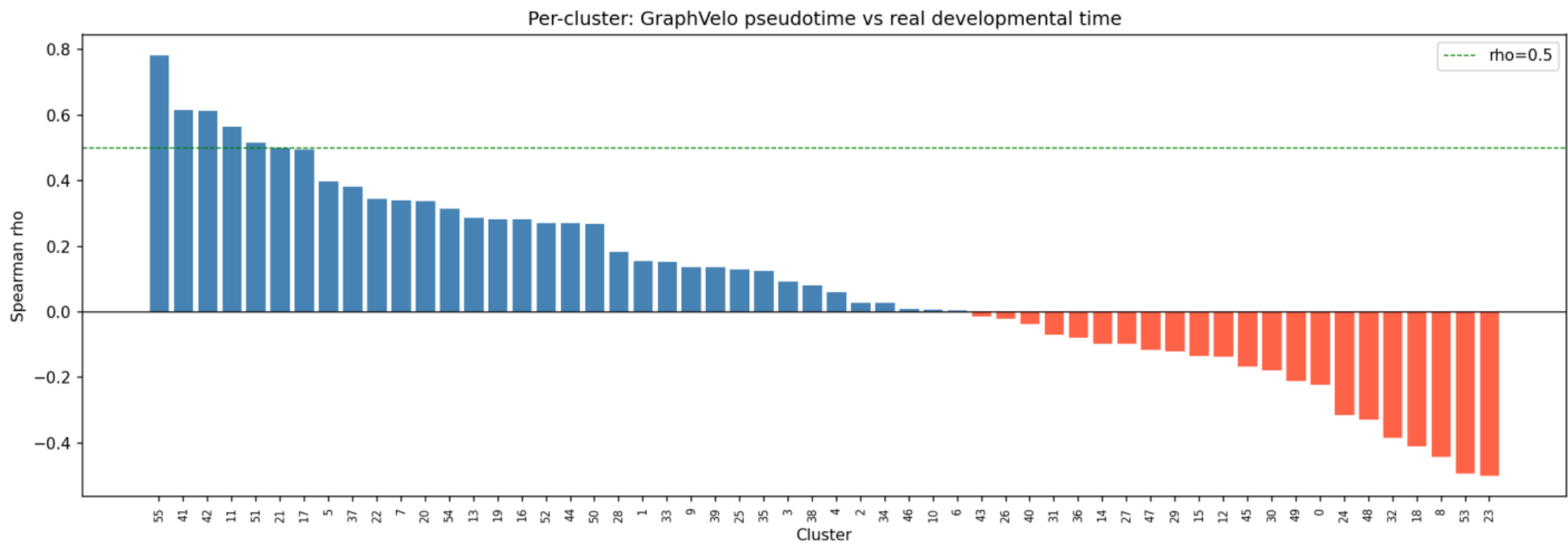
EXPLORATORY WHOLE-ATLAS GRAPHVELO



Whole-atlas velocity visualizations; not independent validation of the final blood cohort.

80. fig5 per cluster rho

EXPLORATORY WHOLE-ATLAS GRAPHVELO



Whole-atlas velocity visualizations; not independent validation of the final blood cohort.